



SSD212

Smart HMI Controller

Preliminary Data Sheet Version 0.5



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REVISION HISTORY

Revision No.	Description	Date
0.1	Initial release	09/11/2020
0.2	- Updated Ambient Operating Temperature Range - Updated Signal Description by adding Power Domain information	01/06/2021
0.3	Added Minimum Order Quantity, Moisture Sensitivity Level, and Storage and Soldering Application Note	01/21/2021
0.4	Added AC/DC Specifications, Hardware Power Sequence Procedure, MSPI Operation Example, and Thermal Resistance	02/26/2021
0.5	Updated UART Interface	06/11/2021

FEATURES

■ High Performance Processor Core

- ARM Cortex-A7 Dual Core up to 1 GHz
- 16KB I-Cache/16KB D-Cache/128KB L2-Cache
- Neon and FPU
- Memory Management Unit for Linux support
- DMA Engine

■ Display Subsystem

- Built-in contrast, brightness, sharpness, and saturation, 3D NR, Gamma control
- TTL output up to 1280x800 60fps with RGB565 or RGB666 or RGB888 format
- BT.656 output up to 720p60
- Serial RGB up to 800x600 60fps
- Supports SPI panel, clock frequency up to 54MHz
- Supports FHD graphic layer with Index 4/8, ARGB1555/ARGB4444/ARGB8888, and RGB565 format
- Supports UI/OSD layer with max. resolution 1280x800

■ 2D Graphics Engine

- Line draw
- Rectangle/gradient rectangle fill
- Bitblt/Stretch Bitblt/Italic Bitblt
- Palette mode (1/2/4/8-bit)
- Format transformation
- Color space conversion
- Clipping
- Alpha blending
- Rotation/Mirror
- Dither

■ Audio Processor

- Three mono ADCs or one mono + one stereo ADC for microphone input
- Two stereo DMIC inputs
- I2S TDM 8-channel, RX 2/4/8 channels, TX 2 channels
- One stereo DAC for lineout
- One HP Driver headphone set
- I2S supports 8K/16K/32K/48K/96KHz sampling rate audio recording

- ADC Pre-Amp gain supports 0dB, 6dB, 13dB, 23dB, 30dB, and 36dB
- ADC boost gain supports -6dB ~ 15dB or 0dB ~ 21dB with interval 3dB
- ADC digital gain supports -63.5dB ~ 33dB with interval 0.5dB, can be muted to zero
- SNR of DR A-Weighted ADC > 90dB (@gain = 0dB)

■ NOR/NAND Flash Interface

- Supports 1/2/4-bit SPI-NOR / NAND flash with two chip selects

■ SDIO 2.0 Interface

- Compatible with SDIO spec. 2.0, data bus 1/4 bit mode
- Compatible with SD spec. 2.0, data bus 1/4 bit mode

■ USB 2.0 Interface

- One USB2.0 configurable host and device
 - Host mode supports EHCI specification
 - Device mode supports 4 end points

■ DRAM Memory

- Embedded 16-bit 64MB DDR2 memory with max. 1333Mbps
- Supports auto-refresh and self-refresh mode

■ Ethernet

- Supports one Ethernet ports
- Supports 10/100Mbps half/full-duplex
- One built-in 10/100M Ethernet PHY
- Supports one RMII to connect external PHY
- Supports two LEDs for ePHY

■ Security Engines

- Supports AES/DES/3DES/RSA/SHA-1/SHA-256
- Supports secure booting

■ Boot options

- ROM
- SPI NOR
- SPI NAND with ECC
- SD Card and USB

■ Peripherals

- Dedicated GPIOs for system control
- Four PWM outputs
- Three generic UARTs and one fast UART with flow control
- Three generic timers and one watchdog timer
- Two SPI masters
- Two I2C masters
- Keypad supports up to 7x7, single mode

■ Miscellaneous

- Built-in eFuse with 1024-bit to store device ID, AES key, chip configurations, etc.
- Built-in power on reset (POR)
- Built-in SAR ADC with 2-channel analog inputs for different kinds of applications

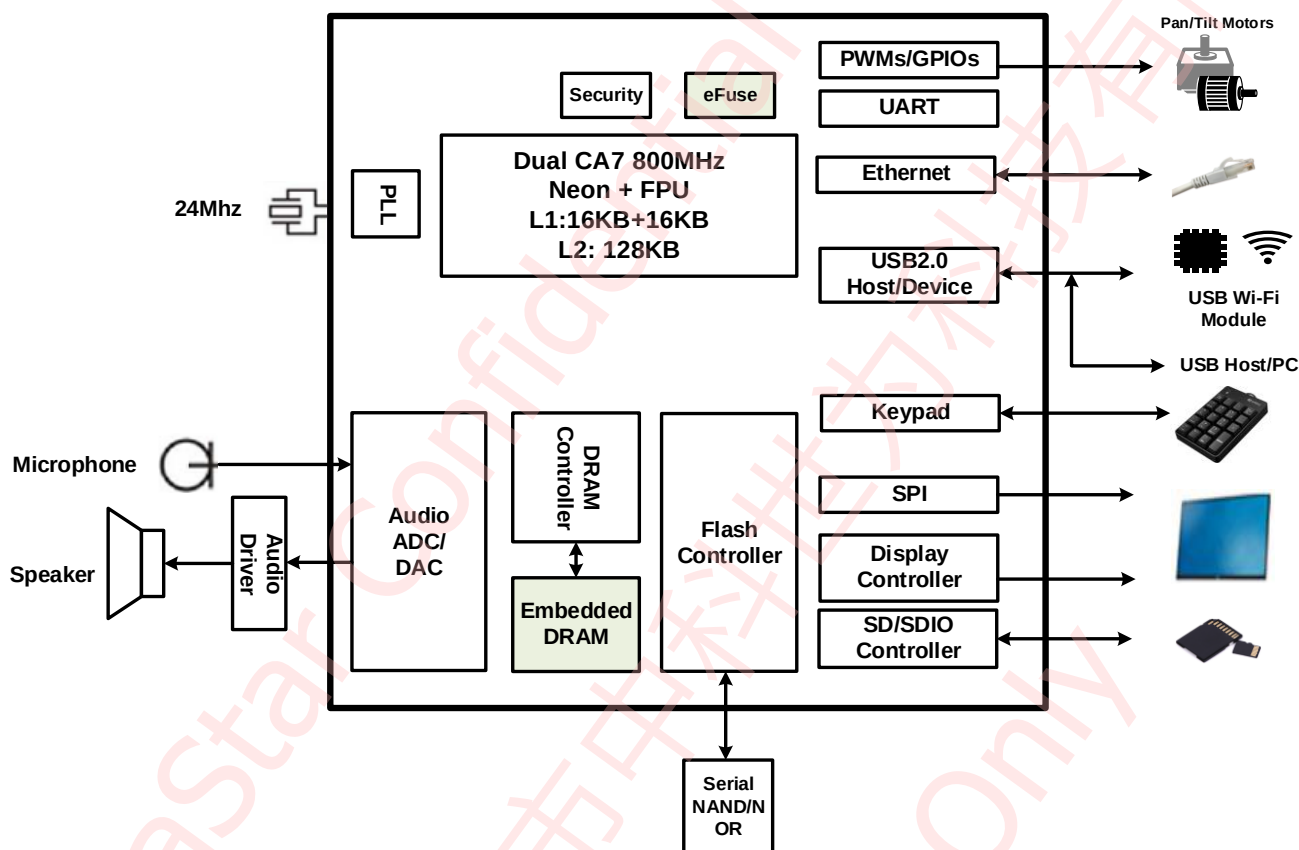
■ Operating Voltage Range

- Core: 0.9V
- I/O: 1.8V ~ 3.3V
- DRAM: 1.8V (DDR2)
- Power Consumption: TBD.
- Operation temperature -20°C ~ 85°C

■ Package

- 128-pin QFN, 12.3mm x 12.3mm
- Moisture Sensitivity Level: 3

BLOCK DIAGRAM



GENERAL DESCRIPTION

The SSD212 is a highly integrated SOC product for VOIP and smart display applications.

Based on ARM Cortex-A7 dual-core, the SSD212 integrates 2D graphics engine, TTL/serial RGB display with adjustable picture quality engine and other useful peripherals.

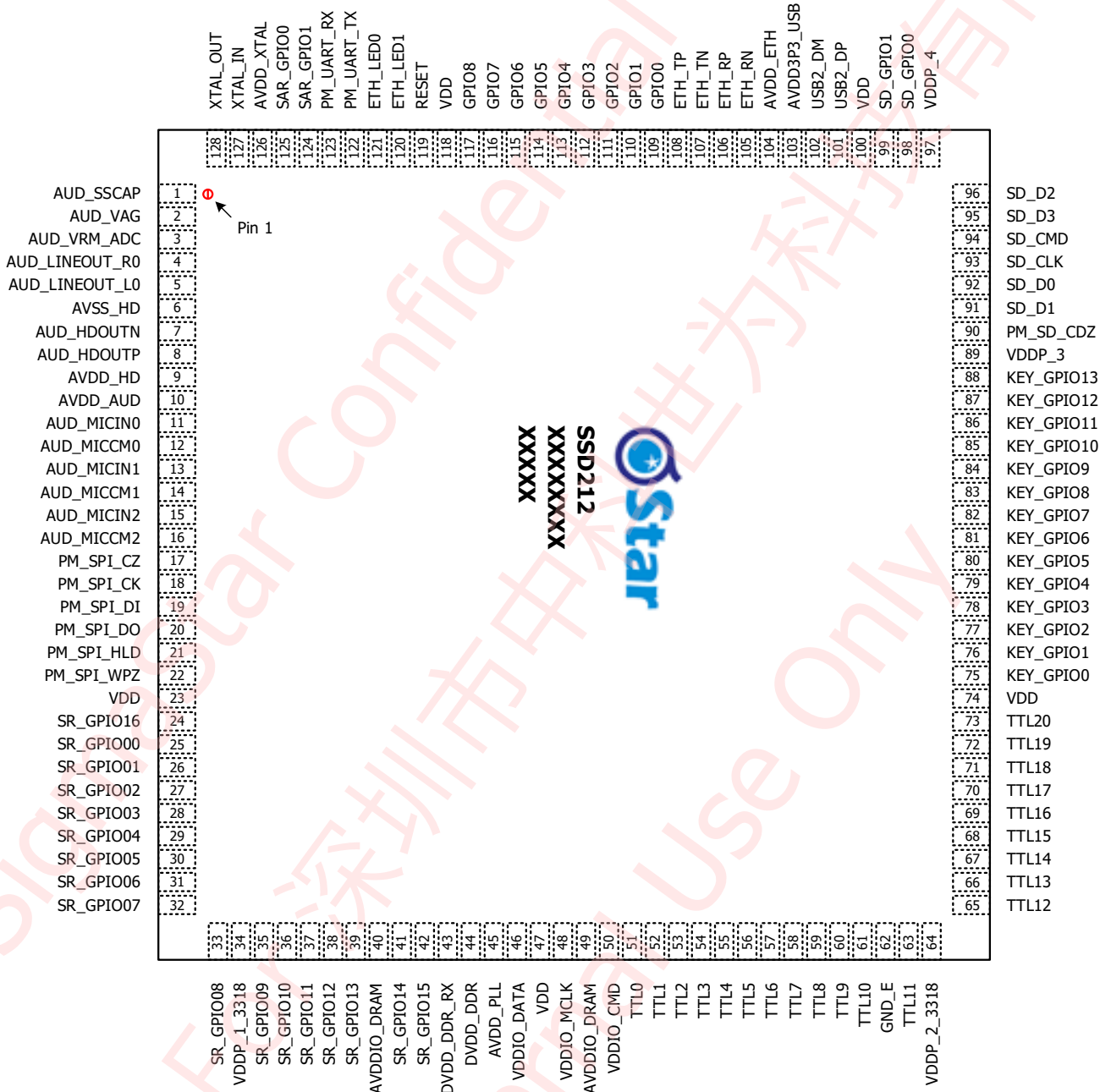
A typical utilization of the SSD212 application processor is demonstrated in the block diagram. The completed system includes NOR/NAND flash, DRAM, SD card, and USB port, and diversified audio connection. Before output to the panel, the images can be enhanced with respect to brightness/contrast/saturation/sharpness to give the best picture quality.

The NOR or NAND flash is usually reserved for operating system and application software. Moreover, other peripherals like SAR ADC, Audio ADC/DAC, UARTs, PWMs, GPIOs and SPI are supported to realize applications with maximal flexibility.

The SSD212 supports secure booting and personalization authentication mechanism for securing system. The AES/DDES/3DES cipher engines could also help encrypt the compressed video/audio streams for privacy protection.

The SSD212, powered by SigmaStar Technology, comes with a complete hardware platform and software SDK, allowing customers to speed up "Time-to-Market."

PIN DIAGRAM



SIGNAL DESCRIPTION

Signal Name	Signal Type	Function	Power Domain	Pin Location
System Reset Interface				
RESET	I	System Reset (Active High)	AVDD_XTAL	119
Debug UART Interface				
PM_UART_RX	I	Debug UART Receive Data Input with Pull Up Resistor / Slave I2C Serial Clock	AVDD_XTAL	123
PM_UART_TX	O	Debug UART Transmit Data Output with Pull Up Resistor / Slave I2C Serial Data	AVDD_XTAL	122
System Interface				
XTAL_IN	I	24MHz Crystal Input	AVDD_XTAL	127
XTAL_OUT	O	24MHz Crystal Output	AVDD_XTAL	128
SPI Flash Interface				
PM_SPI_CZ	O	SPI Chip Select (Active Low)	AVDD_XTAL	17
PM_SPI_CK	O	SPI Clock	AVDD_XTAL	18
PM_SPI_DI	O	SPI Serial Data To Device (MOSI)	AVDD_XTAL	19
PM_SPI_DO	I	SPI Serial Data From Device (MISO)	AVDD_XTAL	20
PM_SPI_WPZ	O	SPI Write Protect	AVDD_XTAL	22
PM_SPI_HLD	O	SPI Hold	AVDD_XTAL	21
GPIO Interface				
GPIO0	I/O	General Purpose Input/Output 0	AVDD_XTAL	109
GPIO1	I/O	General Purpose Input/Output 1	AVDD_XTAL	110
GPIO2	I/O	General Purpose Input/Output 2	AVDD_XTAL	111
GPIO3	I/O	General Purpose Input/Output 3	AVDD_XTAL	112
GPIO4	I/O	General Purpose Input/Output 4	AVDD_XTAL	113
GPIO5	I/O	General Purpose Input/Output 5	AVDD_XTAL	114
GPIO6	I/O	General Purpose Input/Output 6	AVDD_XTAL	115
GPIO7	I/O	General Purpose Input/Output 7	AVDD_XTAL	116
GPIO8	I/O	General Purpose Input/Output 8	AVDD_XTAL	117
SR_GPIO00	I/O	General Purpose Input/Output 0	VDDP_1_3318	25
SR_GPIO01	I/O	General Purpose Input/Output 1	VDDP_1_3318	26
SR_GPIO02	I/O	General Purpose Input/Output 2	VDDP_1_3318	27

Signal Name	Signal Type	Function	Power Domain	Pin Location
SR_GPIO03	I/O	General Purpose Input/Output 3	VDDP_1_3318	28
SR_GPIO04	I/O	General Purpose Input/Output 4	VDDP_1_3318	29
SR_GPIO05	I/O	General Purpose Input/Output 5	VDDP_1_3318	30
SR_GPIO06	I/O	General Purpose Input/Output 6	VDDP_1_3318	31
SR_GPIO07	I/O	General Purpose Input/Output 7	VDDP_1_3318	32
SR_GPIO08	I/O	General Purpose Input/Output 8	VDDP_1_3318	33
SR_GPIO09	I/O	General Purpose Input/Output 9	VDDP_1_3318	35
SR_GPIO10	I/O	General Purpose Input/Output 10	VDDP_1_3318	36
SR_GPIO11	I/O	General Purpose Input/Output 11	VDDP_1_3318	37
SR_GPIO12	I/O	General Purpose Input/Output 12	VDDP_1_3318	38
SR_GPIO13	I/O	General Purpose Input/Output 13	VDDP_1_3318	39
SR_GPIO14	I/O	General Purpose Input/Output 14	VDDP_1_3318	41
SR_GPIO15	I/O	General Purpose Input/Output 15	VDDP_1_3318	42
SR_GPIO16	I/O	General Purpose Input/Output 16	VDDP_1_3318	24
KEY GPIO Interface				
KEY_GPIO0	I/O	KEY General Purpose Input/Output 0	VDDP_3	75
KEY_GPIO1	I/O	KEY General Purpose Input/Output 1	VDDP_3	76
KEY_GPIO2	I/O	KEY General Purpose Input/Output 2	VDDP_3	77
KEY_GPIO3	I/O	KEY General Purpose Input/Output 3	VDDP_3	78
KEY_GPIO4	I/O	KEY General Purpose Input/Output 4	VDDP_3	79
KEY_GPIO5	I/O	KEY General Purpose Input/Output 5	VDDP_3	80
KEY_GPIO6	I/O	KEY General Purpose Input/Output 6	VDDP_3	81
KEY_GPIO7	I/O	KEY General Purpose Input/Output 7	VDDP_3	82
KEY_GPIO8	I/O	KEY General Purpose Input/Output 8	VDDP_3	83
KEY_GPIO9	I/O	KEY General Purpose Input/Output 9	VDDP_3	84
KEY_GPIO10	I/O	KEY General Purpose Input/Output 10	VDDP_3	85
KEY_GPIO11	I/O	KEY General Purpose Input/Output 11	VDDP_3	86
KEY_GPIO12	I/O	KEY General Purpose Input/Output 12	VDDP_3	87
KEY_GPIO13	I/O	KEY General Purpose Input/Output 13	VDDP_3	88
SAR ADC Interface				
SAR_GPIO0	I	General Purpose Input/Output or Muxed to SARADC Input Channel 0	AVDD_XTAL	125
SAR_GPIO1	I	General Purpose Input/Output or Muxed to SARADC Input Channel 1	AVDD_XTAL	124

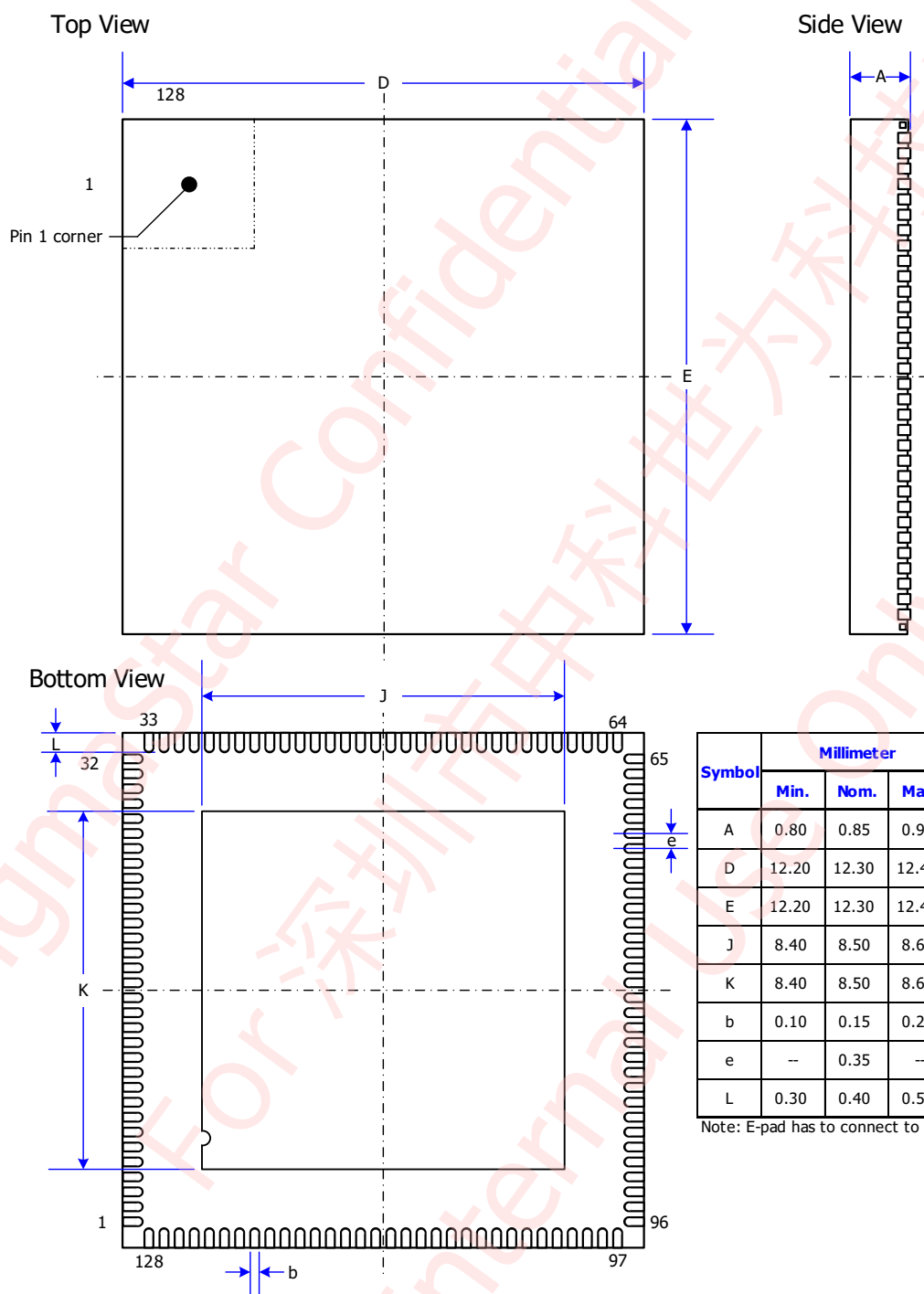
Signal Name	Signal Type	Function	Power Domain	Pin Location
TTL Interface				
TTL0	I/O	Parallel LCD General Purpose Input/Output 0	VDDP_2_3318	51
TTL1	I/O	Parallel LCD General Purpose Input/Output 1	VDDP_2_3318	52
TTL2	I/O	Parallel LCD General Purpose Input/Output 2	VDDP_2_3318	53
TTL3	I/O	Parallel LCD General Purpose Input/Output 3	VDDP_2_3318	54
TTL4	I/O	Parallel LCD General Purpose Input/Output 4	VDDP_2_3318	55
TTL5	I/O	Parallel LCD General Purpose Input/Output 5	VDDP_2_3318	56
TTL6	I/O	Parallel LCD General Purpose Input/Output 6	VDDP_2_3318	57
TTL7	I/O	Parallel LCD General Purpose Input/Output 7	VDDP_2_3318	58
TTL8	I/O	Parallel LCD General Purpose Input/Output 8	VDDP_2_3318	59
TTL9	I/O	Parallel LCD General Purpose Input/Output 9	VDDP_2_3318	60
TTL10	I/O	Parallel LCD General Purpose Input/Output 10	VDDP_2_3318	61
TTL11	I/O	Parallel LCD General Purpose Input/Output 11	VDDP_2_3318	63
TTL12	I/O	Parallel LCD General Purpose Input/Output 12	VDDP_2_3318	65
TTL13	I/O	Parallel LCD General Purpose Input/Output 13	VDDP_2_3318	66
TTL14	I/O	Parallel LCD General Purpose Input/Output 14	VDDP_2_3318	67
TTL15	I/O	Parallel LCD General Purpose Input/Output 15	VDDP_2_3318	68
TTL16	I/O	Parallel LCD General Purpose Input/Output 16	VDDP_2_3318	69
TTL17	I/O	Parallel LCD General Purpose Input/Output 17	VDDP_2_3318	70
TTL18	I/O	Parallel LCD General Purpose Input/Output 18	VDDP_2_3318	71
TTL19	I/O	Parallel LCD General Purpose Input/Output 19	VDDP_2_3318	72
TTL20	I/O	Parallel LCD General Purpose Input/Output 20	VDDP_2_3318	73
10/100M Ethernet Interface				
ETH_RN	I	10/100M Ethernet Differential Pair of Receiver Signal Negative	AVDD_ETH	105

Signal Name	Signal Type	Function	Power Domain	Pin Location
ETH_RP	I	10/100M Ethernet Differential Pair of Receiver Signal Positive	AVDD_ETH	106
ETH_TN	O	10/100M Ethernet Differential Pair of Transmitter Signal Negative	AVDD_ETH	107
ETH_TP	O	10/100M Ethernet Differential Pair of Transmitter Signal Positive	AVDD_ETH	108
ETH_LED0	O	10/100M Ethernet LED0 Control Driven Active When Linked	AVDD_XTAL	121
ETH_LED1	O	10/100M Ethernet LED1 Control Driven Active When Linked in 100 Base-TX and Blinking When Transmitting or Receiving Data	AVDD_XTAL	120
SD 2.0 Card Interface				
SD_CLK	O	SD 2.0 Clock	VDDP_4	93
SD_CMD	O	SD 2.0 Command	VDDP_4	94
SD_D0	I/O	SD 2.0 Data Bus 0	VDDP_4	92
SD_D1	I/O	SD 2.0 Data Bus 1	VDDP_4	91
SD_D2	I/O	SD 2.0 Data Bus 2	VDDP_4	96
SD_D3	I/O	SD 2.0 Data Bus 3	VDDP_4	95
PM_SD_CDZ	I	Power Manage SD 2.0 Card Detect	VDDP_4	90
SD_GPIO0	I/O	SD General Purpose Input/Output 0	VDDP_4	98
SD_GPIO1	I/O	SD General Purpose Input/Output 1	VDDP_4	99
Audio Line Out Interface				
AUD_LINEOUT_L0	O	Audio Left Channel Line Output	AVDD_AUD	5
AUD_LINEOUT_R0	O	Audio Right Channel Line Output	AVDD_AUD	4
AUD_VAG	O	Audio Reference Voltage from 1/2 AVDD_AUD	AVDD_AUD	2
AUD_VRM_ADC	I	Audio Reference Voltage for ADC	AVDD_AUD	3
AUD_SSCAP	I	Audio Reference CAP	AVDD_AUD	1
Audio Head Phone Out Interface				
AUD_HDOUTN	O	Audio Head Phone Out Differential Pair, Negative	AVDD_HD	7
AUD_HDOUTP	O	Audio Head Phone Out Differential Pair, Positive	AVDD_HD	8
Analog Microphone Interface				
AUD_MICIN0	I	Audio Left Channel Microphone Positive	AVDD_AUD	11

Signal Name	Signal Type	Function	Power Domain	Pin Location
		Input		
AUD_MICCM0	I	Audio Left Channel Microphone Negative Input	AVDD_AUD	12
AUD_MICIN1	I	Audio Right Channel Microphone Positive Input	AVDD_AUD	13
AUD_MICCM1	I	Audio Right Channel Microphone Negative Input	AVDD_AUD	14
AUD_MICIN2	I	Audio Left Channel Microphone Positive Input	AVDD_AUD	15
AUD_MICCM2	I	Audio Left Channel Microphone Negative Input	AVDD_AUD	16
USB 2.0 Interface				
USB2_DM	I/O	USB 2.0 Differential Pair, Negative	AVDD3P3_USB	102
USB2_DP	I/O	USB 2.0 Differential Pair, Positive	AVDD3P3_USB	101
Power pins				
VDD	Core Power	Digital Core Power		23, 47, 74, 100, 118
DVDD_DDR	Core Power	Digital Power for DDR TX		44
DVDD_DDR_RX	Core Power	Digital Power for DDR RX LDO (0.1uF CAP to GND)		43
VDDIO_DATA	DDR Power	IO Power for DDR Data		46
VDDIO_MCLK	DDR Power	IO Power for DDR Clock		48
VDDIO_CMD	DDR Power	IO Power for DDR Command		50
AVDDIO_DRAM	DDR Power	IO Power for embedded DRAM		40, 49
VDDP_1_3318	1.8V/3.3V Power	Digital Input/Output Power for Domain 1		34
VDDP_2_3318	1.8V/3.3V Power	Digital Input/Output Power for Domain 2		64
VDDP_3	3.3V Power	Digital Input/Output Power for Domain 3		89
VDDP_4	3.3V Power	Digital Input/Output Power for Domain 4		97
AVDD_PLL	3.3V Power	Analog Power for PLL		45
AVDD_XTAL	3.3V Power	Analog Power for XTAL		126
AVDD3P3_USB	3.3V Power	Analog Power for USB2.0		103
AVDD_ETH	3.3V Power	Analog Power for Ethernet		104
AVDD_AUD	3.3V Power	Analog Power for Audio		10
AVDD_HD	3.3V Power	Analog Power for Audio Head Phone		9

Signal Name	Signal Type	Function	Power Domain	Pin Location
GND_E	GND	Ground		62
AVSS_HD	GND	Analog Ground for Audio Head Phone		6

MECHANICAL DIMENSIONS



Symbol	Millimeter			Inch		
	Min.	Nom.	Max.	Min.	Nom.	Max.
A	0.80	0.85	0.90	0.031	0.033	0.035
D	12.20	12.30	12.40	0.480	0.484	0.488
E	12.20	12.30	12.40	0.480	0.484	0.488
J	8.40	8.50	8.60	0.331	0.335	0.339
K	8.40	8.50	8.60	0.331	0.335	0.339
b	0.10	0.15	0.20	0.004	0.006	0.008
e	--	0.35	--	--	0.014	--
L	0.30	0.40	0.50	0.012	0.016	0.020

Note: E-pad has to connect to system GND net.

STORAGE AND SOLDERING APPLICATION NOTE

Storage and SMT Reflow Information of Pb-free Parts

A. IC Storage Condition (refer to the caution label on packing box)

Item	Pb-free Part
Vacuum packing bag	Yes
Humidity indicator inside	Yes
Desiccant inside	Yes
Moisture sensitivity	MSL 3
Shelf life in sealed bag	12 months @ <40°C / 90% RH
Usage life after vacuum packing is unpacked	7 days below @ 30°C / 60% RH (see Note 2)
Re-baked IC when over usage life	24hrs @ 125°C

Note:

- The lead frame, substrate, Pb-free plating (Sn/Bi or Pure Sn), and solder ball must be prepared according to SigmaStar's qualification report.
- If the factory environment is over 30°C and 60% RH, the usage life would be reduced.

B. SMT Reflow Process

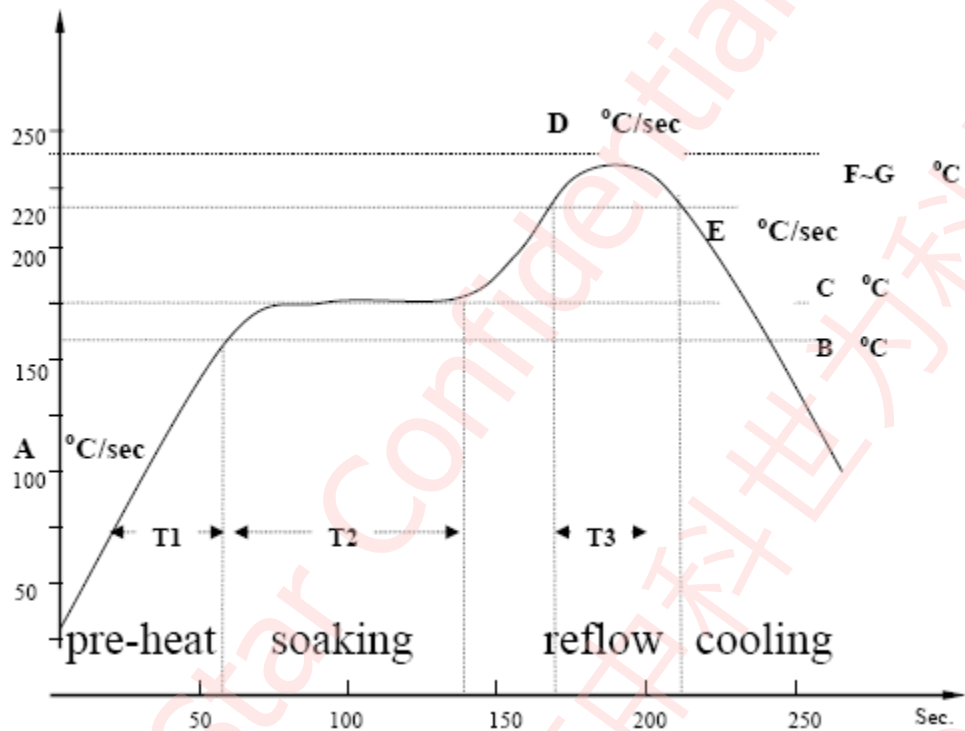
Item	Pb-free Part
Solder paste type	Pb-free (Sn/Ag/Cu)
Pre-heating time (RT~150°C) (T1)	50~80 seconds
Soaking time (150~200°C) (T2)	60~180 seconds
Ramp up rate	3°C/sec max.
Peak temp (guaranteed)	260°C max.
Peak temp (recommended)	245 +/- 5°C
Reflow time (above 217°C) (T3)	60~150 seconds
Retention time at peak temp	20~40 seconds
Cool down rate (175~125°C)	6°C/sec max.
Reflow cycle	3 times

Note:

- The reflow profiles in this document are for classification/preconditioning purpose only and are not meant to specify the board assembly profiles. The actual board assembly profiles should be developed based on specific process needs and board designs without exceeding the parameters given in the above table.

Recommendation for Pb-free Solder Reflow Profile

IR Reflow Temperature Profile (Pb-free)



- Note:
- The reflow profile illustrated above for Pb-free series is provided as a recommendation only. Please refer to the reflow profile provided by your solder paste vendor for reflow profile optimization.

ELECTRICAL SPECIFICATIONS

Interface Characteristics

Parameter	Symbol	Min.	Typ.	Max.	Unit
DIGITAL INPUTS					
Input Voltage, High	V_{IH}	$V_{DDP} \times 0.7$ Note			V
Input Voltage, Low	V_{IL}			0.8	V
Input Current, High	I_{IH}			-1.0	uA
Input Current, Low	I_{IL}			1.0	uA
Input Capacitance			5		pF
DIGITAL OUTPUTS					
Output Voltage, High	V_{OH}	$V_{DDP} - 0.1$ Note			V
Output Voltage, Low	V_{OL}			0.1	V
SAR ADC Input		0		V_{VDD_33}	V
AUDIO OUTPUTS					
Line-Out			2.54		Vp-p
XTAL Specifications					
Input Voltage, High	V_{IH}	2.0		3.6	V
Input Voltage, Low	V_{IL}	-0.3		0.8	V
Clock frequency			24		MHz
Crystal accuracy			+/-30		ppm
Long-term jitter			+/-500		ps

Note: VDDP typical voltage is 3.3V or 1.8V

Recommended Operating Conditions

Parameter	Symbol	Min	Typ.	Max.	Unit
3.3V Supply Voltage	V_{VDD_33}	3.14	3.3	3.46	V
1.8V Supply Voltage (DDR II)	V_{VDD_18}	1.71	1.8	1.89	V
Digital Input/Output Power for Domain 1	$V_{VDDP_1_3318}$	3.14	3.3	3.46	V
		1.71	1.8	1.89	V
Digital Input/Output Power for Domain 2	$V_{VDDP_2_3318}$	3.14	3.3	3.46	V
		1.71	1.8	1.89	V
Core Power Supply Voltage (Core)	V_{VDD_core}	0.87	0.9	0.93	V
Ambient Operation Temperature	T_A	-20		70	°C
Junction Temperature	T_J			125	°C

Absolute Maximum Ratings

Parameter	Symbol	Min	Typ.	Max.	Unit
3.3V Supply Voltage	V _{VDD_33}	2.97	3.3	3.63	V
1.8V Supply Voltage (DDR II)	V _{VDD_18}			1.98	V
3.3/1.8V I/O Supply Voltage	V _{VDDP_1_3318} V _{VDDP_2_3318}			3.63	V
Core Power Supply Voltage (Core)	V _{VDD_core}			1.1	V
Storage Temperature	T _{STG}	-40		150	°C

Note: Stresses above those listed in Absolute Maximum Ratings may cause permanent damage to the device. This is a stress rating only and does not imply functional operation of device. Exposure to absolute maximum ratings for extended periods may affect device reliability.

AC/DC SPECIFICATIONS

USB Interface

USB Full Speed DC Characteristics

Parameter	Min	Typ	Max	Unit
Input Levels				
Differential Receiver Input Sensitivity	0.2			V
Single-Ended Receiver Low Level Input Voltage			0.8	V
Single-Ended Receiver High Level Input Voltage	2.0			V
Output Levels				
Low Level Output Voltage			0.3	V
High Level Output Voltage	2.8			V
Termination				
Driver Output Impedance	40.5		49.5	Ω
Pull-up Resistor Impedance (Idle Bus)	0.9		1.575	k Ω
Pull-up Resistor Impedance (Upstream Port Receiving)	1.425		3.090	k Ω
Pull-down Resistor Impedance	14.25		24.80	k Ω

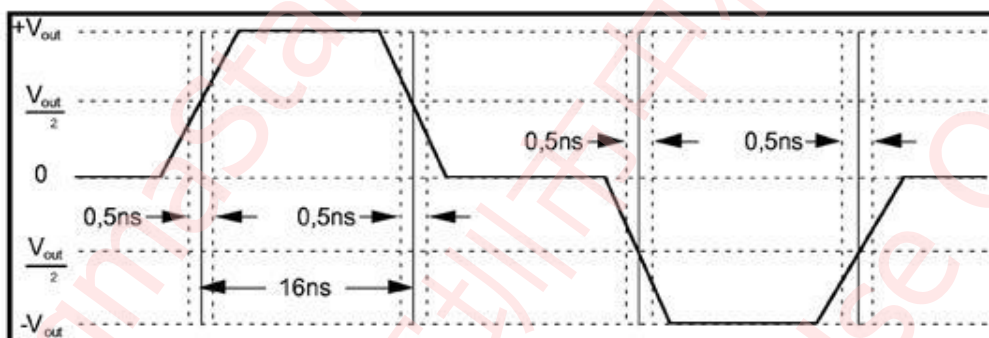
USB High Speed DC Characteristics

Parameter	Min	Typ	Max	Unit
Input Levels				
Differential Receiver Input Sensitivity	100			mV
Squelch Detection Threshold	100		150	mV
Output Levels				
Low Level Output Voltage (45 Ω Load)	-10		10	mV
High Level Output Voltage (45 Ω Load)	360		440	mV
Termination				
Driver Output Impedance	40.5		49.5	Ω

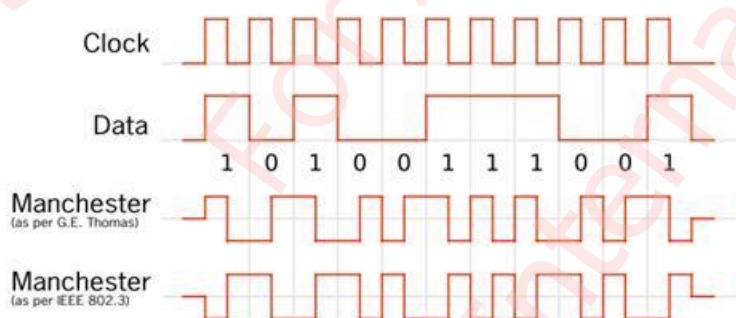
EPHY Interface

Parameter	Min	Typ	Max	Unit
ETHERNET ANALOG INTERFACE (10BASE-T)				
Analog Input Range	4.4	5	5.6	Vdp-p
Differential Input Impedance		100		ohm
ETHERNET ANALOG INTERFACE (100BASE-TX)				
Analog Input Range	1.9	2	2.1	Vdp-p
Differential Input Impedance		100		ohm
Rise/Fall Time	3	4	5	ns
Rise/Fall Time Symmetry			0.5	ns
Duty Cycle Distortion	-0.25		0.25	ns
Amplitude Symmetry	98	100	102	%
Overshoot			5	%

100BASE-TX



10BASE-T

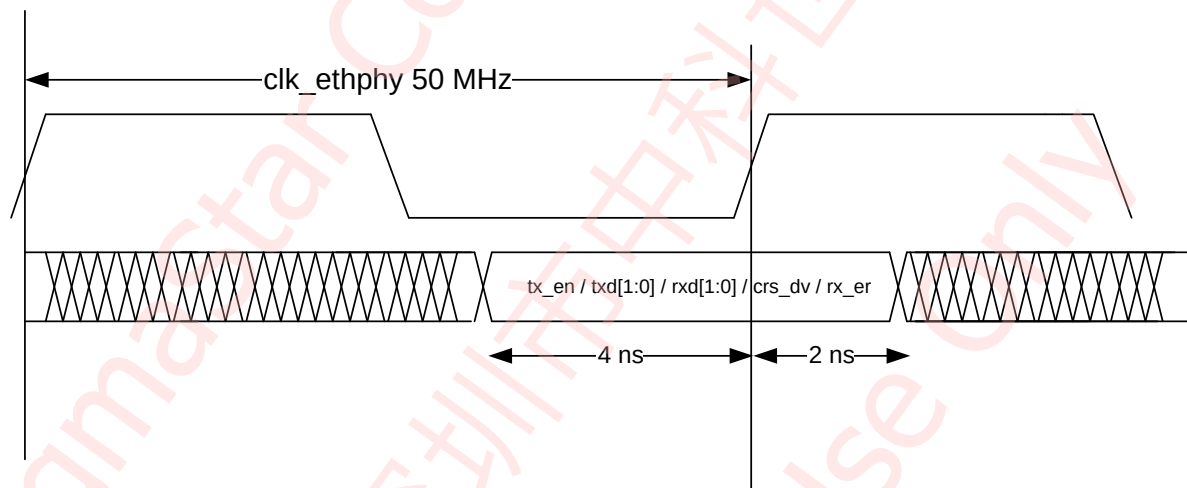


RMII interface

AC Characteristics

Symbol	Parameter	Min	Typ	Max	Unit
	REF_CLK Frequency		50		MHz
	REF_CLK Duty Cycle	35		65	%
Tsu	TXD[1:0], TX_EN, RXD[1:0], CRS_DV, RX_ER Data Setup to REF_CLK Rising Edge	4			ns
Thold	TXD[1:0], TX_EN, RXD[1:0], CRS_DV, RX_ER Data Hold from REF_CLK Rising Edge	2			ns

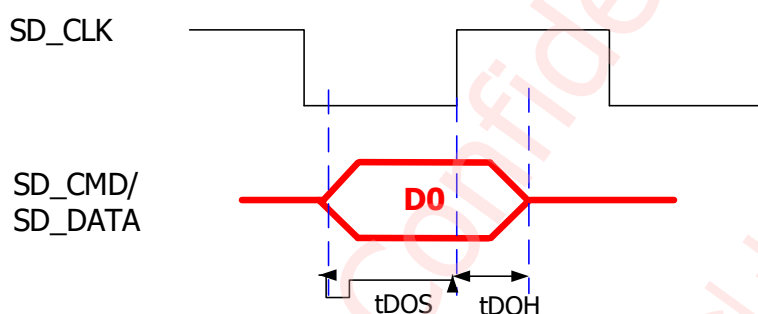
RMII Timing



SDIO Interface

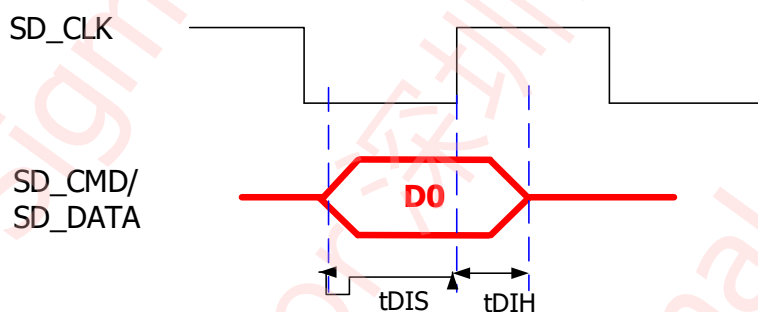
Data Output (TX) Timing

Symbol	Parameter	Min	Max	Unit	Remarks
tDOS	Data Output Setup Time	6	-	ns	
tDOH	Data Output Hold Time	6	-	ns	



Data Input (RX) Timing

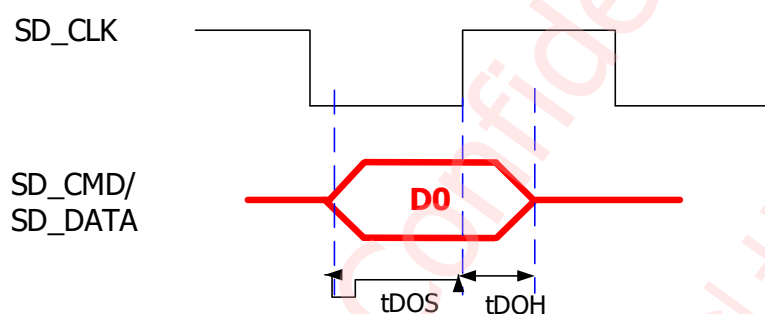
Symbol	Parameter	Min	Max	Unit	Remarks
tDIS	Data Input Setup Time	6	-	ns	
tDIH	Data Input Hold Time	1.5	-	ns	



SD Card Interface

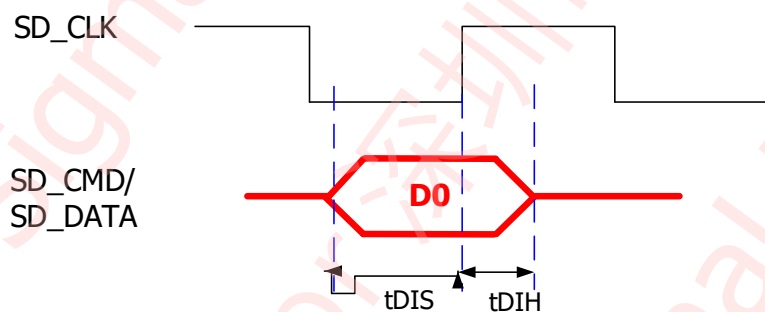
Data Output (TX) Timing

Symbol	Parameter	Min	Max	Unit	Remarks
tDOS	Data Output Setup Time	6	-	ns	
tDOH	Data Output Hold Time	6	-	ns	



Data Input (RX) Timing

Symbol	Parameter	Min	Max	Unit	Remarks
tDIS	Data Input Setup Time	6	-	ns	
tDIH	Data Input Hold Time	1.5	-	ns	



Audio Interface

Stereo Audio-DAC Interface

Parameter	Min	Typ	Max	Unit
Frequency Response (20KHz)	-1.0	0	+0.1	dB
THD+N (1KHz sine-wave; -3dBFS) [dB, non A-weighted]		-80		dB
S/N-ratio [dB, A-weighted]		92		dB
Dynamic-range [dB, A-weighted]		92		dB
Crosstalk (1KHz sine-wave: full scale)		-90		dB
Analog Output Level (1KHz sine-wave: full scale)		1		Vrms
Load Impedance		1000		ohm

Stereo Audio-ADC Interface

Parameter	Min	Typ	Max	Unit
THD+N (1KHz sine-wave; -3dBFS) [dB, non A-weighted]		-80		dB
S/N-ratio [dB, A-weighted]		90		dB
Dynamic Range [dB, A-weighted]		90		dB
Crosstalk (1KHz sine-wave: full scale)		-85		dB
Analog Input Range (-1.4dBFS)		1		Vrms

Headphone Interface

Parameter	Min	Typ	Max	Unit
THD+N (1KHz sine-wave; -3dBFS) [dB, non A-weighted]		-80		dB
S/N-ratio [dB, A-weighted]		96		dB
Dynamic-range [dB, A-weighted]		96		dB
Analog Output Level (1KHz sine-wave: full scale)		1.6		Vrms

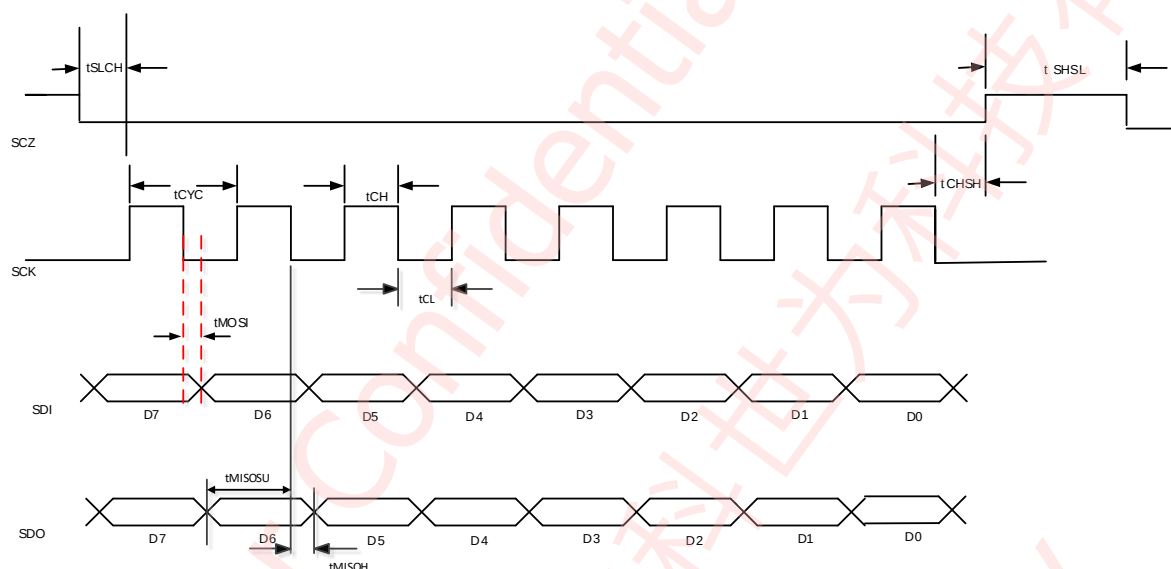
SAR Interface

SAR ADC DC Spec

Parameter	Min	Typ	Max	Unit
SAR ADC Input	0		AVDD_NODIE	V

SPI NOR Interface

SPI Data Timing - CSZ, SCK, SDI and SDO



SPI AC Characteristics for Operation

Parameter	Symbol	Min	Typ	Max	Unit	Remarks
CS High	tSHSL	1		15	tCYC	
CS Setup	tSLCH	1		15	tCYC	
CS Hold	tCHSH	1		15	tCYC	
SCK period	tCYC	9.3			ns	
SCK High Time	tCH	45	50	55	%	
SCK Low Time	tCL	45	50	55	%	
Master Out Slave In	tMOSI	-2		(tCYC/2)-2	ns	Relative to the falling edge of SCK
Master In Slave Out Setup time	tMISOSU	4.6			ns	Relative to the falling edge of SCK
Master In Slave Out Hold time	tMISOH	0.3			ns	Relative to the falling edge of SCK

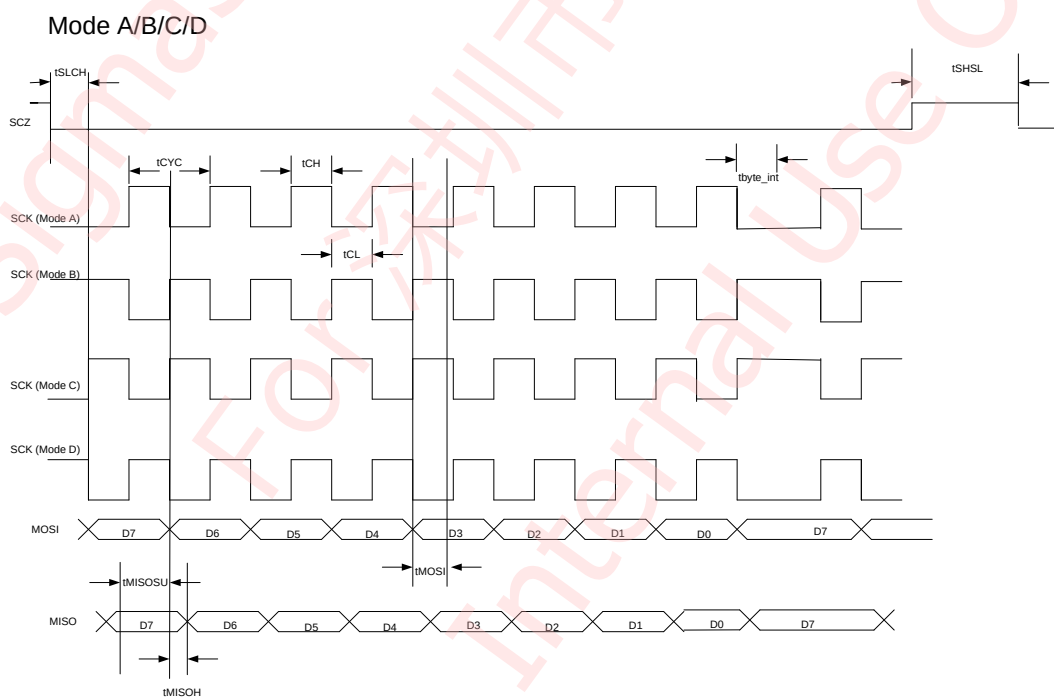
SPI DC Characteristics for Operation

Parameter	Symbol	Min	Typ	Max	Unit	Remarks
Input Low Voltage	VIL			0.9	V	
Input High Voltage	VIH	2.0			V	

MSPI Interface

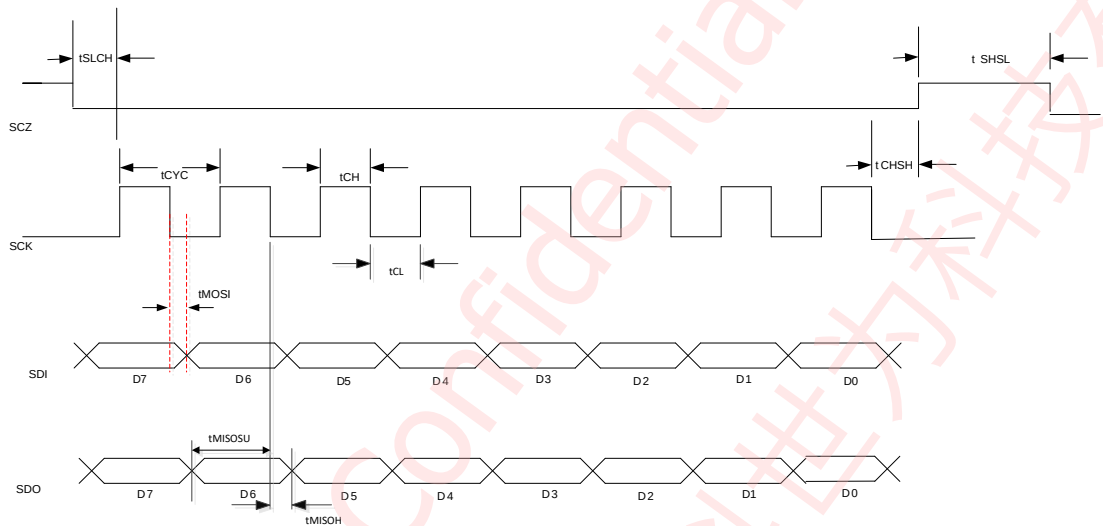
AC Characteristics

Parameter	Symbol	Min	Max	Unit	Remarks
SCZ Active Setup Time Relative to SCK	tSLCH	4	255	cycle	1 cycle = tCYC
SCZ Deselect Time	tSHSL	4.5	255	cycle	1 cycle = tCYC
SCK Period Time	tCYC	37.1		ns	
SCK High Time	tCH	tCYC/2		ns	
SCK Low Time	tCL	tCYC/2		ns	
Master Out Slave In	tMOSI	5-(tCYC/2)	(tCYC/2)-2	ns	Relative to the falling edge of SCK
Master In Slave Out Setup Time	tMISOSU	6		ns	Relative to the falling edge of SCK
Master In Slave Out Hold Time	tMISOH	0.3		ns	Relative to the falling edge of SCK
Byte Interval	tbyte_int	0	255	cycle	1 cycle = tCYC



SPI NAND Interface

SPI Data Timing - CSZ, SCK, SDI and SDO



SPI AC Characteristics for Operation

Parameter	Symbol	Min	Typ	Max	Unit	Remarks
CS High	tSHSL	1		15	tCYC	
CS Setup	tSLCH	1		15	tCYC	
CS Hold	tCHSH	1		15	tCYC	
SCK period	tCYC	9.3			ns	
SCK High Time	tCH	45	50	55	%	
SCK Low Time	tCL	45	50	55	%	
Master Out Slave In	tMOSI	-2		(tCYC/2)-2	ns	Relative to the falling edge of SCK
Master In Slave Out Setup time	tMISOSU	4.6			ns	Relative to the falling edge of SCK
Master In Slave Out Hold time	tMISOH	0.3			ns	Relative to the falling edge of SCK

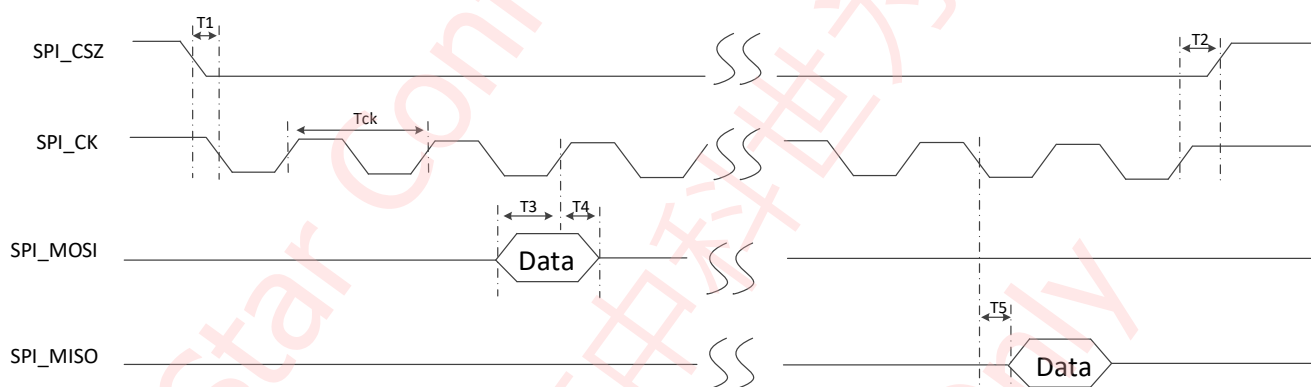
SPI DC Characteristics for Operation

Parameter	Symbol	Min	Typ	Max	Unit	Remarks
Input Low Voltage	VIL			0.9	V	
Input High Voltage	VIH	2.0			V	

PSPI Interface

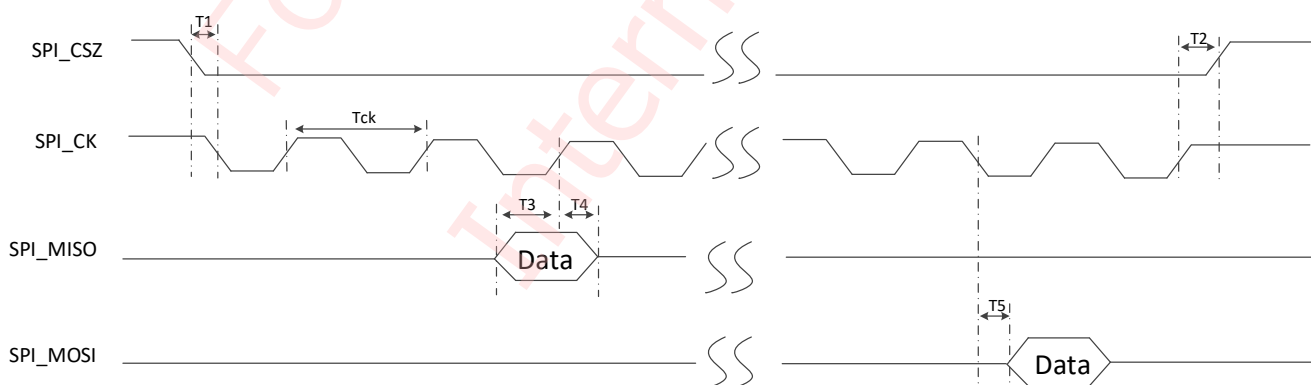
Slave AC Characteristics

Description	Symbol	Min	Typ	Max	Unit
SPI_CK Frequency	Tck	T		54	MHz
SPI_CSZ go low to SPI_CK go low	T1	$>1000/T$			ns
SPI_CK go high to SPI_CSZ go high	T2	$>1000/T$			ns
SPI_MOSI setup time from SPI_CK go high	T3	4			ns
SPI_MOSI hold time after SPI_CK go high	T4	1			ns
SPI_MISO output delay after SPI_CK go low	T5			11	ns



Master AC Characteristics

Description	Symbol	Min	Typ	Max	Unit
SPI_CK Frequency	Tck	T		54	MHz
SPI_CSZ go low to SPI_CK go low	T1	$1000/T$			ns
SPI_CK go high to SPI_CSZ go high	T2	$1000/T$			ns
SPI_MOSI setup time from SPI_CK go high	T3	5			ns
SPI_MOSI hold time after SPI_CK go high	T4	2			ns
SPI_MISO output delay after SPI_CK go low	T5			8	ns



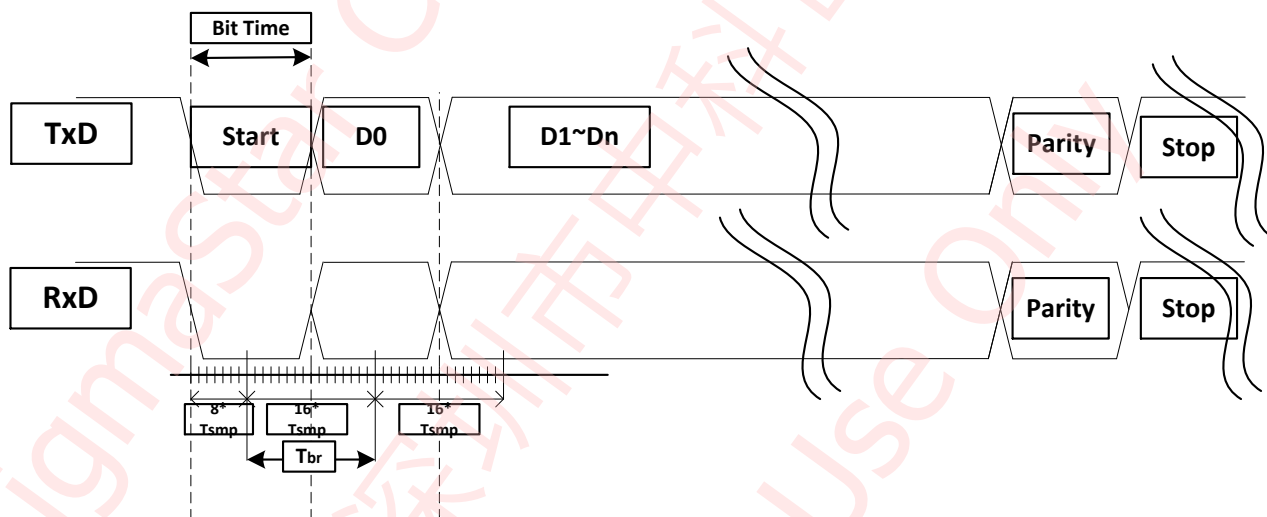
UART Interface

AC Characteristics

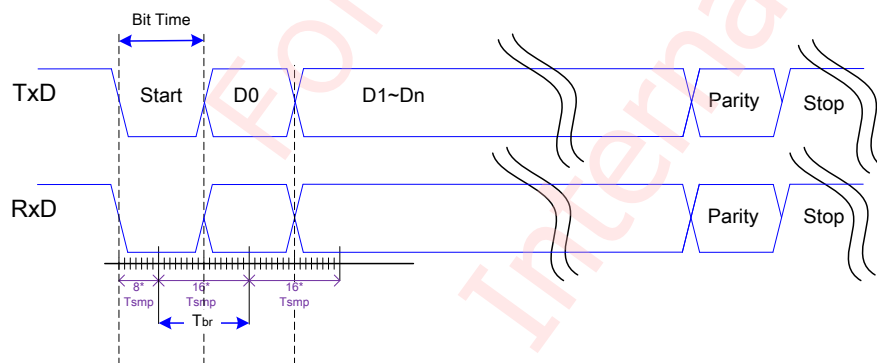
Parameter	Min	Typ	Max	Unit
Baud Rate Period (Tbr)		115200		bps
UART Sampling Period (Tsmpl)		1/16		Tbr
RTS low to RxD on (t1)	0.1	2		u sec
RTS high to RxD off (t2)	1		30	byte
CTS low to TxD on (t3)	0.1	2		u sec
CTS high to TxD off (t4)			1	byte

(t2: 1 character in the FIFO, FIFO 1/4 full, FIFO 1/2 full, FIFO 2 less than full)

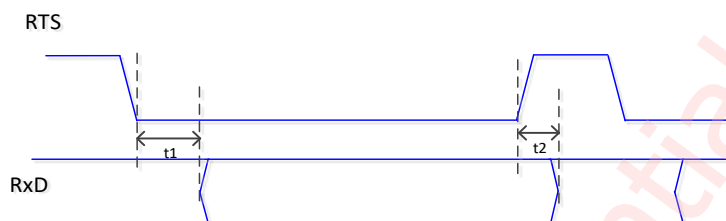
Interface Timing



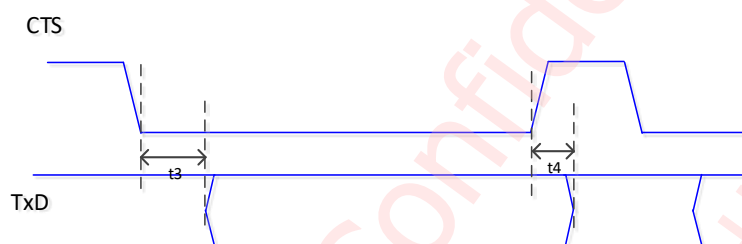
TxD/RxD Timing



RTS Timing



CTS Timing



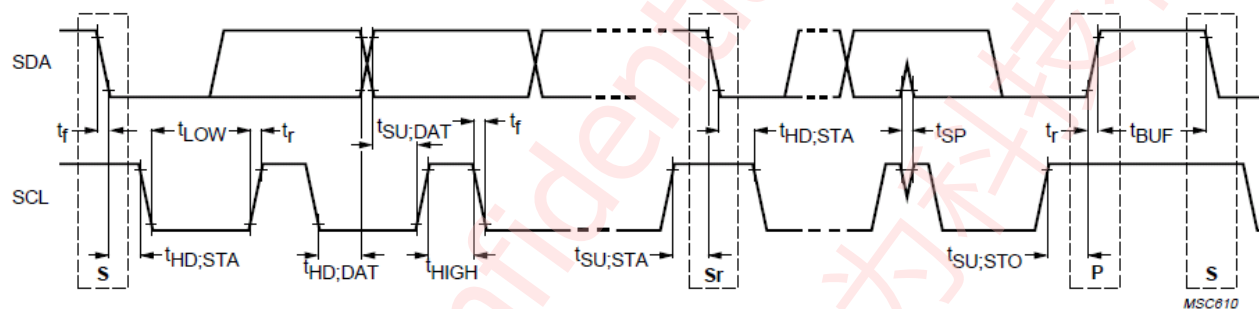
KEYPAD Interface

AC Characteristics (under input clock 12MHz, clk default setting)

Parameter	Min	Max	Unit
Key press to detection delay		2.5	us
Key release to detection delay		2.5	us
Debounce		86	ms
Align time		340	ms

I2C Interface

AC Timing Diagram



Standard Mode

Parameter	Symbol	Standard mode		Unit
		Min	Max	
Clock Frequency	fSCL	-	100	kHz
Re-start Hold Time	tHD;STA	4	-	μs
SCL Low Period	tLOW	4.7	-	μs
SCL High Period	tHIGH	4.0	-	μs
RE-start Set-up Time	tSU;STA	4.7	-	μs
SDA Hold Time	tHD;DAT	5	-	μs
SDA Set-up Time	tSU;DAT	250	-	ns
Rise Time of Signals	tr	-	1000	ns
Fall Time of Signals	tf	-	300	ns
STOP Set-up Time	tSU;STO	4.0	-	μs
Bus Free High Time between STOP and START Condition	tBUF	4.7	-	μs

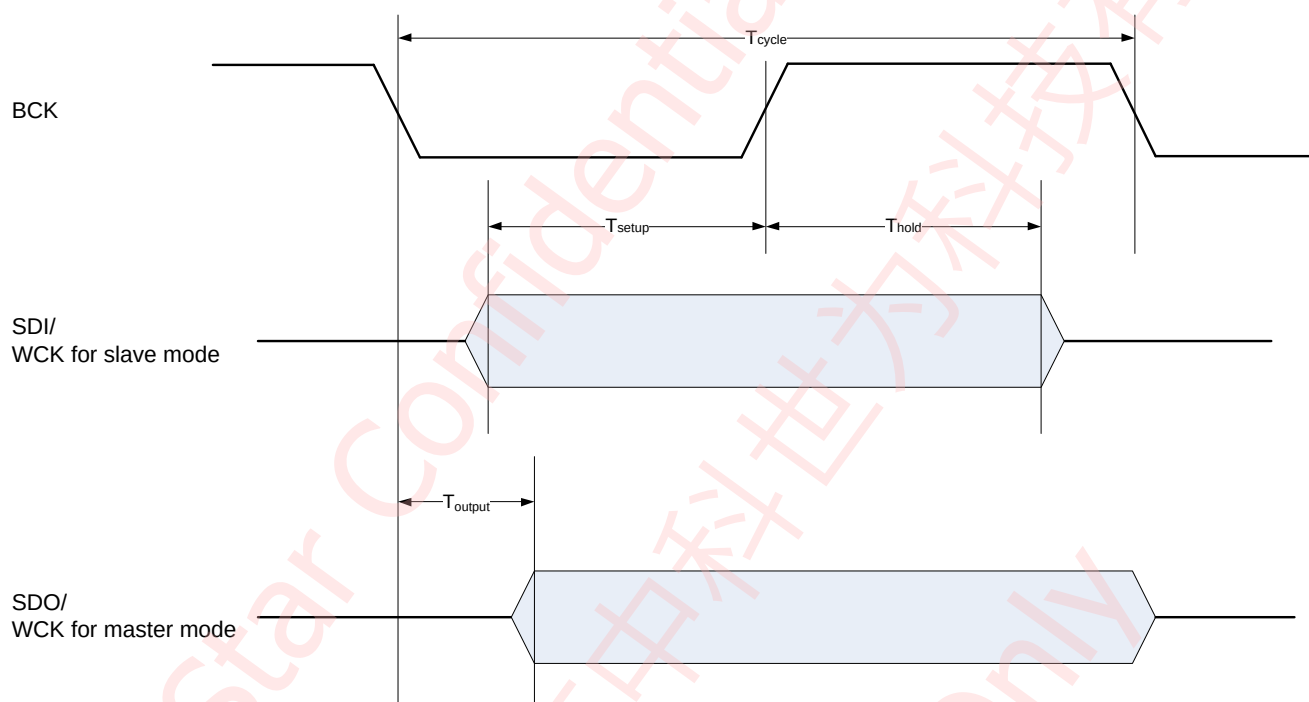
Fast Mode

Parameter	Symbol	Fast mode		Unit
		Min	Max	
Clock Frequency	fSCL	-	400	kHz
Re-start Hold Time	tHD;STA	0.6	-	μs
SCL Low Period	tLOW	1.3	-	μs
SCL High Period	tHIGH	0.6	-	μs
RE-start Set-up Time	tSU;STA	0.6	-	μs
SDA Hold Time	tHD;DAT	0	0.9-	μs
SDA Set-up Time	tSU;DAT	100	-	ns
Rise Time of Signals	tr	20+0.1Cb	300	ns
Fall Time of Signals	tf	20+0.1Cb	300	ns
STOP Set-up Time	tSU;STO	0.6	-	μs
Bus Free High Time between STOP and START Condition	tBUF	1.3	-	μs

Note: Cb = total capacitance of one bus line in pF

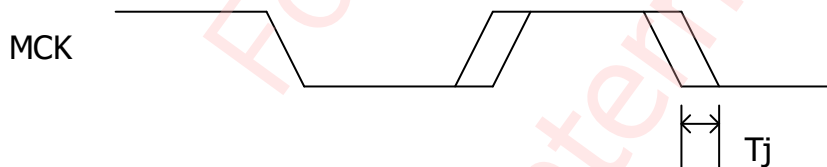
I2S Interface

I2S Audio Timing



Parameter	Symbol	Min	Typ	Max	Unit
BCK period	T_{cycle}	80	-	-	ns
BCK Duty Cycle	-	-	50	-	%
Setup Time	T_{setup}	20	-	-	ns
Hold Time	T_{hold}	20	-	-	ns
Output Delay Time	T_{output}	-	-	20	ns

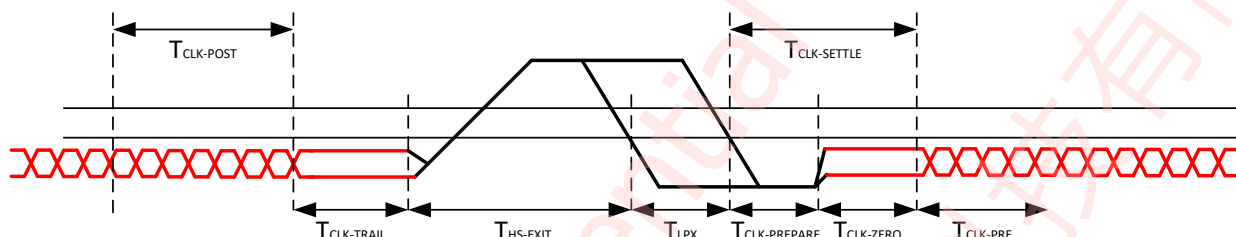
256FS Audio Clock Output Timing



Parameter	Symbol	Min	Typ	Max	Unit
MCK Period (FS = 48KHz)	-	-	81.38	-	ns
MCK Frequency (FS = 48KHz)	-	-	12.288	-	MHz
Jitter	T_j	-	-	2	ns

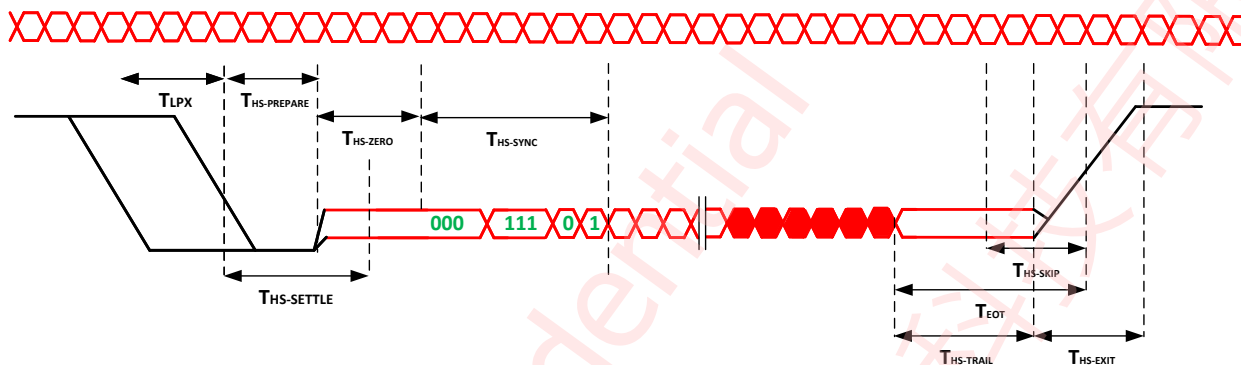
MIPI Interface

Clock Lane



Symbol	Description	Min	Max	Unit
UI	As HS differential data unit	0.83	12.5	ns
$T_{CLK-POST}$	Time that the transmitter continues to send HS clock after the last associated Data Lane has transitioned to LP Mode. Interval is defined as the period from the end of $T_{HS-TRAIL}$ to the beginning of $T_{CLK-TRAIL}$.	-	60	ns
T_{LPX}	Transmitted length of any Low-Power state period	50	-	ns
$T_{CLK_PREPARE}$	Time that the transmitter drives the Clock Lane LP-00 Line state immediately before the HS-0 Line state starting the HS transmission.	38	95	ns
$T_{CLK-ZERO}$	Time that the transmitter drives the HS-0 state prior to starting the Clock.	262	-	ns
$T_{CLK-PRE}$	Time that the HS clock shall be driven by the transmitter prior to any associated Data Lane beginning the transition from LP to HS mode.	8		UI
$T_{CLK-SETTLE}$	Time interval during which the HS receiver should ignore any Clock Lane HS transitions, starting from the beginning of $T_{CLK-prepare}$.	95	300	ns
$T_{HS-EXIT}$	Time that the transmitter drives LP-11 following a HS burst.	100		ns
$T_{CLK-TRAIL}$	Time that the transmitter drives the HS-0 state after the last payload clock bit of a HS transmission burst.	60		ns

Data Lane



Symbol	Description	Min	Max	Unit
UI	As HS single data unit	0.83	12.5	ns
$T_{HS-SKIP}$	Time interval during which the HS-RX should ignore any transitions on the Data Lane, following a HS burst. The end point of the interval is defined as the beginning of the LP-11 state following the HS burst.	40	$55 \text{ ns} + 4 * UI$	ns
T_{LPX}	Transmitted length of any Low-Power state period	50	-	ns
$T_{HS_PREPARE}$	Time that the transmitter drives the Data Lane LP-00 Line state immediately before the HS-0 Line state starting the HS transmission.	$40 \text{ ns} + 4 * UI$	$85 \text{ ns} + 6 * UI$	ns
$T_{HS-ZERO}$	Time that the transmitter drives the HS-0 state prior to transmitting the Sync sequence.	$105 \text{ ns} + 10 * UI$	-	ns
$T_{HS-SYNC}$	Time that the HS Start-of-Transmission (SoT) procedure shall transmit '00011101' as sync pattern.	$8 * UI$		UI
$T_{HS-settle}$	Time interval during which the HS receiver shall ignore any Data Lane HS transitions, starting from the beginning of THS-PREPARE. The HS receiver shall ignore any Data Lane transitions before the minimum value, and the HS receiver shall respond to any Data Lane transitions after the maximum value.	$85 \text{ ns} + 6 * UI$	$145 \text{ ns} + 10 * UI$	ns
$T_{HS-EXIT}$	Time that the transmitter drives LP-11 following a HS burst.	100		ns
$T_{HS-TRAIL}$	Time that the transmitter drives the flipped differential state after last payload data bit of a HS transmission burst.	$\text{Max}(8 * UI, 60 \text{ ns} + 4 * UI)$		ns
T_{EOT}	Transmitted time interval from the start of THS-TRAIL or TCLK-TRAIL, to the start of the LP-11 state following a HS burst.		$105 \text{ ns} + 12 * UI$	ns

HARDWARE POWER SEQUENCE PROCEDURE

The timing requirements of the hardware reset signal are shown as below:

Hardware Reset

HWRESET: Chip Reset; High Reset (Level)

The HWRESET pin is suggested to connect with 3.3V standby as shown in Figure 1. The V_{IH} is 2V (Typ) $\pm$ 10% (2.2V~1.8V); the V_{IL} is 1.2V (Typ) $\pm$ 10% (1.08V~1.32V). The power sequence is as shown in Figure 2.

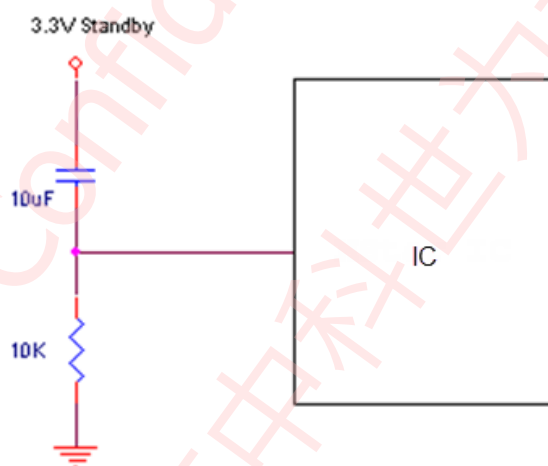
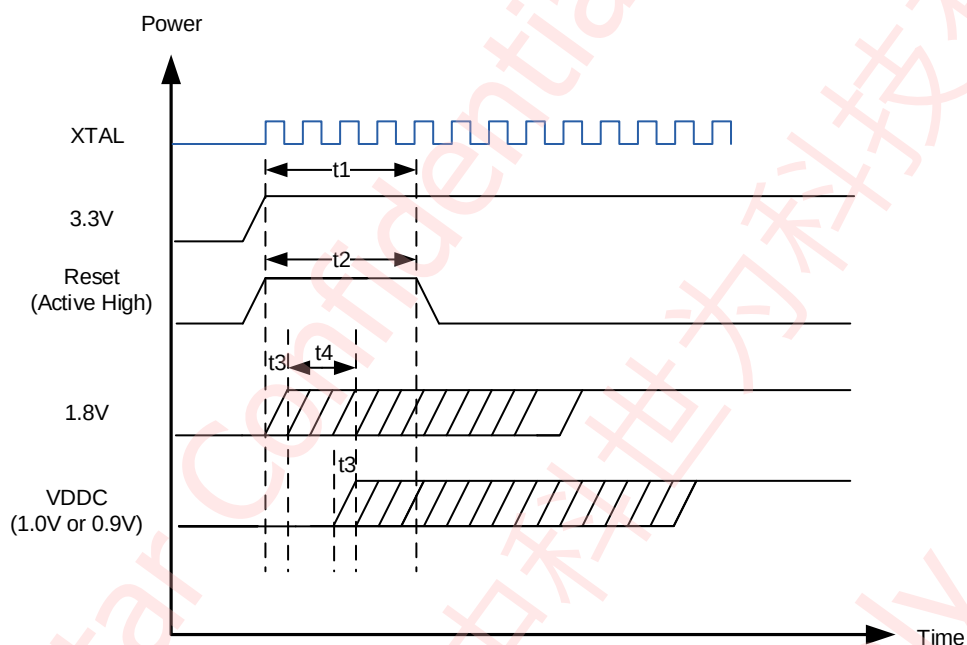


Figure 1: Reset Application Circuit

External Reset IC with External LDO

The timing is shown as Figure 2. The RST and power waveform must satisfy Figure 2 with parameters listed in Table 1.



Note:

- *3.3V (AVDD_XTAL, AVDD_ETH, AVDD3P3_USB, AVDD_PLL, AVDD_AUD, AVDD_HD, VDDP_3, VDDP_4, VDDP_1_3318 (if 3.3V is used), VDDP_2_3318 (if 3.3V is used))
- *1.0V/0.9V (VDD)
- *1.8V (AVDDIO_DRAM, VDDIO_DATA, VDDIO_CMD, VDDIO_MCLK, VDDP_1_3318 (if 1.8V is used), VDDP_2_3318 (if 1.8V is used))

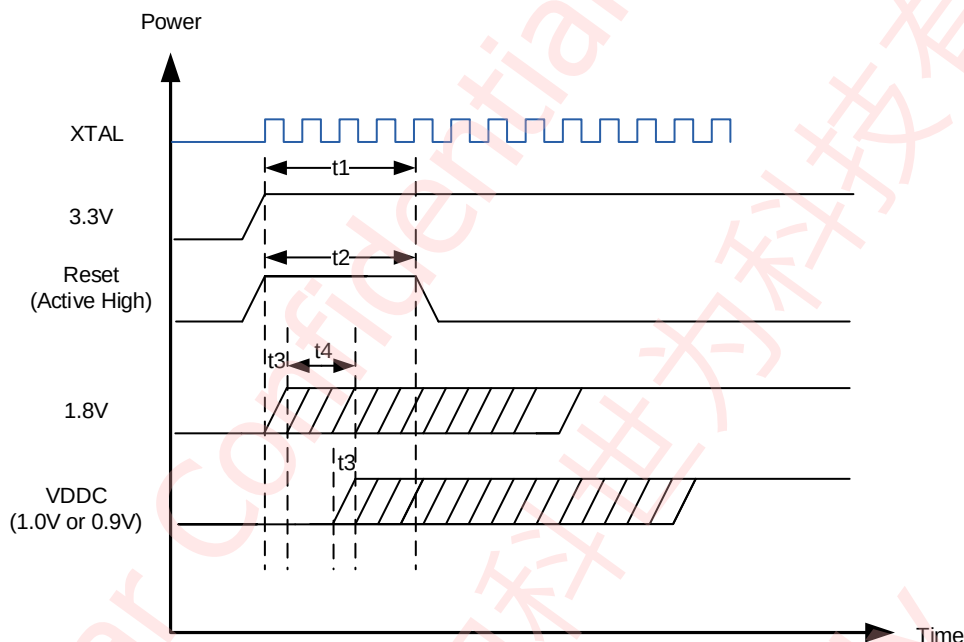
Figure 2: Power on Sequence

Table 1: Power Requirements

Time	Description	Min	Typ.	Max	Unit
t ₁	XTAL stable to Reset falling	5	—	—	ms
t ₂	Reset pulse width	5	—	—	ms
t ₃	Normal 3.3V, 1.8V, and VDDC power rising time (0% to 100%)	—	—	20	ms
t ₄	Normal 3.3V and 1.8V to VDDC lead time	1	—	—	ms

Without External Reset IC with External LDO

The timing is shown as Figure 3. The power waveform must satisfy Figure 3 with parameters listed in Table 2.



Note:

- *3.3V (AVDD_XTAL, AVDD_ETH, AVDD3P3_USB, AVDD_PLL, AVDD_AUD, AVDD_HD, VDDP_3, VDDP_4, VDDP_1_3318 (if 3.3V is used), VDDP_2_3318 (if 3.3V is used))
- *1.0V/0.9V (VDD)
- *1.8V (AVDDIO_DRAM, VDDIO_DATA, VDDIO_CMD, VDDIO_MCLK, VDDP_1_3318 (if 1.8V is used), VDDP_2_3318 (if 1.8V is used))

Figure 3: Power on Sequence

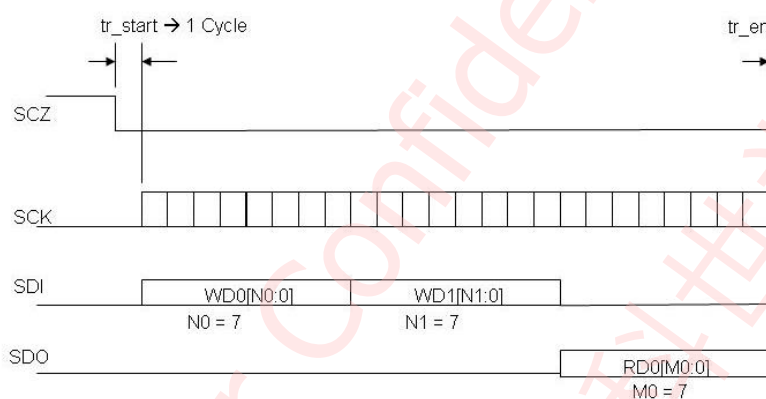
Table 2: Power Requirements

Time	Description	Min	Typ.	Max	Unit
t ₁	Normal 3.3V, 1.8V, and VDDC power rising time (0% to 100%)	—	—	20	ms
t ₂	Normal 3.3V and 1.8V to VDDC lead time	1	—	—	ms

MSPI OPERATION EXAMPLE

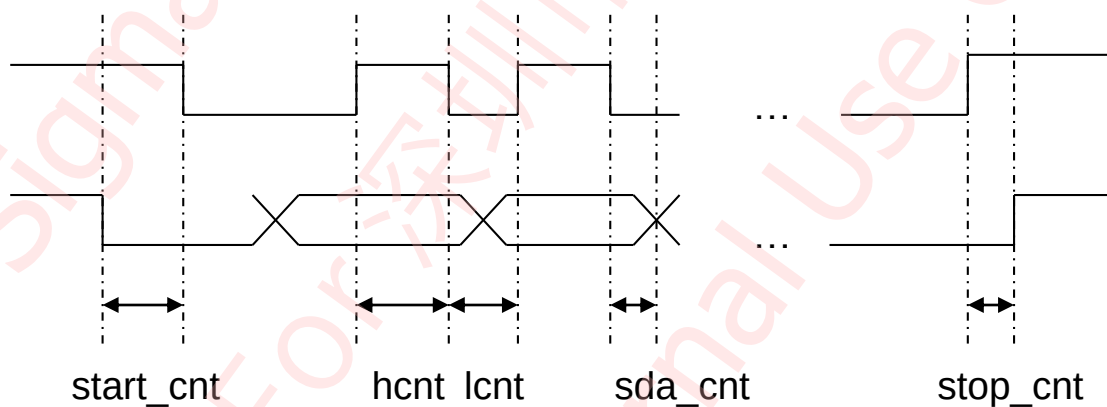
This section describes an example of MSPI operation.

- (0) Initialize
- (1) CS goes low
- (2) Write 2-Byte data
- (3) Read 1-Byte data
- (4) CS goes high



I2C clock frequency is configurable between 100KHz and 400KHz.

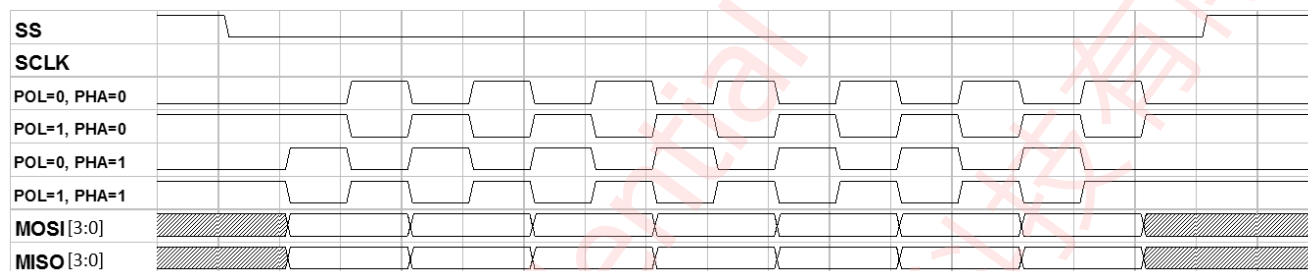
Set MIIC Speed



clk_miic	12MHz	24MHz
lcnt (>1.3us)	>16T	>31T
hcnt (>0.6us)	>8T	>15T
start (>0.6us)	>8T	>15T
stop (>0.6us)	>8T	>15T
between start and stop (>1.3us)	>16T	>31T
data_latch (>0us)	>0T	>0T
sda change (<0.9us)	<11T	<22T

Register Name	Address	Description
reg_stop_cnt	'h08[15:0]	Sets the SCL and SDA count for stop
reg_hcnt	'h09[15:0]	Sets the SCL clock high-period count
reg_lcnt	'h0a[15:0]	Sets the SCL clock low-period count
reg_sda_cnt	'h0b[15:0]	Sets the clock count between falling edge SCL and SDA
reg_start_cnt	'h0c[15:0]	Sets the SCL and SDA count for start
reg_data_lat_cnt	'h0d[15:0]	Sets the data latch timing

PSPI INTERFACE

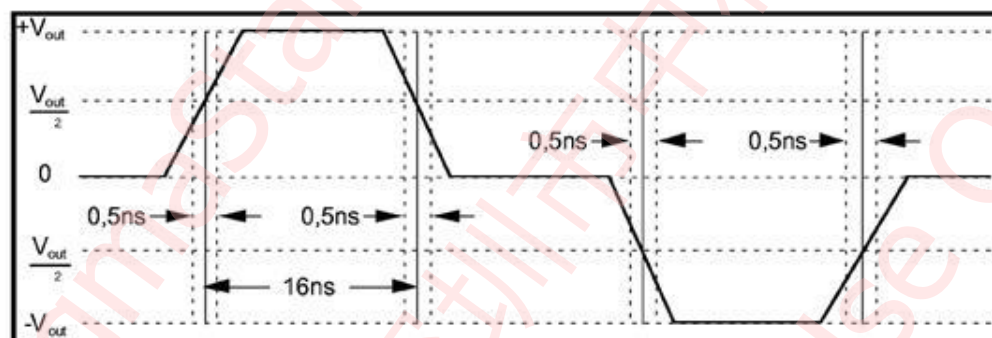


POL	PHA	Transfer
0	0	Receive at rising edge, transmit at falling edge
1	0	Receive at falling edge, transmit at rising edge
0	1	Receive at falling edge, transmit at rising edge
1	1	Receive at rising edge, transmit at falling edge

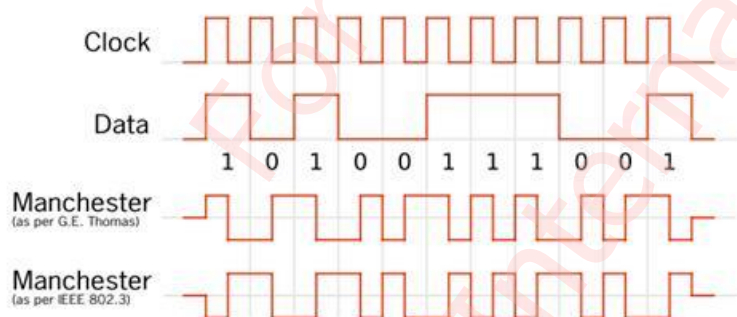
EPHY INTERFACE

Parameter	Min	Typ	Max	Unit
ETHERNET ANALOG INTERFACE (10BASE-T)				
Analog Input Range	4.4	5	5.6	V _{dp-p}
Differential Input Impedance		100		ohm
ETHERNET ANALOG INTERFACE (100BASE-TX)				
Analog Input Range	1.9	2	2.1	V _{dp-p}
Differential Input Impedance		100		ohm
Rise/Fall Time	3	4	5	ns
Rise/Fall Time Symmetry			0.5	ns
Duty Cycle Distortion	-0.25		0.25	ns
Amplitude Symmetry	98	100	102	%
Overshoot			5	%

100BASE-TX



10BASE-T

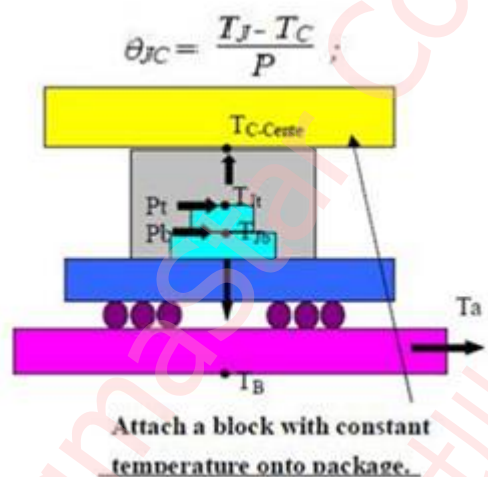


THERMAL RESISTANCE (°C/W)

Thermal simulation mode

1. PCB conditions (JEDEC JESD51-5)
2. PCB layers: 2L (1S1P)
3. PCB dimensions (mm x mm): 76.2 x 114.3
4. PCB thickness (mm): 1.6

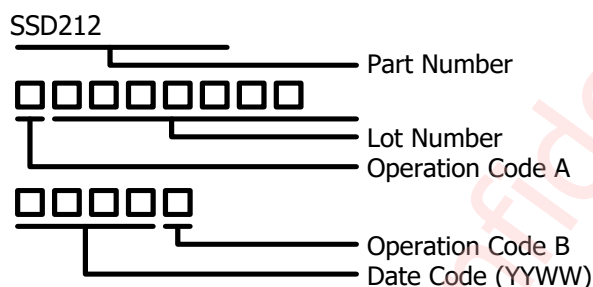
PKG Type	PKG Size (mm) / Pin Count	PCB Layer	Theta jc (C/W)	Theta jb (C/W)	Ta (C)	Tj (C)	Theta ja (C/W)
							0 m/s
QFN	12.3x12.3 / 128L	2L	6.9	10.82	75	125	22.6



ORDERING GUIDE

Part Number	Temperature Range	Package Description	Package Option	Minimum Order Quantity
SSD212	-20°C to +70°C	QFN	128-pin	1216ea

MARKING INFORMATION



DISCLAIMER

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Electrostatic charges accumulate on both test equipment and human body and can discharge without detection. SSD212 comes with ESD protection circuitry; however, the device may be permanently damaged when subjected to high energy discharges. The device should be handled with proper ESD precautions to prevent malfunction and performance degradation.

PWM Register (Bank = 1A)

PWM Register (Bank = 1A)				
Index (Absolute)	Mnemonic	Bit	Description	
00h (1A00h)	REG1A00	7:0	Default: 0x00	Access: R/W
	PWM0_SHIFT[7:0]	7:0	PWM0 rising point shift counter.	
00h (1A01h)	REG1A01	7:0	Default: 0x00	Access: R/W
	PWM0_SHIFT[15:8]	7:0	See description of '1A00h.'	
01h (1A02h)	REG1A02	7:0	Default: 0x00	Access: R/W
	-	7:2	Reserved.	
	PWM0_SHIFT[17:16]	1:0	See description of '1A00h.'	
02h (1A04h)	REG1A04	7:0	Default: 0x00	Access: R/W
	PWM0_DUTY[7:0]	7:0	PWM0 duty.	
02h (1A05h)	REG1A05	7:0	Default: 0x00	Access: R/W
	PWM0_DUTY[15:8]	7:0	See description of '1A04h.'	
03h (1A06h)	REG1A06	7:0	Default: 0x00	Access: R/W
	-	7:2	Reserved.	
	PWM0_DUTY[17:16]	1:0	See description of '1A04h.'	
04h (1A08h)	REG1A08	7:0	Default: 0x00	Access: R/W
	PWM0_PERIOD[7:0]	7:0	PWM0 period.	
04h (1A09h)	REG1A09	7:0	Default: 0x00	Access: R/W
	PWM0_PERIOD[15:8]	7:0	See description of '1A08h.'	
05h (1A0Ah)	REG1A0A	7:0	Default: 0x00	Access: R/W
	-	7:2	Reserved.	
	PWM0_PERIOD[17:16]	1:0	See description of '1A08h.'	
06h (1A0Ch)	REG1A0C	7:0	Default: 0x00	Access: R/W
	PWM0_DIV[7:0]	7:0	PWM0 divider.	
06h (1A0Dh)	REG1A0D	7:0	Default: 0x00	Access: R/W
	PWM0_DIV[15:8]	7:0	See description of '1A0Ch.'	
07h (1A0Eh)	REG1A0E	7:0	Default: 0x01	Access: R/W
	-	7:5	Reserved.	
	PWM0_POLARITY	4	PWM0 polarity.	
	PWM0_SHIFT_GAT	3	PWM0 enable shift counter gating.	
	PWM0_DIFF_P_EN	2	Enable multiple differential pulse width mode.	
	PWM0_DBEN	1	PWM0 double buffer enable.	
	PWM0_VDBEN_SW	0	PWM0 double buffer enabled by software. 0: Disable this function; 1: Enable this function.	

PWM Register (Bank = 1A)				
Index (Absolute)	Mnemonic	Bit	Description	
08h (1A10h)	REG1A10	7:0	Default: 0xFF	Access: R/W
	PWM0_SHIFT2[7:0]	7:0	PWM0 rising point shift2 counter.	
08h (1A11h)	REG1A11	7:0	Default: 0xFF	Access: R/W
	PWM0_SHIFT2[15:8]	7:0	See description of '1A10h.'	
09h (1A12h)	REG1A12	7:0	Default: 0xFF	Access: R/W
	PWM0_DUTY2[7:0]	7:0	PWM0 duty2.	
09h (1A13h)	REG1A13	7:0	Default: 0xFF	Access: R/W
	PWM0_DUTY2[15:8]	7:0	See description of '1A12h.'	
0Ah (1A14h)	REG1A14	7:0	Default: 0xFF	Access: R/W
	PWM0_SHIFT3[7:0]	7:0	PWM0 rising point shift3 counter.	
0Ah (1A15h)	REG1A15	7:0	Default: 0xFF	Access: R/W
	PWM0_SHIFT3[15:8]	7:0	See description of '1A14h.'	
0Bh (1A16h)	REG1A16	7:0	Default: 0xFF	Access: R/W
	PWM0_DUTY3[7:0]	7:0	PWM0 duty3.	
0Bh (1A17h)	REG1A17	7:0	Default: 0xFF	Access: R/W
	PWM0_DUTY3[15:8]	7:0	See description of '1A16h.'	
0Ch (1A18h)	REG1A18	7:0	Default: 0xFF	Access: R/W
	PWM0_SHIFT4[7:0]	7:0	PWM0 rising point shift4 counter.	
0Ch (1A19h)	REG1A19	7:0	Default: 0xFF	Access: R/W
	PWM0_SHIFT4[15:8]	7:0	See description of '1A18h.'	
0Dh (1A1Ah)	REG1A1A	7:0	Default: 0xFF	Access: R/W
	PWM0_DUTY4[7:0]	7:0	PWM0 duty4.	
0Dh (1A1Bh)	REG1A1B	7:0	Default: 0xFF	Access: R/W
	PWM0_DUTY4[15:8]	7:0	See description of '1A1Ah.'	
10h (1A20h)	REG1A20	7:0	Default: 0x00	Access: R/W
	GROUP0_ROUND_NUMBER[7:0]	7:0	Round number for group0.	
10h (1A21h)	REG1A21	7:0	Default: 0x00	Access: R/W
	GROUP0_ROUND_NUMBER[15:8]	7:0	See description of '1A20h.'	
11h (1A22h)	REG1A22	7:0	Default: 0x00	Access: R/W
	GROUP0_PWM0_DELAY_COUNT[7:0]	7:0	GROUP0_PWM0_DELAY_COUNT.	
11h (1A23h)	REG1A23	7:0	Default: 0x00	Access: R/W
	GROUP0_PWM0_DELAY_COUNT[15:8]	7:0	See description of '1A22h.'	
12h (1A24h)	REG1A24	7:0	Default: 0x00	Access: R/W
	-	7:2	Reserved.	
	GROUP0_PWM0_DELAY_COUNT[17:16]	1:0	See description of '1A22h.'	

PWM Register (Bank = 1A)				
Index (Absolute)	Mnemonic	Bit	Description	
13h (1A26h)	REG1A26	7:0	Default: 0x00	Access: R/W
	GROUP0_PWM1_DELAY_COUNT[7:0]	7:0	GROUP0_PWM1_DELAY_COUNT.	
13h (1A27h)	REG1A27	7:0	Default: 0x00	Access: R/W
	GROUP0_PWM1_DELAY_COUNT[15:8]	7:0	See description of '1A26h.'	
14h (1A28h)	REG1A28	7:0	Default: 0x00	Access: R/W
	-	7:2	Reserved.	
	GROUP0_PWM1_DELAY_COUNT[17:16]	1:0	See description of '1A26h.'	
15h (1A2Ah)	REG1A2A	7:0	Default: 0x00	Access: R/W
	GROUP0_PWM2_DELAY_COUNT[7:0]	7:0	GROUP0_PWM2_DELAY_COUNT.	
15h (1A2Bh)	REG1A2B	7:0	Default: 0x00	Access: R/W
	GROUP0_PWM2_DELAY_COUNT[15:8]	7:0	See description of '1A2Ah.'	
16h (1A2Ch)	REG1A2C	7:0	Default: 0x00	Access: R/W
	-	7:2	Reserved.	
	GROUP0_PWM2_DELAY_COUNT[17:16]	1:0	See description of '1A2Ah.'	
17h (1A2Eh)	REG1A2E	7:0	Default: 0x00	Access: R/W
	GROUP0_PWM3_DELAY_COUNT[7:0]	7:0	GROUP0_PWM3_DELAY_COUNT.	
17h (1A2Fh)	REG1A2F	7:0	Default: 0x00	Access: R/W
	GROUP0_PWM3_DELAY_COUNT[15:8]	7:0	See description of '1A2Eh.'	
18h (1A30h)	REG1A30	7:0	Default: 0x00	Access: R/W
	-	7:2	Reserved.	
	GROUP0_PWM3_DELAY_COUNT[17:16]	1:0	See description of '1A2Eh.'	
20h (1A40h)	REG1A40	7:0	Default: 0x00	Access: R/W
	PWM1_SHIFT[7:0]	7:0	PWM1 rising point shift counter.	
20h (1A41h)	REG1A41	7:0	Default: 0x00	Access: R/W
	PWM1_SHIFT[15:8]	7:0	See description of '1A40h.'	
21h (1A42h)	REG1A42	7:0	Default: 0x00	Access: R/W
	-	7:2	Reserved.	
	PWM1_SHIFT[17:16]	1:0	See description of '1A40h.'	
22h (1A44h)	REG1A44	7:0	Default: 0x00	Access: R/W
	PWM1_DUTY[7:0]	7:0	PWM1 duty.	
22h (1A45h)	REG1A45	7:0	Default: 0x00	Access: R/W
	PWM1_DUTY[15:8]	7:0	See description of '1A44h.'	
23h	REG1A46	7:0	Default: 0x00	Access: R/W

PWM Register (Bank = 1A)				
Index (Absolute)	Mnemonic	Bit	Description	
(1A46h)	-	7:2	Reserved.	
	PWM1_DUTY[17:16]	1:0	See description of '1A44h.'	
24h (1A48h)	REG1A48	7:0	Default: 0x00	Access: R/W
	PWM1_PERIOD[7:0]	7:0	PWM1 period.	
24h (1A49h)	REG1A49	7:0	Default: 0x00	Access: R/W
	PWM1_PERIOD[15:8]	7:0	See description of '1A48h.'	
25h (1A4Ah)	REG1A4A	7:0	Default: 0x00	Access: R/W
	-	7:2	Reserved.	
	PWM1_PERIOD[17:16]	1:0	See description of '1A48h.'	
26h (1A4Ch)	REG1A4C	7:0	Default: 0x00	Access: R/W
	PWM1_DIV[7:0]	7:0	PWM1 divider.	
26h (1A4Dh)	REG1A4D	7:0	Default: 0x00	Access: R/W
	PWM1_DIV[15:8]	7:0	See description of '1A4Ch.'	
27h (1A4Eh)	REG1A4E	7:0	Default: 0x01	Access: R/W
	-	7:5	Reserved.	
	PWM1_POLARITY	4	PWM1 polarity.	
	PWM1_SHIFT_GAT	3	PWM1 enable shift counter gating.	
	PWM1_DIFF_P_EN	2	Enable multiple differential pulse width mode.	
	PWM1_DBEN	1	PWM1 double buffer enable.	
	PWM1_VDBEN_SW	0	PWM1 double buffer enabled by software. 0: Disable this function; 1: Enable this function.	
28h (1A50h)	REG1A50	7:0	Default: 0xFF	Access: R/W
	PWM1_SHIFT2[7:0]	7:0	PWM1 rising point shift2 counter.	
28h (1A51h)	REG1A51	7:0	Default: 0xFF	Access: R/W
	PWM1_SHIFT2[15:8]	7:0	See description of '1A50h.'	
29h (1A52h)	REG1A52	7:0	Default: 0xFF	Access: R/W
	PWM1_DUTY2[7:0]	7:0	PWM1 duty2.	
29h (1A53h)	REG1A53	7:0	Default: 0xFF	Access: R/W
	PWM1_DUTY2[15:8]	7:0	See description of '1A52h.'	
2Ah (1A54h)	REG1A54	7:0	Default: 0xFF	Access: R/W
	PWM1_SHIFT3[7:0]	7:0	PWM1 rising point shift3 counter.	
2Ah (1A55h)	REG1A55	7:0	Default: 0xFF	Access: R/W
	PWM1_SHIFT3[15:8]	7:0	See description of '1A54h.'	

PWM Register (Bank = 1A)				
Index (Absolute)	Mnemonic	Bit	Description	
2Bh (1A56h)	REG1A56	7:0	Default: 0xFF	Access: R/W
	PWM1_DUTY3[7:0]	7:0	PWM1 duty3.	
2Bh (1A57h)	REG1A57	7:0	Default: 0xFF	Access: R/W
	PWM1_DUTY3[15:8]	7:0	See description of '1A56h.'	
2Ch (1A58h)	REG1A58	7:0	Default: 0xFF	Access: R/W
	PWM1_SHIFT4[7:0]	7:0	PWM1 rising point shift4 counter.	
2Ch (1A59h)	REG1A59	7:0	Default: 0xFF	Access: R/W
	PWM1_SHIFT4[15:8]	7:0	See description of '1A58h.'	
2Dh (1A5Ah)	REG1A5A	7:0	Default: 0xFF	Access: R/W
	PWM1_DUTY4[7:0]	7:0	PWM1 duty4.	
2Dh (1A5Bh)	REG1A5B	7:0	Default: 0xFF	Access: R/W
	PWM1_DUTY4[15:8]	7:0	See description of '1A5Ah.'	
30h (1A60h)	REG1A60	7:0	Default: 0x00	Access: R/W
	GROUP1_ROUND_NUMBER[7:0]	7:0	Round number for group1.	
30h (1A61h)	REG1A61	7:0	Default: 0x00	Access: R/W
	GROUP1_ROUND_NUMBER[15:8]	7:0	See description of '1A60h.'	
31h (1A62h)	REG1A62	7:0	Default: 0x00	Access: R/W
	GROUP1_PWM0_DELAY_COUNT[7:0]	7:0	GROUP1_PWM0_DELAY_COUNT.	
31h (1A63h)	REG1A63	7:0	Default: 0x00	Access: R/W
	GROUP1_PWM0_DELAY_COUNT[15:8]	7:0	See description of '1A62h.'	
32h (1A64h)	REG1A64	7:0	Default: 0x00	Access: R/W
	-	7:2	Reserved.	
	GROUP1_PWM0_DELAY_COUNT[17:16]	1:0	See description of '1A62h.'	
33h (1A66h)	REG1A66	7:0	Default: 0x00	Access: R/W
	GROUP1_PWM1_DELAY_COUNT[7:0]	7:0	GROUP1_PWM1_DELAY_COUNT.	
33h (1A67h)	REG1A67	7:0	Default: 0x00	Access: R/W
	GROUP1_PWM1_DELAY_COUNT[15:8]	7:0	See description of '1A66h.'	
34h (1A68h)	REG1A68	7:0	Default: 0x00	Access: R/W
	-	7:2	Reserved.	
	GROUP1_PWM1_DELAY_COUNT[17:16]	1:0	See description of '1A66h.'	
35h (1A6Ah)	REG1A6A	7:0	Default: 0x00	Access: R/W
	GROUP1_PWM2_DELAY_COUNT[7:0]	7:0	GROUP1_PWM2_DELAY_COUNT.	
35h (1A6Bh)	REG1A6B	7:0	Default: 0x00	Access: R/W
	GROUP1_PWM2_DELAY_COUNT[15:8]	7:0	See description of '1A6Ah.'	

PWM Register (Bank = 1A)				
Index (Absolute)	Mnemonic	Bit	Description	
36h (1A6Ch)	REG1A6C	7:0	Default: 0x00	Access: R/W
	-	7:2	Reserved.	
	GROUP1_PWM2_DELAY_COUNT[17:16]	1:0	See description of '1A6Ah.'	
37h (1A6Eh)	REG1A6E	7:0	Default: 0x00	Access: R/W
	GROUP1_PWM3_DELAY_COUNT[7:0]	7:0	GROUP1_PWM3_DELAY_COUNT.	
37h (1A6Fh)	REG1A6F	7:0	Default: 0x00	Access: R/W
	GROUP1_PWM3_DELAY_COUNT[15:8]	7:0	See description of '1A6Eh.'	
38h (1A70h)	REG1A70	7:0	Default: 0x00	Access: R/W
	-	7:2	Reserved.	
	GROUP1_PWM3_DELAY_COUNT[17:16]	1:0	See description of '1A6Eh.'	
40h (1A80h)	REG1A80	7:0	Default: 0x00	Access: R/W
	PWM2_SHIFT[7:0]	7:0	PWM2 rising point shift counter.	
40h (1A81h)	REG1A81	7:0	Default: 0x00	Access: R/W
	PWM2_SHIFT[15:8]	7:0	See description of '1A80h.'	
41h (1A82h)	REG1A82	7:0	Default: 0x00	Access: R/W
	-	7:2	Reserved.	
	PWM2_SHIFT[17:16]	1:0	See description of '1A80h.'	
42h (1A84h)	REG1A84	7:0	Default: 0x00	Access: R/W
	PWM2_DUTY[7:0]	7:0	PWM2 duty.	
42h (1A85h)	REG1A85	7:0	Default: 0x00	Access: R/W
	PWM2_DUTY[15:8]	7:0	See description of '1A84h.'	
43h (1A86h)	REG1A86	7:0	Default: 0x00	Access: R/W
	-	7:2	Reserved.	
	PWM2_DUTY[17:16]	1:0	See description of '1A84h.'	
44h (1A88h)	REG1A88	7:0	Default: 0x00	Access: R/W
	PWM2_PERIOD[7:0]	7:0	PWM2 period.	
44h (1A89h)	REG1A89	7:0	Default: 0x00	Access: R/W
	PWM2_PERIOD[15:8]	7:0	See description of '1A88h.'	
45h (1A8Ah)	REG1A8A	7:0	Default: 0x00	Access: R/W
	-	7:2	Reserved.	
	PWM2_PERIOD[17:16]	1:0	See description of '1A88h.'	
46h (1A8Ch)	REG1A8C	7:0	Default: 0x00	Access: R/W
	PWM2_DIV[7:0]	7:0	PWM2 divider.	

PWM Register (Bank = 1A)				
Index (Absolute)	Mnemonic	Bit	Description	
46h (1A8Dh)	REG1A8D	7:0	Default: 0x00	Access: R/W
	PWM2_DIV[15:8]	7:0	See description of '1A8Ch.'	
47h (1A8Eh)	REG1A8E	7:0	Default: 0x01	Access: R/W
	-	7:5	Reserved.	
	PWM2_POLARITY	4	PWM2 polarity.	
	PWM2_SHIFT_GAT	3	PWM2 enable shift counter gating.	
	PWM2_DIFF_P_EN	2	Enable multiple differential pulse width mode.	
	PWM2_DBEN	1	PWM2 double buffer enable.	
	PWM2_VDBEN_SW	0	PWM2 double buffer enabled by software. 0: Disable this function; 1: Enable this function.	
48h (1A90h)	REG1A90	7:0	Default: 0xFF	Access: R/W
	PWM2_SHIFT2[7:0]	7:0	PWM2 rising point shift2 counter.	
48h (1A91h)	REG1A91	7:0	Default: 0xFF	Access: R/W
	PWM2_SHIFT2[15:8]	7:0	See description of '1A90h.'	
49h (1A92h)	REG1A92	7:0	Default: 0xFF	Access: R/W
	PWM2_DUTY2[7:0]	7:0	PWM2 duty2.	
49h (1A93h)	REG1A93	7:0	Default: 0xFF	Access: R/W
	PWM2_DUTY2[15:8]	7:0	See description of '1A92h.'	
4Ah (1A94h)	REG1A94	7:0	Default: 0xFF	Access: R/W
	PWM2_SHIFT3[7:0]	7:0	PWM2 rising point shift3 counter.	
4Ah (1A95h)	REG1A95	7:0	Default: 0xFF	Access: R/W
	PWM2_SHIFT3[15:8]	7:0	See description of '1A94h.'	
4Bh (1A96h)	REG1A96	7:0	Default: 0xFF	Access: R/W
	PWM2_DUTY3[7:0]	7:0	PWM2 duty3.	
4Bh (1A97h)	REG1A97	7:0	Default: 0xFF	Access: R/W
	PWM2_DUTY3[15:8]	7:0	See description of '1A96h.'	
4Ch (1A98h)	REG1A98	7:0	Default: 0xFF	Access: R/W
	PWM2_SHIFT4[7:0]	7:0	PWM2 rising point shift4 counter.	
4Ch (1A99h)	REG1A99	7:0	Default: 0xFF	Access: R/W
	PWM2_SHIFT4[15:8]	7:0	See description of '1A98h.'	
4Dh (1A9Ah)	REG1A9A	7:0	Default: 0xFF	Access: R/W
	PWM2_DUTY4[7:0]	7:0	PWM2 duty4.	
4Dh (1A9Bh)	REG1A9B	7:0	Default: 0xFF	Access: R/W
	PWM2_DUTY4[15:8]	7:0	See description of '1A9Ah.'	

PWM Register (Bank = 1A)				
Index (Absolute)	Mnemonic	Bit	Description	
50h (1AA0h)	REG1AA0	7:0	Default: 0x00	Access: R/W
	GROUP2_ROUND_NUMBER[7:0]	7:0	Round number for group2.	
50h (1AA1h)	REG1AA1	7:0	Default: 0x00	Access: R/W
	GROUP2_ROUND_NUMBER[15:8]	7:0	See description of '1AA0h.'	
51h (1AA2h)	REG1AA2	7:0	Default: 0x00	Access: R/W
	GROUP2_PWM0_DELAY_COUNT[7:0]	7:0	GROUP2_PWM0_DELAY_COUNT.	
51h (1AA3h)	REG1AA3	7:0	Default: 0x00	Access: R/W
	GROUP2_PWM0_DELAY_COUNT[15:8]	7:0	See description of '1AA2h.'	
52h (1AA4h)	REG1AA4	7:0	Default: 0x00	Access: R/W
	-	7:2	Reserved.	
	GROUP2_PWM0_DELAY_COUNT[17:16]	1:0	See description of '1AA2h.'	
53h (1AA6h)	REG1AA6	7:0	Default: 0x00	Access: R/W
	GROUP2_PWM1_DELAY_COUNT[7:0]	7:0	GROUP2_PWM1_DELAY_COUNT.	
53h (1AA7h)	REG1AA7	7:0	Default: 0x00	Access: R/W
	GROUP2_PWM1_DELAY_COUNT[15:8]	7:0	See description of '1AA6h.'	
54h (1AA8h)	REG1AA8	7:0	Default: 0x00	Access: R/W
	-	7:2	Reserved.	
	GROUP2_PWM1_DELAY_COUNT[17:16]	1:0	See description of '1AA6h.'	
55h (1AAAh)	REG1AAA	7:0	Default: 0x00	Access: R/W
	GROUP2_PWM2_DELAY_COUNT[7:0]	7:0	GROUP2_PWM2_DELAY_COUNT.	
55h (1AABh)	REG1AAB	7:0	Default: 0x00	Access: R/W
	GROUP2_PWM2_DELAY_COUNT[15:8]	7:0	See description of '1AAAh.'	
56h (1AACh)	REG1AAC	7:0	Default: 0x00	Access: R/W
	-	7:2	Reserved.	
	GROUP2_PWM2_DELAY_COUNT[17:16]	1:0	See description of '1AAAh.'	
60h (1AC0h)	REG1AC0	7:0	Default: 0x00	Access: R/W
	PWM3_SHIFT[7:0]	7:0	PWM3 rising point shift counter.	
60h (1AC1h)	REG1AC1	7:0	Default: 0x00	Access: R/W
	PWM3_SHIFT[15:8]	7:0	See description of '1AC0h.'	
61h (1AC2h)	REG1AC2	7:0	Default: 0x00	Access: R/W
	-	7:2	Reserved.	
	PWM3_SHIFT[17:16]	1:0	See description of '1AC0h.'	
62h (1AC4h)	REG1AC4	7:0	Default: 0x00	Access: R/W
	PWM3_DUTY[7:0]	7:0	PWM3 duty.	

PWM Register (Bank = 1A)				
Index (Absolute)	Mnemonic	Bit	Description	
62h (1AC5h)	REG1AC5	7:0	Default: 0x00	Access: R/W
	PWM3_DUTY[15:8]	7:0	See description of '1AC4h.'	
63h (1AC6h)	REG1AC6	7:0	Default: 0x00	Access: R/W
	-	7:2	Reserved.	
	PWM3_DUTY[17:16]	1:0	See description of '1AC4h.'	
64h (1AC8h)	REG1AC8	7:0	Default: 0x00	Access: R/W
	PWM3_PERIOD[7:0]	7:0	PWM3 period.	
64h (1AC9h)	REG1AC9	7:0	Default: 0x00	Access: R/W
	PWM3_PERIOD[15:8]	7:0	See description of '1AC8h.'	
65h (1ACAh)	REG1ACA	7:0	Default: 0x00	Access: R/W
	-	7:2	Reserved.	
	PWM3_PERIOD[17:16]	1:0	See description of '1AC8h.'	
66h (1ACCh)	REG1ACC	7:0	Default: 0x00	Access: R/W
	PWM3_DIV[7:0]	7:0	PWM3 divider.	
66h (1ACDh)	REG1ACD	7:0	Default: 0x00	Access: R/W
	PWM3_DIV[15:8]	7:0	See description of '1ACCh.'	
67h (1ACEh)	REG1ACE	7:0	Default: 0x01	Access: R/W
	-	7:5	Reserved.	
	PWM3_POLARITY	4	PWM3 polarity.	
	PWM3_SHIFT_GAT	3	PWM3 enable shift counter gating.	
	PWM3_DIFF_P_EN	2	Enable multiple differential pulse width mode.	
	PWM3_DBEN	1	PWM3 double buffer enable.	
	PWM3_VDBEN_SW	0	PWM3 double buffer enabled by software. 0: Disable this function; 1: Enable this function.	
68h (1AD0h)	REG1AD0	7:0	Default: 0xFF	Access: R/W
	PWM3_SHIFT2[7:0]	7:0	PWM3 rising point shift2 counter.	
68h (1AD1h)	REG1AD1	7:0	Default: 0xFF	Access: R/W
	PWM3_SHIFT2[15:8]	7:0	See description of '1AD0h.'	
69h (1AD2h)	REG1AD2	7:0	Default: 0xFF	Access: R/W
	PWM3_DUTY2[7:0]	7:0	PWM3 duty2.	
69h (1AD3h)	REG1AD3	7:0	Default: 0xFF	Access: R/W
	PWM3_DUTY2[15:8]	7:0	See description of '1AD2h.'	
6Ah (1AD4h)	REG1AD4	7:0	Default: 0xFF	Access: R/W
	PWM3_SHIFT3[7:0]	7:0	PWM3 rising point shift3 counter.	

PWM Register (Bank = 1A)				
Index (Absolute)	Mnemonic	Bit	Description	
6Ah (1AD5h)	REG1AD5	7:0	Default: 0xFF	Access: R/W
	PWM3_SHIFT3[15:8]	7:0	See description of '1AD4h.'	
6Bh (1AD6h)	REG1AD6	7:0	Default: 0xFF	Access: R/W
	PWM3_DUTY3[7:0]	7:0	PWM3 duty3.	
6Bh (1AD7h)	REG1AD7	7:0	Default: 0xFF	Access: R/W
	PWM3_DUTY3[15:8]	7:0	See description of '1AD6h.'	
6Ch (1AD8h)	REG1AD8	7:0	Default: 0xFF	Access: R/W
	PWM3_SHIFT4[7:0]	7:0	PWM3 rising point shift4 counter.	
6Ch (1AD9h)	REG1AD9	7:0	Default: 0xFF	Access: R/W
	PWM3_SHIFT4[15:8]	7:0	See description of '1AD8h.'	
6Dh (1ADAh)	REG1ADA	7:0	Default: 0xFF	Access: R/W
	PWM3_DUTY4[7:0]	7:0	PWM3 duty4.	
6Dh (1ADBh)	REG1ADB	7:0	Default: 0xFF	Access: R/W
	PWM3_DUTY4[15:8]	7:0	See description of '1ADAh.'	
71h (1AE2h)	REG1AE2	7:0	Default: 0x00	Access: R/W
	-	7:3	Reserved.	
	HOLD_MODE[2:0]	2:0	[0] 1: Group0 hold mode enable. 0: Group0 hold mode disable.	
72h (1AE4h)	REG1AE4	7:0	Default: 0x00	Access: R/W
	-	7:3	Reserved.	
	STOP_MODE[2:0]	2:0	[0] 1: Group0 stop mode enable. 0: Group0 stop mode disable.	
73h (1AE6h)	REG1AE6	7:0	Default: 0x00	Access: R/W
	-	7:3	Reserved.	
	PWM_ENABLE[2:0]	2:0	[0] 1: PWM group0 enable. 0: PWM group0 disable.	
74h (1AE8h)	REG1AE8	7:0	Default: 0x00	Access: R/W
	SYNC_MODE_EN[7:0]	7:0	[0] 1: PWM0 sync mode enable. [1] 1: PWM1 sync mode enable. ... [7] 1: PWM7 sync mode enable. for PWM3~0.	

PWM Register (Bank = 1A)				
Index (Absolute)	Mnemonic	Bit	Description	
74h (1AE9h)	REG1AE9	7:0	Default: 0x00	Access: R/W
	-	7:3	Reserved.	
	SYNC_MODE_EN[10:8]	2:0	See description of '1AE8h.'	
75h (1AEA h)	REG1AEA	7:0	Default: 0x00	Access: RO
	-	7:6	Reserved.	
	PWM_INT[5:0]	5:0	[0]: PWM group0 hold int. [1]: PWM group0 round int.	
76h (1AEC h)	REG1AEC	7:0	Default: 0x01	Access: R/W
	-	7:1	Reserved.	
	PWM_DUTY_QE0	0	If duty = 0, need to open this bit.	
77h (1AEEh)	REG1AEE	7:0	Default: 0x00	Access: R/W
	-	7:1	Reserved.	
	PWM_HOLD_MODE1	0	Hold mode 1 enable (keep value low).	
7Eh (1AFCh)	REG1AFC	7:0	Default: 0x00	Access: RO
	PWM_OUT[7:0]	7:0	PWM output for 3~0.	
7Eh (1AFDh)	REG1AFD	7:0	Default: 0x00	Access: RO
	-	7:3	Reserved.	
	PWM_OUT[10:8]	2:0	See description of '1AFCh.'	
7Fh (1AFEh)	REG1AFE	7:0	Default: 0x00	Access: R/W
	-	7:4	Reserved.	
	PWM3_SW_RST	3	PWM3 software reset.	
	PWM2_SW_RST	2	PWM2 software reset.	
	PWM1_SW_RST	1	PWM1 software reset.	
	PWM0_SW_RST	0	PWM0 software reset.	
7Fh (1AFFh)	REG1AFF	7:0	Default: 0x00	Access: R/W
	-	7:4	Reserved.	
	GROUP0_SW_RST	3	Group0 software reset.	
	-	2:0	Reserved.	

FSP Register (Bank = 16)

FSP Register (Bank = 16)				
Index (Absolute)	Mnemonic	Bit	Description	
60h (16C0h)	REG16C0	7:0	Default: 0x66	Access: R/W
	FSP_WD0[7:0]	7:0	Write data buffer0.	
60h (16C1h)	REG16C1	7:0	Default: 0x99	Access: R/W
	FSP_WD1[7:0]	7:0	Write data buffer1.	
61h (16C2h)	REG16C2	7:0	Default: 0x05	Access: R/W
	FSP_WD2[7:0]	7:0	Write data buffer2.	
61h (16C3h)	REG16C3	7:0	Default: 0x00	Access: R/W
	FSP_WD3[7:0]	7:0	Write data buffer3.	
62h (16C4h)	REG16C4	7:0	Default: 0x00	Access: R/W
	FSP_WD4[7:0]	7:0	Write data buffer4.	
62h (16C5h)	REG16C5	7:0	Default: 0x00	Access: R/W
	FSP_WD5[7:0]	7:0	Write data buffer5.	
63h (16C6h)	REG16C6	7:0	Default: 0x00	Access: R/W
	FSP_WD6[7:0]	7:0	Write data buffer6.	
63h (16C7h)	REG16C7	7:0	Default: 0x00	Access: R/W
	FSP_WD7[7:0]	7:0	Write data buffer7.	
64h (16C8h)	REG16C8	7:0	Default: 0x00	Access: R/W
	FSP_WD8[7:0]	7:0	Write data buffer8.	
64h (16C9h)	REG16C9	7:0	Default: 0x00	Access: R/W
	FSP_WD9[7:0]	7:0	Write data buffer9.	
65h (16CAh)	REG16CA	7:0	Default: 0x00	Access: RO
	FSP_RD0[7:0]	7:0	Read data buffer0.	
65h (16CBh)	REG16CB	7:0	Default: 0x00	Access: RO
	FSP_RD1[7:0]	7:0	Read data buffer1.	
66h (16CCh)	REG16CC	7:0	Default: 0x00	Access: RO
	FSP_RD2[7:0]	7:0	Read data buffer2.	
66h (16CDh)	REG16CD	7:0	Default: 0x00	Access: RO
	FSP_RD3[7:0]	7:0	Read data buffer3.	
67h (16CEh)	REG16CE	7:0	Default: 0x00	Access: RO
	FSP_RD4[7:0]	7:0	Read data buffer4.	
67h (16CFh)	REG16CF	7:0	Default: 0x00	Access: RO
	FSP_RD5[7:0]	7:0	Read data buffer5.	

FSP Register (Bank = 16)				
Index (Absolute)	Mnemonic	Bit	Description	
68h (16D0h)	REG16D0	7:0	Default: 0x00	Access: RO
	FSP_RD6[7:0]	7:0	Read data buffer6.	
68h (16D1h)	REG16D1	7:0	Default: 0x00	Access: RO
	FSP_RD7[7:0]	7:0	Read data buffer7.	
69h (16D2h)	REG16D2	7:0	Default: 0x00	Access: RO
	FSP_RD8[7:0]	7:0	Read data buffer8.	
69h (16D3h)	REG16D3	7:0	Default: 0x00	Access: RO
	FSP_RD9[7:0]	7:0	Read data buffer9.	
6Ah (16D4h)	REG16D4	7:0	Default: 0x11	Access: R/W
	FSP_WBF_SIZE1[3:0]	7:4	Set the number of bytes to be transmitted in the second command. Max size is 15 bytes. Min size is 1 byte.	
	FSP_WBF_SIZE0[3:0]	3:0	Set the number of bytes to be transmitted in the first command. Max size is 15 bytes. Min size is 1 byte.	
6Ah (16D5h)	REG16D5	7:0	Default: 0x01	Access: R/W
	-	7:4	Reserved.	
	FSP_WBF_SIZE2[3:0]	3:0	Set the number of bytes to be transmitted in the third command. Max size is 15 bytes. Min size is 1 byte.	
6Bh (16D6h)	REG16D6	7:0	Default: 0x00	Access: R/W
	FSP_RBF_SIZE1[3:0]	7:4	Set the number of bytes to be received in the second command. Max size is 10 bytes. Min size is 0 byte.	
	FSP_RBF_SIZE0[3:0]	3:0	Set the number of bytes to be received in the first command. Max size is 10 bytes. Min size is 0 byte.	
6Bh (16D7h)	REG16D7	7:0	Default: 0x01	Access: R/W
	-	7:4	Reserved.	
	FSP_RBF_SIZE2[3:0]	3:0	Set the number of bytes to be received in the third command.	

FSP Register (Bank = 16)				
Index (Absolute)	Mnemonic	Bit	Description	
			Max size is 10 bytes. Min size is 0 byte.	
6Ch (16D8h)	REG16D8	7:0	Default: 0x07	Access: R/W
	FSP_CTRL0[7:0]	7:0	Control Register0. Bit [7:4]: Reserved. Bit [3]: Enable auto check error flag. Bit [2]: Enable FSP interrupt. 0: Disable. 1: Enable. Bit [1]: Reset. 0: Reset. 1: Not Reset. Bit [0]: Enable FSP. 0: Disable. 1: Enable.	
6Ch (16D9h)	REG16D9	7:0	Default: 0xF0	Access: R/W
	FSP_CTRL1[7:0]	7:0	Control Register1. Bit [7]: Enable Second command. 0: Disable. 1: Enable. Bit [6]: Enable third command. 0: Disable. 1: Enable. Bit [5]: Enable auto check flash status. 0: Disable. 1: Enable. Bit [4:3]: Indicate which command is RDSR. 00: First command. 01: Second command. 10: Third command. 11: Reserved. Bit [2:0]: Indicate which bit is flash ready bit (for auto check flash status). If Bit [2:0] = 0x0 – 0x7, HW will auto check REG_FSP_RD10[7:0].	
6Dh (16DAh)	REG16DA	7:0	Default: 0x00	Access: WO
	-	7:1	Reserved.	
	FSP_TRIGGER	0	Start flash self-programming.	
6Eh	REG16DC	7:0	Default: 0x00	Access: RO

FSP Register (Bank = 16)				
Index (Absolute)	Mnemonic	Bit	Description	
(16DCh)	-	7:2	Reserved.	
	FSP_AUTO_CHK_ERROR	1	Reserved.	
	FSP_DONE_FLAG	0	Busy status, HW set to one when self-programming is completed. 1: Self-programming done or interrupt pending. 0: Self-programming busy or interrupt not pending.	
6Fh (16DEh)	REG16DE	7:0	Default: 0x00	Access: WO
	-	7:1	Reserved.	
	FSP_CLEAR_DONE_FLAG	0	SW needs to set this bit to clear done flag or interrupt.	
70h (16E0h)	REG16E0	7:0	Default: 0x00	Access: R/W
	FSP_WD10[7:0]	7:0	Write data buffer10.	
70h (16E1h)	REG16E1	7:0	Default: 0x00	Access: R/W
	FSP_WD11[7:0]	7:0	Write data buffer11.	
71h (16E2h)	REG16E2	7:0	Default: 0x00	Access: R/W
	FSP_WD12[7:0]	7:0	Write data buffer12.	
71h (16E3h)	REG16E3	7:0	Default: 0x00	Access: R/W
	FSP_WD13[7:0]	7:0	Write data buffer13.	
72h (16E4h)	REG16E4	7:0	Default: 0x00	Access: R/W
	FSP_WD14[7:0]	7:0	Write data buffer14.	
72h (16E5h)	REG16E5	7:0	Default: 0x00	Access: R/W
	FSP_WD15[7:0]	7:0	Write data buffer15.	
73h (16E6h)	REG16E6	7:0	Default: 0x00	Access: R/W
	FSP_WD16[7:0]	7:0	Write data buffer16.	
75h (16EAh)	REG16EA	7:0	Default: 0x00	Access: R/W
	FSP_CTRL2[7:0]	7:0	FSP_CTRL2. [0]: FSP_QUAD_MODE.	
75h (16EBh)	REG16EB	7:0	Default: 0x00	Access: R/W
	FSP_CTRL3[7:0]	7:0	FSP_CTRL3. [0]: OUTSIDE_TRIG_EN_0. [1]: OUTSIDE_TRIG_EN_1. [2]: OUTSIDE_TRIG_EN_2. [3]: OUTSIDE_TRIG_EN_3. [4]: OUTSIDE_TRIG_EN_4. [5]: OUTSIDE_TRIG_EN_5. [6]: OUTSIDE_TRIG_EN_6.	

FSP Register (Bank = 16)				
Index (Absolute)	Mnemonic	Bit	Description	
			[7]: OUTSIDE_TRIG_EN_7.	
76h (16ECh)	REG16EC	7:0	Default: 0x00	Access: R/W
	FSP_CTRL4[7:0]	7:0	FSP_CTRL4. [0]: OUTSIDE_TRIG_EN_8. [1]: OUTSIDE_TRIG_EN_9. [2]: OUTSIDE_TRIG_EN_a. [3]: OUTSIDE_TRIG_EN_b. [4]: OUTSIDE_TRIG_EN_c. [5]: OUTSIDE_TRIG_EN_d. [6]: OUTSIDE_TRIG_EN_e. [7]: OUTSIDE_TRIG_EN_f.	
76h (16EDh)	REG16ED	7:0	Default: 0x00	Access: R/W
	SPARE_0[7:0]	7:0		
77h (16EEh)	REG16EE	7:0	Default: 0xFF	Access: R/W
	SPARE_1[7:0]	7:0		
77h (16EFh)	REG16EF	7:0	Default: 0xFF	Access: R/W
	SPARE_1[15:8]	7:0	See description of '16EEh.'	
78h (16F0h)	REG16F0	7:0	Default: 0x00	Access: R/W
	FSP_WBF_SIZE_OUTSIDE[7:0]	7:0		
78h (16F1h)	REG16F1	7:0	Default: 0x00	Access: R/W
	-	7:4	Reserved.	
	FSP_WBF_SIZE_OUTSIDE[11:8]	3:0	See description of '16F0h.'	
79h (16F2h)	REG16F2	7:0	Default: 0x00	Access: R/W
	FSP_WBF_REPLACED[7:0]	7:0	Decide which wbyte will be replaced with outside write data.	
79h (16F3h)	REG16F3	7:0	Default: 0x00	Access: R/W
	FSP_WBF_OUTSIDE_SRC[1:0]	7:6		
	FSP_WBF_OUTSIDE_TX_TIMEO UT_EN	5		
	FSP_WBF_OUTSIDE_EN	4		
	FSP_WBF_MODE[3:0]	3:0	0: WBF0 = REG_FSP_WBF_SIZE_OUTSIDE + REG_FSP_WBF_SIZE0. 1: WBF1 = REG_FSP_WBF_SIZE_OUTSIDE + REG_FSP_WBF_SIZE1. 2: WBF1 = REG_FSP_WBF_SIZE_OUTSIDE + REG_FSP_WBF_SIZE2.	
7Ah	REG16F4	7:0	Default: 0x00	Access: RO

FSP Register (Bank = 16)				
Index (Absolute)	Mnemonic	Bit	Description	
(16F4h)	FSP_WBF_OUTSIDE_CNT[7:0]	7:0		
7Ah (16F5h)	REG16F5	7:0	Default: 0x00	Access: RO
	-	7:4	Reserved.	
	FSP_WBF_OUTSIDE_CNT[11:8]	3:0	See description of '16F4h.'	
7Bh (16F6h)	REG16F6	7:0	Default: 0x00	Access: R/W
	FSP_WBF_OUTSIDE_TX_TIMEO UT_MAX[7:0]	7:0	When timer == {FSP_WBF_OUTSIDE_TX_TIMEOUT_MAX,4'b0} / overflow.	
7Bh (16F7h)	REG16F7	7:0	Default: 0x00	Access: R/W
	FSP_WBF_OUTSIDE_TX_TIMEO UT_MAX[15:8]	7:0	See description of '16F6h.'	
7Ch (16F8h)	REG16F8	7:0	Default: 0x00	Access: RO
	FSP_WBF_OUTSIDE_TX_TIMEO UT_CNT[7:0]	7:0	CNT == TIME_CNT [20:4].	
7Ch (16F9h)	REG16F9	7:0	Default: 0x00	Access: RO
	FSP_WBF_OUTSIDE_TX_TIMEO UT_CNT[15:8]	7:0	See description of '16F8h.'	
7Dh (16FAh)	REG16FA	7:0	Default: 0x00	Access: RO
	FSP_DEBUG_INFO0[7:0]	7:0		
7Dh (16FBh)	REG16FB	7:0	Default: 0x00	Access: RO
	FSP_DEBUG_INFO0[15:8]	7:0	See description of '16FAh.'	
7Eh (16FCh)	REG16FC	7:0	Default: 0x00	Access: RO
	FSP_DEBUG_INFO1[7:0]	7:0		
7Eh (16FDh)	REG16FD	7:0	Default: 0x00	Access: RO
	FSP_DEBUG_INFO1[15:8]	7:0	See description of '16FCh.'	
7Fh (16FEh)	REG16FE	7:0	Default: 0x00	Access: RO
	FSP_DEBUG_INFO2[7:0]	7:0		
7Fh (16FFh)	REG16FF	7:0	Default: 0x00	Access: RO
	FSP_DEBUG_INFO2[15:8]	7:0	See description of '16FEh.'	

QSPI Register (Bank = 17)

QSPI Register (Bank = 17)				
Index (Absolute)	Mnemonic	Bit	Description	
00h (1700h)	REG1700	7:0	Default: 0x00	Access: R/W
	MASK_GRANT[7:0]	7:0		
01h (1702h)	REG1702	7:0	Default: 0x00	Access: R/W
	MASK_TIME_OUT_CTRL[7:0]	7:0		
02h (1705h)	REG1705	7:0	Default: 0x00	Access: RO
	MASK_TIME_OUT_STATUS[7:0]	7:0		
03h (1706h)	REG1706	7:0	Default: 0x00	Access: R/W
	MASK_TIME_OUT_LEN[7:0]	7:0		
03h (1707h)	REG1707	7:0	Default: 0x00	Access: R/W
	MASK_TIME_OUT_LEN[15:8]	7:0	See description of '1706h.'	
04h (1708h)	REG1708	7:0	Default: 0x00	Access: R/W
	MASK_TIME_OUT_LEN[23:16]	7:0	See description of '1706h.'	
04h (1709h)	REG1709	7:0	Default: 0x10	Access: R/W
	MASK_TIME_OUT_LEN[31:24]	7:0	See description of '1706h.'	
05h (170Ah)	REG170A	7:0	Default: 0x00	Access: RO
	MASK_TIME_OUT_CNT[7:0]	7:0		
05h (170Bh)	REG170B	7:0	Default: 0x00	Access: RO
	MASK_TIME_OUT_CNT[15:8]	7:0	See description of '170Ah.'	
06h (170Ch)	REG170C	7:0	Default: 0x00	Access: RO
	MASK_TIME_OUT_CNT[23:16]	7:0	See description of '170Ah.'	
06h (170Dh)	REG170D	7:0	Default: 0x00	Access: RO
	MASK_TIME_OUT_CNT[31:24]	7:0	See description of '170Ah.'	
07h (170Eh)	REG170E	7:0	Default: 0x00	Access: R/W
	SPI_ARB_CTRL[7:0]	7:0	[0]: NON_PM_ACK TIMEOUT_EN.	
07h (170Fh)	REG170F	7:0	Default: 0x00	Access: RO
	SPI_ARB_STATUS[7:0]	7:0	[0]: REG_NON_PM_ACK TIMEOUT_FLAG.	
08h (1710h)	REG1710	7:0	Default: 0xFF	Access: R/W
	NON_PM_ACK_TIMEOUT_LEN[7:0]	7:0		
08h (1711h)	REG1711	7:0	Default: 0x01	Access: R/W
	NON_PM_ACK_TIMEOUT_LEN	7:0	See description of '1710h.'	

QSPI Register (Bank = 17)				
Index (Absolute)	Mnemonic	Bit	Description	
	[15:8]			
09h (1712h)	REG1712	7:0	Default: 0x00	Access: RO
	NON_PM_ACK_TIMEOUT_CNT [7:0]	7:0		
09h (1713h)	REG1713	7:0	Default: 0x00	Access: RO
	NON_PM_ACK_TIMEOUT_CNT [15:8]	7:0	See description of '1712h.'	
0Ah (1714h)	REG1714	7:0	Default: 0x02	Access: R/W
	SPI_SW_MODE[7:0]	7:0	[0]: Cs SW mode enable. [1]: Cs SW ctrl 0: cs = 0; 1: cs = 1.	
40h (1780h)	REG1780	7:0	Default: 0x04	Access: R/W
	-	7:3	Reserved.	
	DELAY_TREE_SEL[2:0]	2:0	The value of the delay tree.	
50h (17A0h)	REG17A0	7:0	Default: 0x00	Access: R/W
	CMD_111_M0[7:0]	7:0	The user defined command for 1-1-1 normal read mode.	
50h (17A1h)	REG17A1	7:0	Default: 0x00	Access: R/W
	CMD_111_M1[7:0]	7:0	The user defined command for 1-1-1 fast read mode.	
51h (17A2h)	REG17A2	7:0	Default: 0x00	Access: R/W
	CMD_112[7:0]	7:0	The user defined command for 1-1-2 read mode.	
51h (17A3h)	REG17A3	7:0	Default: 0x00	Access: R/W
	CMD_122[7:0]	7:0	The user defined command for 1-2-2 read mode.	
52h (17A4h)	REG17A4	7:0	Default: 0x00	Access: R/W
	CMD_114[7:0]	7:0	The user defined command for 1-1-4 read mode.	
52h (17A5h)	REG17A5	7:0	Default: 0x00	Access: R/W
	CMD_144[7:0]	7:0	The user defined command for 1-4-4 read mode.	
53h (17A6h)	REG17A6	7:0	Default: 0x00	Access: R/W
	CMD_444_M0[7:0]	7:0	The user defined command for 4-4-4 read mode (dummy cycle = 4).	
53h (17A7h)	REG17A7	7:0	Default: 0x00	Access: R/W
	CMD_444_M1[7:0]	7:0	The user defined command for 4-4-4 read mode (dummy cycle = 6).	
54h (17A8h)	REG17A8	7:0	Default: 0x00	Access: R/W
	DUMMY_CYC_VAL[7:0]	7:0	The user defined dummy cycle value.	

QSPI Register (Bank = 17)				
Index (Absolute)	Mnemonic	Bit	Description	
54h (17A9h)	REG17A9	7:0	Default: 0x00	Access: R/W
	-	7:4	Reserved.	
	WRAP_VAL[3:0]	3:0	The user defined wrap value for SPI NAND.	
58h (17B0h)	REG17B0	7:0	Default: 0x00	Access: R/W
	2_CMD_111_M0[7:0]	7:0	The user defined command for 1-1-1 normal read mode for CS1.	
58h (17B1h)	REG17B1	7:0	Default: 0x00	Access: R/W
	2_CMD_111_M1[7:0]	7:0	The user defined command for 1-1-1 fast read mode for CS1.	
59h (17B2h)	REG17B2	7:0	Default: 0x00	Access: R/W
	2_CMD_112[7:0]	7:0	The user defined command for 1-1-2 read mode for CS1.	
59h (17B3h)	REG17B3	7:0	Default: 0x00	Access: R/W
	2_CMD_122[7:0]	7:0	The user defined command for 1-2-2 read mode for CS1.	
5Ah (17B4h)	REG17B4	7:0	Default: 0x00	Access: R/W
	2_CMD_114[7:0]	7:0	The user defined command for 1-1-4 read mode for CS1.	
5Ah (17B5h)	REG17B5	7:0	Default: 0x00	Access: R/W
	2_CMD_144[7:0]	7:0	The user defined command for 1-4-4 read mode for CS1.	
5Bh (17B6h)	REG17B6	7:0	Default: 0x00	Access: R/W
	2_CMD_444_M0[7:0]	7:0	The user defined command for 4-4-4 read mode for CS1 (dummy cycle = 4).	
5Bh (17B7h)	REG17B7	7:0	Default: 0x00	Access: R/W
	2_CMD_444_M1[7:0]	7:0	The user defined command for 4-4-4 read mode for CS1 (dummy cycle = 6).	
5Ch (17B8h)	REG17B8	7:0	Default: 0x00	Access: R/W
	2_DUMMY_CYC_VAL[7:0]	7:0	The user defined dummy cycle value for CS1.	
5Ch (17B9h)	REG17B9	7:0	Default: 0x00	Access: R/W
	-	7:4	Reserved.	
	2_WRAP_VAL[3:0]	3:0	The user defined wrap value for SPI NAND.	
60h (17C0h)	REG17C0	7:0	Default: 0x00	Access: R/W
	-	7	Reserved.	
	SECOND_CKG_SPI[6:0]	6:0	Bit [3:0]: User defined dummy cycle number for CS1. Bit [4]: User defined dummy cycle mode enable for CS1. 0: Disable. 1: Enable.	

QSPI Register (Bank = 17)				
Index (Absolute)	Mnemonic	Bit	Description	
60h (17C1h)	REG17C1	7:0	Default: 0x00	Access: R/W
	SECOND_CFG_QSPI[7:0]	7:0	Bit [8]: CMD_BYPASS_MODE for CS1. Bit [11]: 3/4 byte address mode. 0: 3 byte. 1: 4 byte.	
61h (17C2h)	REG17C2	7:0	Default: 0x1A	Access: R/W
	SECOND_CSZ_SETUP[3:0]	7:4	CSZ setup time for CS1 (relative to SCK). 4'h0: 1 SPI clock cycle. 4'h1: 2 SPI clock cycles. 4'hf: 16 SPI clock cycles.	
	SECOND_CSZ_HIGH[3:0]	3:0	CSZ deselect time for CS1 (SCZ = high). 4'h0: 1 SPI clock cycle. 4'h1: 2 SPI clock cycles. 4'hf: 16 SPI clock cycles.	
61h (17C3h)	REG17C3	7:0	Default: 0x01	Access: R/W
	-	7:4	Reserved.	
	SECOND_CSZ_HOLD[3:0]	3:0	CSZ hold time for CS1 (relative to SCK). 4'h0: 1 SPI clock cycle. 4'h1: 2 SPI clock cycles. 4'hf: 16 SPI clock cycles.	
62h (17C4h)	REG17C4	7:0	Default: 0x00	Access: R/W
	-	7:4	Reserved.	
	SECOND_MODE_SEL[3:0]	3:0	Second SPI mode select for CS1. 0x0: Normal mode (1-1-1), SPI command is 0x03. 0x1: Enable fast read mode (1-1-1), SPI command is 0x0B. 0x2: Enable (1-1-2) mode, SPI command is 0x3B. 0x3: Enable (1-2-2) mode, SPI command is 0xBB. 0xa: Enable (1-1-4) mode, SPI command is 0x6B. 0xb: Enable (1-4-4) mode, SPI command is 0xEB. 0xc: Enable (4-4-4) mode with 4 dummy cycles, SPI command is 0x0B. 0xd: Enable (4-4-4) mode with 6 dummy cycles, SPI command is 0xEB.	
63h (17C6h)	REG17C6	7:0	Default: 0x00	Access: R/W
	SECOND_REPLACED_CMD[7:0]	7:0	The second replaced command for CS1.	
64h	REG17C8	7:0	Default: 0x00	Access: R/W

QSPI Register (Bank = 17)				
Index (Absolute)	Mnemonic	Bit	Description	
(17C8h)	-	7:4	Reserved.	
	CLK_DIV_CNT_EN	3	SPI receive div. counter enable.	
	CLK_DIV_CNT_SEL[2:0]	2:0	SPI receive div. counter output mux select.	
66h (17CCh)	REG17CC	7:0	Default: 0xFF	Access: R/W
	TIMEOUT_CNT_VALUE[7:0]	7:0	Timeout counter value.	
66h (17CDh)	REG17CD	7:0	Default: 0xFF	Access: R/W
	TIMEOUT_CNT_VALUE[15:8]	7:0	See description of '17CCh.'	
67h (17CEh)	REG17CE	7:0	Default: 0xFF	Access: R/W
	TIMEOUT_CNT_VALUE[23:16]	7:0	See description of '17CCh.'	
67h (17CFh)	REG17CF	7:0	Default: 0x40	Access: R/W
	TIMEOUT_CNT_EN	7	Timeout counter enable.	
	TIMEOUT_CNT_RST	6	Timeout counter reset.	
	-	5:0	Reserved.	
68h (17D0h)	REG17D0	7:0	Default: 0x00	Access: R/W
	CSZ_REPLACE_VAL[3:0]	7:4	Replace CSZ signal by register value. B0: Replace value of CS0. B1: Replace value of CS1. B2: Replace value of CS2. B3: Replace value of CS3.	
	CSZ_REPLACE_EN[3:0]	3:0	Enable replacement of CSZ signal by register value. B0: CS0 replace enable. B1: CS1 replace enable. B2: CS2 replace enable. B3: CS3 replace enable.	
6Dh (17DAh)	REG17DA	7:0	Default: 0x1A	Access: R/W
	FSP_CSZ_SETUP[3:0]	7:4	CSZ setup time for FSP (relative to SCK). 4'h0: 1 SPI clock cycle. 4'h1: 2 SPI clock cycles. 4'hf: 16 SPI clock cycles.	
	FSP_CSZ_HIGH[3:0]	3:0	CSZ deselect time for FSP (SCZ = high). 4'h0: 1 SPI clock cycle. 4'h1: 2 SPI clock cycles. 4'hf: 16 SPI clock cycles.	
6Dh (17DBh)	REG17DB	7:0	Default: 0x01	Access: R/W
	-	7:4	Reserved.	
	FSP_CSZ_HOLD[3:0]	3:0	CSZ hold time for FSP (relative to SCK).	

QSPI Register (Bank = 17)				
Index (Absolute)	Mnemonic	Bit	Description	
			4'h0: 1 SPI clock cycle. 4'h1: 2 SPI clock cycles. 4'hf: 16 SPI clock cycles.	
6Eh (17DCh)	REG17DC	7:0	Default: 0x1A	Access: R/W
	FSP2_CSZ_SETUP[3:0]	7:4	CSZ setup time for FSP2 (relative to SCK). 4'h0: 1 SPI clock cycle. 4'h1: 2 SPI clock cycles. 4'hf: 16 SPI clock cycles.	
	FSP2_CSZ_HIGH[3:0]	3:0	CSZ deselect time for FSP2 (SCZ = high). 4'h0: 1 SPI clock cycle. 4'h1: 2 SPI clock cycles. 4'hf: 16 SPI clock cycles.	
6Eh (17DDh)	REG17DD	7:0	Default: 0x01	Access: R/W
	-	7:4	Reserved.	
	FSP2_CSZ_HOLD[3:0]	3:0	CSZ hold time for FSP2 (relative to SCK). 4'h0: 1 SPI clock cycle. 4'h1: 2 SPI clock cycles. 4'hf: 16 SPI clock cycles.	
70h (17E0h)	REG17E0	7:0	Default: 0x00	Access: R/W
	-	7	Reserved.	
	CKG_SPI[6:0]	6:0	Bit [3:0]: User defined dummy cycle number. Bit [4]: User defined dummy cycle mode enable. 0: Disable. 1: Enable. Bit [5]: Force to disable Address Continue at FSP mode. 0: Disable. 1: Enable.	
70h (17E1h)	REG17E1	7:0	Default: 0x00	Access: R/W
	CFG_QSPI[7:0]	7:0	Bit [8]: CMD_BYPASS_MODE. Bit [9]: Disable Address Continue. Bit [10]: Wait FSP done. Bit [11]: 3/4 byte address mode. 0: 3 byte. 1: 4 byte.	
71h (17E2h)	REG17E2	7:0	Default: 0x1A	Access: R/W
	CSZ_SETUP[3:0]	7:4	CSZ setup time (relative to SCK). 4'h0: 1 SPI clock cycle. 4'h1: 2 SPI clock cycles.	

QSPI Register (Bank = 17)				
Index (Absolute)	Mnemonic	Bit	Description	
			4'hf: 16 SPI clock cycles.	
	CSZ_HIGH[3:0]	3:0	CSZ deselect time (SCZ = high). 4'h0: 1 SPI clock cycle. 4'h1: 2 SPI clock cycles. 4'hf: 16 SPI clock cycles.	
71h (17E3h)	REG17E3	7:0	Default: 0x01	Access: R/W
	-	7:4	Reserved.	
	CSZ_HOLD[3:0]	3:0	CSZ hold time (relative to SCK). 4'h0: 1 SPI clock cycle. 4'h1: 2 SPI clock cycles. 4'hf: 16 SPI clock cycles.	
72h (17E4h)	REG17E4	7:0	Default: 0x00	Access: R/W
	-	7:4	Reserved.	
	MODE_SEL[3:0]	3:0	SPI mode select (command-address-data). 0x0: Normal mode (1-1-1), SPI command is 0x03. 0x1: Enable fast read mode (1-1-1), SPI command is 0x0B. 0x2: Enable (1-1-2) mode, SPI command is 0x3B. 0x3: Enable (1-2-2) mode, SPI command is 0xBB. 0xa: Enable (1-1-4) mode, SPI command is 0x6B. 0xb: Enable (1-4-4) mode, SPI command is 0xEB. 0xc: Enable (4-4-4) mode with 4 dummy cycles, SPI command is 0x0B. 0xd: Enable (4-4-4) mode with 6 dummy cycles, SPI command is 0xEB.	
73h (17E6h)	REG17E6	7:0	Default: 0x00	Access: R/W
	REPLACED_CMD[7:0]	7:0	The replaced command.	
74h (17E8h)	REG17E8	7:0	Default: 0x00	Access: R/W
	SPARE_0[7:0]	7:0	Bit [0]: Wrap mode for CS0 flash, for SW to write when the flash enters wrap mode. 0: Not wrap mode. 1: Wrap mode. Bit [1]: Wrap 16 bytes for CS0 flash, for SW to write. 0: Not 16 byte. 1: 16 byte. Bit [2]: Wrap 32 bytes for CS0 flash, for SW to write. 0: Not 32 byte. 1: 32 byte. Bit [3]: Wrap 64 bytes for CS0 flash, for SW to write.	

QSPI Register (Bank = 17)				
Index (Absolute)	Mnemonic	Bit	Description	
			0: Not 64 byte. 1: 64 byte. Bit [4]: Wrap 128 bytes for CS0 flash, for SW to write. 0: Not 128 byte. 1: 128 byte. Bit [7:5]: Reserved, keep the data at default value. Bit [8]: Wrap mode for CS1 flash, for SW to write when the flash enters wrap mode. 0: Not wrap mode. 1: Wrap mode. Bit [9]: Wrap 16 bytes for CS1 flash, for SW to write. 0: Not 16 byte. 1: 16 byte. Bit [10]: Wrap 32 bytes for CS1 flash, for SW to write. 0: Not 32 byte. 1: 32 byte. Bit [11]: Wrap 64 bytes for CS1 flash, for SW to write. 0: Not 64 byte. 1: 64 byte. Bit [12]: Wrap 128 bytes for CS1 flash, for SW to write. 0: Not 128 byte. 1: 128 byte. Bit [15:13]: Reserved, keep the data at default value.	
74h	REG17E9	7:0	Default: 0x00	Access: R/W
(17E9h)	SPARE_0[15:8]	7:0	See description of '17E8h.'	
76h	REG17EC	7:0	Default: 0x00	Access: RO
(17ECh)	DEBUG_BUS_0[7:0]	7:0	DEBUG_BUS_0.	
76h	REG17ED	7:0	Default: 0x00	Access: RO
(17EDh)	DEBUG_BUS_0[15:8]	7:0	See description of '17ECh.'	
77h	REG17EE	7:0	Default: 0x00	Access: RO
(17EEh)	DEBUG_BUS_1[7:0]	7:0	DEBUG_BUS_1.	
77h	REG17EF	7:0	Default: 0x00	Access: RO
(17EFh)	DEBUG_BUS_1[15:8]	7:0	See description of '17EEh.'	
78h	REG17F0	7:0	Default: 0x00	Access: RO
(17F0h)	DEBUG_BUS_2[7:0]	7:0	DEBUG_BUS_2.	
78h	REG17F1	7:0	Default: 0x00	Access: RO
(17F1h)	DEBUG_BUS_2[15:8]	7:0	See description of '17F0h.'	

QSPI Register (Bank = 17)				
Index (Absolute)	Mnemonic	Bit	Description	
79h (17F2h)	REG17F2	7:0	Default: 0x00	Access: RO
	DEBUG_BUS_3[7:0]	7:0	DEBUG_BUS_3.	
79h (17F3h)	REG17F3	7:0	Default: 0x00	Access: RO
	DEBUG_BUS_3[15:8]	7:0	See description of '17F2h.'	
7Ah (17F4h)	REG17F4	7:0	Default: 0x00	Access: R/W
	-	7:2	Reserved.	
	CHIP_SELECT[1:0]	1:0	00: Select external #1 SPI Flash. 01: Select external #2 SPI Flash. 10: Select external #3 SPI Flash. 11: Reserved.	
7Bh (17F6h)	REG17F6	7:0	Default: 0x00	Access: RO
	-	7:1	Reserved.	
	SWITCH_CS_BUSY	0	1: Switch SPI CS is busy. (In this stage, access to SPI Flash is forbidden.) 0: Switch SPI CS is done.	
7Ch (17F8h)	REG17F8	7:0	Default: 0xFF	Access: R/W
	FUNC_SETTING_DEF1[7:0]	7:0	Bit [0]: Reserved. Bit [1]: Use comb. CSZ setting (setup and high). 0: Disable. 1: Enable. Bit [2]: Reserved. Bit [4:3]: Embedded flash size. 00: 64Mb. 01: 128Mb. 10: 16Mb. 11: 32Mb. Bit [15:5]: Reserved, keep the data at default value.	
7Ch (17F9h)	REG17F9	7:0	Default: 0xFF	Access: R/W
	FUNC_SETTING_DEF1[15:8]	7:0	See description of '17F8h.'	
7Dh (17FAh)	REG17FA	7:0	Default: 0x00	Access: R/W
	FUNC_SETTING_DEF0[7:0]	7:0	Bit [0]: Use index to support 4-byte address. 0: Disable. 1: Enable. Bit [1]: One burst data over two flash enable. 0: Disable. 1: Enable. Bit [2]: 4-byte address mode enable.	

QSPI Register (Bank = 17)				
Index (Absolute)	Mnemonic	Bit	Description	
			0: Disable. 1: Enable. Bit [3]: Use user defined command. 0: Use default command. 1: Use user defined command. Bit [4]: Use user defined dummy cycle value at command bypass mode. 0: Use 0xa5. 1: Use user defined value. Bit [5]: Support enable switch for incremental command at flash wrap mode. 0: Disable. 1: Enable. Bit [6]: Support enable switch for wrap command at flash non-wrap mode. 0: Disable. 1: Enable. Bit [7]: Reserved, keep the data at default value. Bit [8]: SPI IO pin mode after FSP read. 0: Output mode. 1: Input mode. Bit [9]: CS select by address enable. Bit [10]: Reserved. Bit [11]: ADDR_2BYTE_EN. Bit [12]: FORCE_DUMMY_CYC_EN. Bit [13]: ADDR_OVER_WRITE_EN. Bit [15:12]: Reserved, keep the data at default value.	
7Dh (17FBh)	REG17FB	7:0	Default: 0x00	Access: R/W
	FUNC_SETTING_DEF0[15:8]	7:0	See description of '17FAh.'	
7Fh (17FEh)	REG17FE	7:0	Default: 0x00	Access: R/W
	-	7:1	Reserved.	
	ENDIA	0	For 32-bit CPU read data.	

WDT Register (Bank = 30)

WDT Register (Bank = 30)				
Index (Absolute)	Mnemonic	Bit	Description	
00h (3000h)	REG3000	7:0	Default: 0x00	Access: WO
	-	7:1	Reserved.	
	WDT_CLR	0	Write "1" to re-start WDT.	
02h (3004h)	REG3004	7:0	Default: 0x00	Access: R/W
	-	7:1	Reserved.	
	WDT_RST_FLAG	0	Assert: WDT reset has occurred. Write "1" to clear.	
02h (3005h)	REG3005	7:0	Default: 0x09	Access: R/W
	WDT_RST_LEN[7:0]	7:0	Length of WDT reset. 0: One xtal clock; 1: two xtal clocks; 3	
03h (3006h)	REG3006	7:0	Default: 0xFF	Access: R/W
	WDT_INT[7:0]	7:0	WDT interrupt period. Interrupt is asserted when "WDT counter [31:16]" is equal to WDT_INT and "WDT counter [15:0]" is equal to 0x0000.	
03h (3007h)	REG3007	7:0	Default: 0xFF	Access: R/W
	WDT_INT[15:8]	7:0	See description of '3006h.'	
04h (3008h)	REG3008	7:0	Default: 0xFF	Access: R/W
	WDT_MAX[7:0]	7:0	WDT period maximum value. WDT is enabled if WDT_MAX is not equal to 0x00000000.	
04h (3009h)	REG3009	7:0	Default: 0xFF	Access: R/W
	WDT_MAX[15:8]	7:0	See description of '3008h.'	
05h (300Ah)	REG300A	7:0	Default: 0xFF	Access: R/W
	WDT_MAX[23:16]	7:0	See description of '3008h.'	
05h (300Bh)	REG300B	7:0	Default: 0xFF	Access: R/W
	WDT_MAX[31:24]	7:0	See description of '3008h.'	



NONPM CH4 REGISTER TABLE

CHIPTOP Register (Bank = 101E)

CHIPTOP Register (Bank = 101E)				
Index (Absolute)	Mnemonic	Bit	Description	
00h (101E00h)	REG101E00	7:0	Default: 0x00	Access: R/W
	-	7:1	Reserved.	
	CKG_ALLDFT	0	MIU clock select DFT clock. MPLL_SYN clock select DFT clock. MIU_REC clock select DFT clock. GE clock select DFT clock.	
01h (101E03h)	REG101E03	7:0	Default: 0x02	Access: R/W
	-	7:6	Reserved.	
	SETL	5	Digital pads set low.	
	SETH	4	Digital pads set high.	
	-	3:2	Reserved.	
	FT_MODE	1	FT mode. 1: Enable FT mode. 0: Disable FT mode.	
12h (101E24h)	REG101E24	7:0	Default: 0x00	Access: R/W
	-	7:6	Reserved.	
	TEST_OUT_MODE[1:0]	5:4	Select TEST_OUT mode. 2'd0: TEST_OUT functions are not enabled. 2'd1: TEST_OUT [23:0] use FUART/SR/SPI0 pads. 2'd2: TEST_OUT [23:0] use I2C0/SD/USB/SAR/PM/ETH pads. 2'd3: TEST_OUT [15:0] use SR pads.	
	-	3:2	Reserved.	
	TEST_IN_MODE[1:0]	1:0	Select TEST_IN mode. 2'd0: TEST_IN functions are not enabled. 2'd1: TEST_IN [23:0] use FUART/SR/SPI0 pads. 2'd2: TEST_IN [23:0] use I2C0/SD/USB/SAR/PM/ETH pads. 2'd3: TEST_IN [14:0] use SAR/ETH/FUART/SPI0/SD/USB pads.	
1Ch (101E39h)	REG101E39	7:0	Default: 0x00	Access: R/W
	BIST_MODE_EXT	7	BIST mode enable (REG_HK51_UART0_EN).	
	BIST_START_EXT	6	BIST mode start.	
	-	5	Reserved.	
	FORCE_ALLSRAM_ON	4	Force power-on against all of the SRAM on the chip.	

CHIPTOP Register (Bank = 101E)				
Index (Absolute)	Mnemonic	Bit	Description	
	-	3:0	Reserved.	
1Dh (101E3Ah)	REG101E3A	7:0	Default: 0x00	Access: RO
	BIST_DONE[7:0]	7:0	Indicate SRAM done. [0]: DIG_GP. [1]: PM_GP. [2]: SC_GP. [3]: DEC_GP. [4] – [14]: N/A. [15]: ALL.	
1Dh (101E3Bh)	REG101E3B	7:0	Default: 0x00	Access: RO
	BIST_DONE[15:8]	7:0	See description of '101E3Ah.'	
1Eh (101E3Ch)	REG101E3C	7:0	Default: 0x00	Access: RO
	BIST_FAIL[7:0]	7:0	Indicate SRAM fail. [0]: DIG_GP. [1]: PM_GP. [2]: SC_GP. [3]: DEC_GP. [4] – [14]: N/A. [15]: ALL.	
1Eh (101E3Dh)	REG101E3D	7:0	Default: 0x00	Access: RO
	BIST_FAIL[15:8]	7:0	See description of '101E3Ch.'	
20h (101E40h)	REG101E40	7:0	Default: 0xFF	Access: R/W
	CHIPTOP_DUMMY_0[7:0]	7:0	Dummy registers for CHIPTOP.	
20h (101E41h)	REG101E41	7:0	Default: 0xFF	Access: R/W
	CHIPTOP_DUMMY_0[15:8]	7:0	See description of '101E40h.'	
21h (101E42h)	REG101E42	7:0	Default: 0x00	Access: R/W
	CHIPTOP_DUMMY_1[7:0]	7:0	Dummy registers for CHIPTOP. [0]: CLK_MIU_XD2MIU ICG control. 0: Disable. 1: Enable.	
21h (101E43h)	REG101E43	7:0	Default: 0x00	Access: R/W
	CHIPTOP_DUMMY_1[15:8]	7:0	See description of '101E42h.'	
22h (101E44h)	REG101E44	7:0	Default: 0xFF	Access: R/W
	CHIPTOP_DUMMY_2[7:0]	7:0	Dummy registers for CHIPTOP.	
22h (101E45h)	REG101E45	7:0	Default: 0xFF	Access: R/W
	CHIPTOP_DUMMY_2[15:8]	7:0	See description of '101E44h.'	

CHIPTOP Register (Bank = 101E)				
Index (Absolute)	Mnemonic	Bit	Description	
23h (101E46h)	REG101E46	7:0	Default: 0x00	Access: R/W
	CHIPTOP_DUMMY_3[7:0]	7:0	Dummy registers for CHIPTOP.	
23h (101E47h)	REG101E47	7:0	Default: 0x00	Access: R/W
	CHIPTOP_DUMMY_3[15:8]	7:0	See description of '101E46h.'	
30h (101E60h)	REG101E60	7:0	Default: 0x00	Access: RO, R/W
	WAKEUP_XTAL_IRQ_EN	7	Enable wakeup xtal interrupt.	
	WAKEUP_XTAL_IRQ	6	Wakeup xtal interrupt status.	
	WAKEUP_XTAL_IRQ_CLR	5	Wakeup xtal interrupt clear.	
	-	4	Reserved.	
	GATE_XTAL12_NODIE	3	Gated xtal 12MHz nodie. Write 1 to gate off, wakeup by GPIO or USB.	
	GATE_XTAL12_CORE	2	Gated xtal 12MHz core. Write 1 to gate off, wakeup by GPIO or USB.	
	GATE_XTAL24_NODIE	1	Gated xtal 24MHz nodie. Write 1 to gate off, wakeup by GPIO or USB.	
30h (101E61h)	REG101E61	7:0	Default: 0x00	Access: RO, R/W
	-	7:5	Reserved.	
	WAKEUP_XTAL_USB_EVENT	4	Xtal wakeup by USB event.	
	-	3:1	Reserved.	
	WAKEUP_XTAL_USB_EN	0	Enable USB wakeup xtal gating.	
32h (101E64h)	REG101E64	7:0	Default: 0x00	Access: R/W
	WAKEUP_XTAL_GPIO_EN[7:0]	7:0	Enable GPIO wakeup xtal gating.	
32h (101E65h)	REG101E65	7:0	Default: 0x00	Access: R/W
	WAKEUP_XTAL_GPIO_EN[15:8]	7:0	See description of '101E64h.'	
33h (101E66h)	REG101E66	7:0	Default: 0x00	Access: R/W
	WAKEUP_XTAL_GPIO_EN[23:16]	7:0	See description of '101E64h.'	
33h (101E67h)	REG101E67	7:0	Default: 0x00	Access: R/W
	WAKEUP_XTAL_GPIO_EN[31:24]	7:0	See description of '101E64h.'	
34h	REG101E68	7:0	Default: 0x00	Access: R/W

CHIPTOP Register (Bank = 101E)				
Index (Absolute)	Mnemonic	Bit	Description	
(101E68h)	WAKEUP_XTAL_GPIO_EN[39:32]	7:0	See description of '101E64h.'	
34h (101E69h)	REG101E69	7:0	Default: 0x00	Access: R/W
	WAKEUP_XTAL_GPIO_EN[47:40]	7:0	See description of '101E64h.'	
35h (101E6Ah)	REG101E6A	7:0	Default: 0x00	Access: R/W
	WAKEUP_XTAL_GPIO_EN[55:48]	7:0	See description of '101E64h.'	
35h (101E6Bh)	REG101E6B	7:0	Default: 0x00	Access: R/W
	WAKEUP_XTAL_GPIO_EN[63:56]	7:0	See description of '101E64h.'	
36h (101E6Ch)	REG101E6C	7:0	Default: 0x00	Access: R/W
	-	7	Reserved.	
	WAKEUP_XTAL_GPIO_EN[70:64]	6:0	See description of '101E64h.'	
38h (101E70h)	REG101E70	7:0	Default: 0x00	Access: RO
	WAKEUP_XTAL_GPIO_EVENT[7:0]	7:0	Xtal wakeup by GPIO event.	
38h (101E71h)	REG101E71	7:0	Default: 0x00	Access: RO
	WAKEUP_XTAL_GPIO_EVENT[15:8]	7:0	See description of '101E70h.'	
39h (101E72h)	REG101E72	7:0	Default: 0x00	Access: RO
	WAKEUP_XTAL_GPIO_EVENT[23:16]	7:0	See description of '101E70h.'	
39h (101E73h)	REG101E73	7:0	Default: 0x00	Access: RO
	WAKEUP_XTAL_GPIO_EVENT[31:24]	7:0	See description of '101E70h.'	
3Ah (101E74h)	REG101E74	7:0	Default: 0x00	Access: RO
	WAKEUP_XTAL_GPIO_EVENT[39:32]	7:0	See description of '101E70h.'	
3Ah (101E75h)	REG101E75	7:0	Default: 0x00	Access: RO
	WAKEUP_XTAL_GPIO_EVENT[47:40]	7:0	See description of '101E70h.'	
3Bh (101E76h)	REG101E76	7:0	Default: 0x00	Access: RO
	WAKEUP_XTAL_GPIO_EVENT	7:0	See description of '101E70h.'	

CHIPTOP Register (Bank = 101E)

Index (Absolute)	Mnemonic	Bit	Description
	T[55:48]		
3Bh (101E77h)	REG101E77	7:0	Default: 0x00 Access: RO
	WAKEUP_XTAL_GPIO_EVEN T[63:56]	7:0	See description of '101E70h.'
3Ch (101E78h)	REG101E78	7:0	Default: 0x00 Access: RO
	-	7	Reserved.
	WAKEUP_XTAL_GPIO_EVEN T[70:64]	6:0	See description of '101E70h.'
3Dh (101E7Ah)	REG101E7A	7:0	Default: 0xFF Access: R/W
	WAKEUP_DEBOUNCE_TIME[7:0]	7:0	Xtal wakeup debounce time (unit: 24MHz).
3Dh (101E7Bh)	REG101E7B	7:0	Default: 0x03 Access: R/W
	WAKEUP_DEBOUNCE_TIME[15:8]	7:0	See description of '101E7Ah.'
40h (101E80h)	REG101E80	7:0	Default: 0x03 Access: R/W
	-	7:2	Reserved.
	MCU_BRIDGE_EN_MODE[1: 0]	1:0	Clock MCU gating control. 00: Use MCU_BRIDGE_EN (HW saving mode 0). 01: Use MCU_BRIDGE_EN_D (HW saving mode 1, preferred). 10: SW saving power mode. 11: Always enable (default).
44h (101E88h)	REG101E88	7:0	Default: 0xFF Access: R/W
	RIU_WCLK_MASK[7:0]	7:0	RIU write clock mask. [0]: SC_GP. [1]: VHE_GP. [2]: HEMCU_GP. [3]: MIPI_GP. [4]: MCU_IF_GP. [5]: Others.
44h (101E89h)	REG101E89	7:0	Default: 0xFF Access: R/W
	RIU_WCLK_MASK[15:8]	7:0	See description of '101E88h.'
45h (101E8Ah)	REG101E8A	7:0	Default: 0xFF Access: R/W
	RESERVED3[7:0]	7:0	Reserved register.
45h (101E8Bh)	REG101E8B	7:0	Default: 0xFF Access: R/W
	RESERVED3[15:8]	7:0	See description of '101E8Ah.'

CHIPTOP Register (Bank = 101E)

Index (Absolute)	Mnemonic	Bit	Description
46h (101E8Ch)	REG101E8C	7:0	Default: 0xFF Access: R/W
	RESERVED4[7:0]	7:0	Reserved register.
46h (101E8Dh)	REG101E8D	7:0	Default: 0xFF Access: R/W
	RESERVED4[15:8]	7:0	See description of '101E8Ch.'
49h (101E92h)	REG101E92	7:0	Default: 0x01 Access: R/W
	-	7:1	Reserved.
	XIU_PIPE_EN	0	Enable XIU pipe function.
50h (101EA1h)	REG101EA1	7:0	Default: 0x80 Access: R/W
	ALLPAD_IN	7	1: Set all pads (except PM) as input.
	-	6:0	Reserved.
53h (101EA6h)	REG101EA6	7:0	Default: 0x10 Access: R/W
	UART_SEL1[3:0]	7:4	Select controller for PAD_FUART_RX and PAD_FUART_TX.
	UART_SEL0[3:0]	3:0	Select controller for PAD_PM_UART_RX and PAD_PM_UART_TX. 0000: N/A. 0001: FUART. 0010: UART0. 0011: UART1. 0100: UART2. 0101: UART_DEC. Note: For PAD_PM_UART_RX and PAD_PM_UART_TX, please refer to the "REG_HK51_UART0_EN" and "REG_UART_RX_ENABLE" in PM_SLEEP register. (a) "REG_HK51_UART0_EN" == 0. (b) "REG_UART_RX_ENABLE" == 1.
53h (101EA7h)	REG101EA7	7:0	Default: 0x32 Access: R/W
	UART_SEL3[3:0]	7:4	Select controller for PAD_UART1_RX and PAD_UART1_TX.
	UART_SEL2[3:0]	3:0	Select controller for PAD_UART0_RX and PAD_UART0_TX.
54h (101EA8h)	REG101EA8	7:0	Default: 0x54 Access: R/W
	UART_SEL5[3:0]	7:4	Select controller for PAD_PM_LCD1/LCD0.
	UART_SEL4[3:0]	3:0	Select controller for PAD_UART2_RX and PAD_UART2_TX.
55h (101EAAh)	REG101EAA	7:0	Default: 0x00 Access: R/W
	-	7:4	Reserved.
	JTAG_SEL[3:0]	3:0	JTAG selection.
55h	REG101EAB	7:0	Default: 0x00 Access: R/W

CHIPTOP Register (Bank = 101E)				
Index (Absolute)	Mnemonic	Bit	Description	
(101EABh)	UART_PAD_INVERSE[7:0]	7:0	Invert PAD UART TX/RX.	
56h (101EACH)	REG101EAC	7:0	Default: 0x00	Access: R/W
	UART_INNER_LOOPBACK[7:0]	7:0	Enable of inner loopback test for 3 sets of UART controller. [0]: N/A. [1]: FUART enable. [2]: UART0 enable. [3]: UART1 enable. [4] – [7]: N/A.	
56h (101EADh)	REG101EAD	7:0	Default: 0x00	Access: R/W
	UART_OUTER_LOOPBACK[7:0]	7:0	Enable of outer loopback test for 4 sets of UART pad. [0]: PM_UART enable. [1]: FUART enable. [2]: UART0 enable. [3]: UART1 enable. [4] – [7]: N/A.	
57h (101EAEh)	REG101EAE	7:0	Default: 0x00	Access: R/W
	FORCE_RX_DISABLE[7:0]	7:0	Disable RX signals from pads.	
57h (101EAFh)	REG101EAF	7:0	Default: 0x00	Access: R/W
	FORCE_RX_DISABLE[15:8]	7:0	See description of '101EAEh.'	
58h (101EB0h)	REG101EB0	7:0	Default: 0x00	Access: R/W
	FPGA_MIU_OPTION[7:0]	7:0	FPGA_MIU_OPTION.	
58h (101EB1h)	REG101EB1	7:0	Default: 0x00	Access: R/W
	FPGA_MIU_OPTION[15:8]	7:0	See description of '101EB0h.'	
65h (101ECAh)	REG101ECA	7:0	Default: 0x00	Access: RO
	-	7:5	Reserved.	
	CHIP_CONFIG_STAT[4:0]	4:0	CHIP_CONFIG status.	
65h (101ECBh)	REG101ECB	7:0	Default: 0x00	Access: RO
	-	7:5	Reserved.	
	POWERGOOD_AVDD	4	POWERGOOD_AVDD status.	
	-	3:2	Reserved.	
	IN_SEL_DBUS	1	IN_SEL_DBUS.	
	IN_SEL_SBUS	0	IN_SEL_SBUS.	
69h (101ED3h)	REG101ED3	7:0	Default: 0x00	Access: R/W
	-	7:4	Reserved.	
	BOOT_FROM_SDRAM	3	Boot from SDRAM.	

CHIPTOP Register (Bank = 101E)				
Index (Absolute)	Mnemonic	Bit	Description	
			1: Enable boot from SDRAM. 0: Disable boot from SDRAM.	
	-	2:0	Reserved.	
6Ah (101ED4h)	REG101ED4	7:0	Default: 0x00	Access: R/W
	BOOT_FROM_SDRAM_OFFSET[7:0]	7:0	The booting address of SDRAM.	
6Ah (101ED5h)	REG101ED5	7:0	Default: 0x00	Access: R/W
	BOOT_FROM_SDRAM_OFFSET[15:8]	7:0	See description of '101ED4h.'	
6Bh (101ED6h)	REG101ED6	7:0	Default: 0x00	Access: R/W
	BOOT_FROM_SDRAM_OFFSET[23:16]	7:0	See description of '101ED4h.'	
6Bh (101ED7h)	REG101ED7	7:0	Default: 0x00	Access: R/W
	-	7:2	Reserved.	
	BOOT_FROM_SDRAM_OFFSET[25:24]	1:0	See description of '101ED4h.'	
70h (101EE1h)	REG101EE1	7:0	Default: 0x00	Access: R/W
	CLK_CALC_EN	7	CLK_CALC_EN. 1: Enable. 0: Disable.	
	-	6:3	Reserved.	
	ROSC_OUT_SEL[2:0]	2:0	Ring OSC output select. 000: Select delay chain 0. 001: Select delay chain 1. 010: Select delay chain 2. 011: Select delay chain 3. 100: Select delay chain 4. 101: Select delay chain 5. 110: Select delay chain 6. 111: Select delay chain 7.	
71h (101EE2h)	REG101EE2	7:0	Default: 0x00	Access: RO
	CALC_CNT_REPORT[7:0]	7:0	CALC_CNT_REPORT.	
71h (101EE3h)	REG101EE3	7:0	Default: 0x00	Access: RO
	CALC_CNT_REPORT[15:8]	7:0	See description of '101EE2h.'	
73h (101EE6h)	REG101EE6	7:0	Default: 0xFF	Access: R/W
	RESERVED[7:0]	7:0	Reserved register.	

CHIPTOP Register (Bank = 101E)

Index (Absolute)	Mnemonic	Bit	Description
73h (101EE7h)	REG101EE7	7:0	Default: 0xFF Access: R/W
	RESERVED[15:8]	7:0	See description of '101EE6h.'
74h (101EE8h)	REG101EE8	7:0	Default: 0x00 Access: R/W
	RESERVED[23:16]	7:0	See description of '101EE6h.'
74h (101EE9h)	REG101EE9	7:0	Default: 0x00 Access: R/W
	RESERVED[31:24]	7:0	See description of '101EE6h.'
75h (101EEAh)	REG101EEA	7:0	Default: 0x00 Access: R/W
	TEST_RB	7	Setting for the data arrangement on test bus.
	TEST_GB	6	Setting for the data arrangement on test bus.
	TEST_RG	5	Setting for the data arrangement on test bus.
	-	4	Reserved.
	SWAPTEST12BIT	3	Swap MSB 12 bits with LSB 12 bits on test bus.
	CLK_OUT_SEL[2:0]	2:0	Select TEST_CLK_OUT source. 3'd0: TEST_CLK_OUT= TEST_BUS_GB [0]. 3'd1: TEST_CLK_OUT= TEST_BUS_GB [1]. 3'd2: TEST_CLK_OUT= TEST_BUS_GB [2]. 3'd3: TEST_CLK_OUT= TEST_BUS_GB [3]. 3'd4: TEST_CLK_OUT= TEST_BUS_GB [4]. 3'd5: TEST_CLK_OUT= TEST_BUS_GB [5]. 3'd6: TEST_CLK_OUT= TEST_BUS_GB [6]. 3'd7: TEST_CLK_OUT= TEST_BUS_GB [7].
75h (101EEBh)	REG101EEB	7:0	Default: 0x00 Access: R/W
	ROSC_IN_SEL	7	Select the input source of ring oscillator in CHIP_CONF. 1: Close-loop (enable ring oscillator). 0: Open-loop (input from external digital input).
	TESTBUS_EN	6	Enable test bus output (disabled by default).
	TESTCLK_MODE	5	TESTCLK_MODE used in TEST_CTRL.
	-	4:2	Reserved.
	SEL_CLK_TEST_OUT[1:0]	1:0	Select CLK_TEST_OUT. 2'd0: Select CLK_TEST_OUT [47:0]. 2'd1: Select CLK_TEST_OUT [95:48]. 2'd2: Select CLK_TEST_OUT [143:96]. 2'd3: Reserved.
76h (101EECh)	REG101EEC	7:0	Default: 0x00 Access: R/W
	-	7:3	Reserved.
	SINGLE_CLK_OUT_SEL[2:0]	2:0	Select single CLK_OUT.

CHIPTOP Register (Bank = 101E)

Index (Absolute)	Mnemonic	Bit	Description
			3'd1: TEST_BUS [11] = TEST_CLK_OUT_d2. TEST_BUS [10] = TEST_CLK_OUT. 3'd2: TEST_BUS [11] = TEST_CLK_OUT_d2. TEST_BUS [10] = TEST_CLK_OUT_d4. 3'd3: TEST_BUS [11] = TEST_CLK_OUT_d2. TEST_BUS [10] = TEST_CLK_OUT_d8. 3'd4: TEST_BUS [11] = TEST_CLK_OUT_d2. TEST_BUS [10] = TEST_CLK_OUT_d16. 3'd5: TEST_BUS [11] = TEST_CLK_OUT_d2. TEST_BUS [10] = TEST_CLK_OUT_d32. 3'd6: TEST_BUS [11] = TEST_CLK_OUT_d2. TEST_BUS [10] = TEST_CLK_OUT_d64. Others: No TEST_CLK_OUT.
77h (101EEEh)	REG101EEE	7:0	Default: 0x00 Access: R/W
	-	7:6	Reserved.
	TEST_BUS24B_SEL[5:0]	5:0	Select TEST_BUS [23:0] source. 6'd2: TEST_BUS = ANA_MISC_TEST_OUT. 6'd3: TEST_BUS = MIU_TEST_OUT. 6'd4: TEST_BUS = DIAMOND_TOP_WP_TEST_OUT. 6'd6: TEST_BUS = UTMI_P1_TEST_OUT. 6'd7: TEST_BUS = UTMI_P2_TEST_OUT. 6'd8: TEST_BUS = AUSDM_TEST_OUT. 6'd9: TEST_BUS = DIG_PM_TEST_OUT. 6'd10: TEST_BUS = MCU_IF_TEST_OUT. 6'd11: TEST_BUS = UTMI_P3_TEST_OUT. 6'd16: TEST_BUS = SC_GP_TEST_OUT. 6'd17: TEST_BUS = DEC_GP_TEST_OUT. 6'd22: TEST_BUS = SATA_PHY_TEST_OUT. 6'd25: TEST_BUS = CLKGEN_TEST_OUT. 6'd26: TEST_BUS = CLKGEN_TEST_OUT2. Others: No TEST_OUT.
7Bh (101EF6h)	REG101EF6	7:0	Default: 0x00 Access: R/W
	CHIPTOP_RESERVED[7:0]	7:0	
7Bh (101EF7h)	REG101EF7	7:0	Default: 0x00 Access: R/W
	CHIPTOP_RESERVED[15:8]	7:0	See description of '101EF6h.'
7Ch (101EF8h)	REG101EF8	7:0	Default: 0xFF Access: R/W
	CHK_CLK_HEMCU_FREQ_CM P_DATA[7:0]	7:0	

CHIPTOP Register (Bank = 101E)

Index (Absolute)	Mnemonic	Bit	Description
7Ch (101EF9h)	REG101EF9	7:0	Default: 0xFF Access: R/W
	CHK_CLK_HEMCU_FREQ_CM P_DATA[15:8]	7:0	See description of '101EF8h.'
7Dh (101EFAh)	REG101EFA	7:0	Default: 0x03 Access: R/W
	-	7:4	Reserved.
	256BUS_2X_DIV_EN[3:0]	3:0	256BUS MIU 2X DIV enable.
7Eh (101EFCh)	REG101EFC	7:0	Default: 0x0F Access: R/W
	-	7:4	Reserved.
	MIU2X_DIV_RSTZ[3:0]	3:0	CLK_MIU2X_DIV SW RSTZ. [0]: MIU0. [3:1]: Reserved.

PADTOP Register (Bank = 103C)

PADTOP Register (Bank = 103C)				
Index (Absolute)	Mnemonic	Bit	Description	
30h (103C60h)	REG103C60	7:0	Default: 0x00	Access: R/W
	KEY_EN[7:0]	7:0	Key pads en.	
30h (103C61h)	REG103C61	7:0	Default: 0x00	Access: R/W
	-	7:6	Reserved.	
	KEY_EN[13:8]	5:0	See description of '103C60h.'	
50h (103CA0h)	REG103CA0	7:0	Default: 0x00	Access: RO
	TEST_OUT[7:0]	7:0	Test bus.	
50h (103CA1h)	REG103CA1	7:0	Default: 0x00	Access: RO
	TEST_OUT[15:8]	7:0	See description of '103CA0h.'	
51h (103CA2h)	REG103CA2	7:0	Default: 0x00	Access: RO
	TEST_OUT[23:16]	7:0	See description of '103CA0h.'	
60h (103CC0h)	REG103CC0	7:0	Default: 0x00	Access: R/W
	-	7:4	Reserved.	
	DMIC_MODE[3:0]	3:0	DMIC mode.	
60h (103CC1h)	REG103CC1	7:0	Default: 0x00	Access: R/W
	-	7:3	Reserved.	
	EJ_MODE[2:0]	2:0	EJ mode.	
62h (103CC4h)	REG103CC4	7:0	Default: 0x00	Access: R/W
	-	7	Reserved.	
	I2S_RX_MODE[2:0]	6:4	I2S RX mode.	
	-	3	Reserved.	
	I2S_MCK_MODE[2:0]	2:0	I2S MCLK mode.	
62h (103CC5h)	REG103CC5	7:0	Default: 0x00	Access: R/W
	-	7	Reserved.	
	I2S_RXTX_MODE[2:0]	6:4	I2S RX TX mode.	
	-	3	Reserved.	
	I2S_TX_MODE[2:0]	2:0	I2S TX mode.	
64h (103CC8h)	REG103CC8	7:0	Default: 0x00	Access: R/W
	-	7:2	Reserved.	
	TX_MIPI_MODE[1:0]	1:0	MIPI TX mode.	
65h (103CCAh)	REG103CCA	7:0	Default: 0x00	Access: R/W
	PWM1_MODE[3:0]	7:4	PWM1 mode.	

PADTOP Register (Bank = 103C)				
Index (Absolute)	Mnemonic	Bit	Description	
	PWM0_MODE[3:0]	3:0	PWM0 mode.	
65h (103CCBh)	REG103CCB	7:0	Default: 0x00	Access: R/W
	PWM3_MODE[3:0]	7:4	PWM3 mode.	
	PWM2_MODE[3:0]	3:0	PWM2 mode.	
67h (103CCEh)	REG103CCE	7:0	Default: 0x00	Access: R/W
	-	7:2	Reserved.	
	SD_CDZ_MODE[1:0]	1:0	SD CDZ mode.	
67h (103CCFh)	REG103CCF	7:0	Default: 0x00	Access: R/W
	-	7:2	Reserved.	
	SDIO_MODE[1:0]	1:0	SDIO mode.	
68h (103CD0h)	REG103CD0	7:0	Default: 0x00	Access: R/W
	-	7:3	Reserved.	
	SPI0_MODE[2:0]	2:0	SPI0 mode.	
68h (103CD1h)	REG103CD1	7:0	Default: 0x00	Access: R/W
	-	7	Reserved.	
	SPI1_CS2_MODE[2:0]	6:4	SPI1 CS2 mode.	
	SPI1_MODE[3:0]	3:0	SPI1 mode.	
69h (103CD2h)	REG103CD2	7:0	Default: 0x00	Access: R/W
	-	7:6	Reserved.	
	PSPI0_G_MODE[1:0]	5:4	PSPI0 general mode.	
	-	3	Reserved.	
	PSPI0_SR_MODE[2:0]	2:0	PSPI0 sensor mode.	
6Ah (103CD4h)	REG103CD4	7:0	Default: 0x00	Access: R/W
	PSPI1_CS2_MODE[3:0]	7:4	PSPI1 CS2 mode.	
	PSPI1_PL_MODE[3:0]	3:0	PSPI1 panel mode.	
6Ah (103CD5h)	REG103CD5	7:0	Default: 0x00	Access: R/W
	-	7:6	Reserved.	
	PSPI1_G_MODE[1:0]	5:4	PSPI1 general mode.	
	PSPI1_TE_MODE[3:0]	3:0	PSPI1 TE Mode.	
6Bh (103CD6h)	REG103CD6	7:0	Default: 0x00	Access: R/W
	-	7:2	Reserved.	
	RGB8_MODE[1:0]	1:0	TTL 8-bit mode (Serial RGB).	
6Bh	REG103CD7	7:0	Default: 0x00	Access: R/W

PADTOP Register (Bank = 103C)				
Index (Absolute)	Mnemonic	Bit	Description	
(103CD7h)	-	7:5	Reserved.	
	BT656_LOOPBACK_MODE	4	BT656 loopback mode.	
	-	3:0	Reserved.	
6Ch (103CD8h)	REG103CD8	7:0	Default: 0x00	Access: R/W
	-	7:6	Reserved.	
	TTL18_MODE[1:0]	5:4	TTL 18-bit mode.	
	-	3	Reserved.	
	TTL16_MODE[2:0]	2:0	TTL 16-bit mode.	
6Ch (103CD9h)	REG103CD9	7:0	Default: 0x00	Access: R/W
	-	7:6	Reserved.	
	BT656_OUT_MODE[1:0]	5:4	BT656 8-bit mode.	
	-	3	Reserved.	
	TTL24_MODE[2:0]	2:0	TTL 24-bit mode.	
6Dh (103CDAh)	REG103CDA	7:0	Default: 0x00	Access: R/W
	UART1_MODE[3:0]	7:4	UART1 mode.	
	-	3	Reserved.	
	UART0_MODE[2:0]	2:0	UART0 mode.	
6Dh (103CDBh)	REG103CDB	7:0	Default: 0x00	Access: R/W
	-	7:3	Reserved.	
	UART2_MODE[2:0]	2:0	UART2 mode.	
6Eh (103CDCh)	REG103CDC	7:0	Default: 0x00	Access: R/W
	-	7:4	Reserved.	
	ETH0_MODE[3:0]	3:0	ETH0 mode.	
6Eh (103CDDh)	REG103CDD	7:0	Default: 0x00	Access: R/W
	-	7:4	Reserved.	
	FUART_MODE[3:0]	3:0	FUART mode.	
6Fh (103CDEh)	REG103CDE	7:0	Default: 0x00	Access: R/W
	I2C1_MODE[3:0]	7:4	I2C1 mode.	
	I2C0_MODE[3:0]	3:0	I2C0 mode.	
70h (103CE0h)	REG103CE0	7:0	Default: 0x00	Access: R/W
	-	7:2	Reserved.	
	PM_SPICZ2_MODE[1:0]	1:0	PM SPI CZ2 mode.	
72h	REG103CE4	7:0	Default: 0x00	Access: R/W

PADTOP Register (Bank = 103C)				
Index (Absolute)	Mnemonic	Bit	Description	
(103CE4h)	-	7:2	Reserved.	
	BT1120_MODE[1:0]	1:0	Keypad mode ECO.	
73h (103CE6h)	REG103CE6	7:0	Default: 0x00	Access: R/W
	-	7:2	Reserved.	
	ISP_IR_MODE[1:0]	1:0	ISP IR mode.	
74h (103CE8h)	REG103CE8	7:0	Default: 0x00	Access: R/W
	-	7:6	Reserved.	
	SR1_MIPI_MODE[1:0]	5:4	SR1 MIPI mode.	
	-	3:1	Reserved.	
	SR0_MIPI_MODE	0	SR0 MIPI mode.	
74h (103CE9h)	REG103CE9	7:0	Default: 0x00	Access: R/W
	-	7:6	Reserved.	
	SR1_CTRL_MODE[1:0]	5:4	SR1 CTRL mode.	
	-	3:2	Reserved.	
	SR0_CTRL_MODE[1:0]	1:0	SR0 CTRL mode.	
75h (103CEAh)	REG103CEA	7:0	Default: 0x00	Access: R/W
	-	7	Reserved.	
	SR0_BT656_MODE[2:0]	6:4	SR0 BT656 mode.	
	-	3:2	Reserved.	
	SR0_BT601_MODE[1:0]	1:0	SR0 DVP mode.	
76h (103CECh)	REG103CEC	7:0	Default: 0x00	Access: R/W
	-	7:6	Reserved.	
	SR0_RST_MODE[1:0]	5:4	SR0 Reset mode.	
	-	3:2	Reserved.	
	SR0_MCLK_MODE[1:0]	1:0	SR0 MCLK mode.	
76h (103CEDh)	REG103CED	7:0	Default: 0x00	Access: R/W
	-	7:1	Reserved.	
	SR0_PDN_MODE	0	SR0 PDN mode.	
77h (103CEEh)	REG103CEE	7:0	Default: 0x00	Access: R/W
	-	7:6	Reserved.	
	SR1_RST_MODE[1:0]	5:4	SR1 Reset mode.	
	-	3:2	Reserved.	
	SR1_MCLK_MODE[1:0]	1:0	SR1 MCLK mode.	

PADTOP Register (Bank = 103C)				
Index (Absolute)	Mnemonic	Bit	Description	
77h (103CEFh)	REG103CEF	7:0	Default: 0x00	Access: R/W
	-	7:2	Reserved.	
	SR1_PDN_MODE[1:0]	1:0	SR1 PDN mode.	
78h (103CF0h)	REG103CF0	7:0	Default: 0x00	Access: R/W
	KEY_READ1_MODE[3:0]	7:4	Keypad read1 mode.	
	KEY_READ0_MODE[3:0]	3:0	Keypad read0 mode.	
78h (103CF1h)	REG103CF1	7:0	Default: 0x00	Access: R/W
	KEY_READ3_MODE[3:0]	7:4	Keypad read3 mode.	
	KEY_READ2_MODE[3:0]	3:0	Keypad read2 mode.	
79h (103CF2h)	REG103CF2	7:0	Default: 0x00	Access: R/W
	KEY_READ5_MODE[3:0]	7:4	Keypad read5 mode.	
	KEY_READ4_MODE[3:0]	3:0	Keypad read4 mode.	
79h (103CF3h)	REG103CF3	7:0	Default: 0x00	Access: R/W
	-	7:5	Reserved.	
	KEY_FIX_MODE	4	Keypad fix mode.	
	KEY_READ6_MODE[3:0]	3:0	Keypad read6 mode.	
7Ah (103CF4h)	REG103CF4	7:0	Default: 0x00	Access: R/W
	KEY_SCAN1_MODE[3:0]	7:4	Keypad scan1 mode.	
	KEY_SCAN0_MODE[3:0]	3:0	Keypad scan0 mode.	
7Ah (103CF5h)	REG103CF5	7:0	Default: 0x00	Access: R/W
	KEY_SCAN3_MODE[3:0]	7:4	Keypad scan3 mode.	
	KEY_SCAN2_MODE[3:0]	3:0	Keypad scan2 mode.	
7Bh (103CF6h)	REG103CF6	7:0	Default: 0x00	Access: R/W
	KEY_SCAN5_MODE[3:0]	7:4	Keypad scan5 mode.	
	KEY_SCAN4_MODE[3:0]	3:0	Keypad scan4 mode.	
7Bh (103CF7h)	REG103CF7	7:0	Default: 0x00	Access: R/W
	-	7:4	Reserved.	
	KEY_SCAN6_MODE[3:0]	3:0	Keypad scan6 mode.	

UART2 Register (Bank = 110A)

UART2 Register (Bank = 110A)				
Index (Absolute)	Mnemonic	Bit	Description	
00h (110A00h)	REG110A00	7:0	Default: 0x00	Access: R/W
	THR_RBR_DLL[7:0]	7:0	1. When "REG_LCR_DL_ACCESS" = 0: Write: Transmitter Holding Register. Write transmit FIFO; note that writing data to a full FIFO results in the write data being lost. Read: Receiver Buffer. Read receive FIFO; note that any incoming data are lost when FIFO is full and an overrun error occurs. 2. When "REG_LCR_DL_ACCESS" = 1: Divisor Latch LSB.	
02h (110A04h)	REG110A04	7:0	Default: 0x00	Access: R/W
	IER_DLH[7:0]	7:0	1. When "REG_LCR_DL_ACCESS" = 0: Interrupt Enable Registers (IER); 1: Enabled. Bit [0]: Received Data Available Interrupt and Character Timeout Interrupt. Bit [1]: Transmitter Holding Register Empty Interrupt. Bit [2]: Receiver Line Status Interrupt. Bit [3]: Modem Status Interrupt. Bit [7]: Programmable THRE Interrupt. 2. When "REG_LCR_DL_ACCESS" = 1: Divisor Latch MSB. Baud rate = (serial clock freq.) / (16 * divisor).	
04h (110A08h)	REG110A08	7:0	Default: 0x00	Access: R/W
	FCR_IIR[7:0]	7:0	1. Write: FIFO Control Register (FCR). Bit [0]: FIFO enable. Bit [1]: Write "1" to clear RX FIFO. Bit [2]: Write "1" to clear TX FIFO. Bit [5:4]: Transmit FIFO Empty trigger level. "00": FIFO empty; "01": 2 characters in the FIFO; "10": FIFO 1/4 full; "11": FIFO 1/2 full; Bit [7:6]: Receiver FIFO Interrupt trigger level. "00": 1 character in the FIFO; "01": FIFO 1/4 full; "10": FIFO 1/2 full; "11": FIFO 2 less than full.	

UART2 Register (Bank = 110A)				
Index (Absolute)	Mnemonic	Bit	Description	
			2. Read: Interrupt Identification Registers (IIR). Bit [0]: 1: No interrupt is pending. Bit [3:1]: Interrupt identified. "110": Character timeout. "011": Receiver Line Status. "010": Receiver Data Available. "001": Transmitter Holding Register empty. "000": Modem Status.	
06h (110A0Ch)	REG110A0C	7:0	Default: 0x03	Access: R/W
	LCR_DL_ACCESS	7	Divisor Latch Access; 1: The divisor latches can be accessed.	
	LCR_BREAK_CTRL	6	Break control bit.	
	-	5	Reserved.	
	LCR_EVEN_PARITY_SEL	4	1: Select even parity.	
	LCR_PARITY_EN	3	1: Generate parity bit on serial out.	
	LCR_STOP_BITS	2	Specify the number of stop bits. "0": 1 stop bit; "1": 1.5 stop bits when 5-bit character length is selected and 2 bits otherwise.	
	LCR_CHAR_BITS[1:0]	1:0	Select number of bits in each character. "00": 5 bits; "01": 6 bits; "10": 7 bits; "11": 8 bits.	
08h (110A10h)	REG110A10	7:0	Default: 0x00	Access: R/W
	-	7:6	Reserved.	
	MCR_AFCE	5	Auto Flow Control Enable; 1: Enable.	
	MCR_LOOPBACK	4	1: Loopback mode. SOUT -> SIN (pad "STX_PAD_O" will be set to "1"). DTR -> DSR. RTS -> CTS. OUT1 -> RI. OUT2 -> DCD.	
	MCR_OUT2	3	In loopback mode, connect to Data Carrier Detect (DCD) signal input.	
	MCR_OUT1	2	In loopback mode, connect to Ring Indicator (RI) signal input.	
	MCR_RTS	1	Request To Send (RTS) signal control. "0": RTS is "1"; "1": RTS is "0."	

UART2 Register (Bank = 110A)				
Index (Absolute)	Mnemonic	Bit	Description	
	MCR_DTR	0	Data Terminal Ready (DTR) signal control. "0": DTR is "1"; "1": DTR is "0."	
0Ah (110A14h)	REG110A14	7:0	Default: 0x00	Access: RO
	LSR_ERROR	7	Receiver FIFO Error bit.	
	LSR_TX_EMPTY	6	1: Transmitter (TX FIFO and shift registers) Empty indicator. Cleared after writing data into TX FIFO.	
	LSR_TXFIFO_EMPTY	5	1: Transmit FIFO is empty. Cleared after writing data into TX FIFO; Gen a Transmitter Holding Register Empty interrupt.	
	LSR_BI	4	Break Interrupt bit.	
	LSR_FE	3	1: Framing Error indicator. Cleared when reading; Gen a Receiver Line Status interrupt.	
	LSR_PE	2	1: Parity Error indicator. Cleared when reading; Gen a Receiver Line Status interrupt.	
	LSR_OE	1	1: RX Overrun Error indicator. Cleared when reading; Gen a Receiver Line Status interrupt.	
	LSR_DR	0	1: Received Data Ready indicator.	
0Ch (110A18h)	REG110A18	7:0	Default: 0x00	Access: RO
	MSR_DCD_COMP	7	Complement of "DCD" or equal to "OUT2" in loopback.	
	MSR_RI_COMP	6	Complement of "RI" or equal to "OUT1" in loopback.	
	MSR_DSR_COMP	5	Complement of "DSR" or equal to "DTR" in loopback.	
	MSR_CTS_COMP	4	Complement of "CTS" or equal to "RTS" in loopback.	
	MSR_DDCD	3	Delta Data Carrier Detect (DDCD) indicator. "1": The "DCD" line has changed its state. Cleared when reading.	
	MSR_TERI	2	Trailing Edge of Ring Indicator (TERI) detector. The "RI" line has changed its state from low to high. Cleared when reading.	
	MSR_DDSD	1	Delta Data Set Ready (DDSD) indicator. "1": The "DSR" line has changed its state. Cleared when reading.	
	MSR_DCTS	0	Delta Clear To Send (DCTS) indicator.	

UART2 Register (Bank = 110A)				
Index (Absolute)	Mnemonic	Bit	Description	
			"1": The "CTS" line has changed its state. Clear when reading.	
0Eh (110A1Ch)	REG110A1C	7:0	Default: 0x00	Access: RO
	-	7:5	Reserved.	
	USR_RFF	4	RX FIFO Full.	
	USR_RFNE	3	RX FIFO Not Empty.	
	USR_TFE	2	TX FIFO Empty.	
	USR_TFNF	1	TX FIFO Not Full.	
	USR_BUSY	0	UART busy.	
10h (110A20h)	REG110A20	7:0	Default: 0x00	Access: RO
	-	7:6	Reserved.	
	TFL[5:0]	5:0	TX FIFO level.	
12h (110A24h)	REG110A24	7:0	Default: 0x00	Access: RO
	-	7:6	Reserved.	
	RFL[5:0]	5:0	RX FIFO level.	
14h (110A28h)	REG110A28	7:0	Default: 0x01	Access: R/W
	-	7:1	Reserved.	
	SW_RST	0	SW_RST.	

FUART Register (Bank = 1102)

FUART Register (Bank = 1102)				
Index (Absolute)	Mnemonic	Bit	Description	
00h (110200h)	REG110200	7:0	Default: 0x00	Access: R/W
	THR_RBR_DLL[7:0]	7:0	1. When "REG_LCR_DL_ACCESS" = 0: Write: Transmitter Holding Register. Write transmit FIFO; note that writing data to a full FIFO results in the write data being lost. Read: Receiver Buffer. Read receive FIFO; note that any incoming data are lost when FIFO is full and an overrun error occurs. 2. When "REG_LCR_DL_ACCESS" = 1: Divisor Latch LSB.	
02h (110204h)	REG110204	7:0	Default: 0x00	Access: R/W
	IER_DLH[7:0]	7:0	1. When "REG_LCR_DL_ACCESS" = 0: Interrupt Enable Registers (IER); 1: Enabled. Bit [0]: Received Data Available Interrupt and Character Timeout Interrupt. Bit [1]: Transmitter Holding Register Empty Interrupt. Bit [2]: Receiver Line Status Interrupt. Bit [3]: Modem Status Interrupt. Bit [7]: Programmable THRE Interrupt. 2. When "REG_LCR_DL_ACCESS" = 1: Divisor Latch MSB. Baud rate = (serial clock freq.) / (16 * divisor).	
04h (110208h)	REG110208	7:0	Default: 0x00	Access: R/W
	FCR_IIR[7:0]	7:0	1. Write: FIFO Control Register (FCR). Bit [0]: FIFO enable. Bit [1]: Write "1" to clear RX FIFO. Bit [2]: Write "1" to clear TX FIFO. Bit [5:4]: Transmit FIFO Empty trigger level. "00": FIFO empty; "01": 2 characters in the FIFO; "10": FIFO 1/4 full; "11": FIFO 1/2 full. Bit [7:6]: Receiver FIFO Interrupt trigger level. "00": 1 character in the FIFO; "01": FIFO 1/4 full; "10": FIFO 1/2 full; "11": FIFO 2 less than full.	

FUART Register (Bank = 1102)

Index (Absolute)	Mnemonic	Bit	Description
			2. Read: Interrupt Identification Registers (IIR). Bit [0]: 1: No interrupt is pending. Bit [3:1]: Interrupt identified. "110": Character timeout. "011": Receiver Line Status. "010": Receiver Data Available. "001": Transmitter Holding Register empty. "000": Modem Status.
06h (11020Ch)	REG11020C	7:0	Default: 0x03 Access: R/W
	LCR_DL_ACCESS	7	Divisor Latch Access; 1: The divisor latches can be accessed.
	LCR_BREAK_CTRL	6	Break control bit.
	-	5	Reserved.
	LCR_EVEN_PARITY_SEL	4	1: Select even parity.
	LCR_PARITY_EN	3	1: Generate parity bit on serial out.
	LCR_STOP_BITS	2	Specify the number of stop bits. "0": 1 stop bit; "1": 1.5 stop bits when 5-bit character length is selected and 2 bits otherwise.
	LCR_CHAR_BITS[1:0]	1:0	Select number of bits in each character. "00": 5 bits; "01": 6 bits; "10": 7 bits; "11": 8 bits.
08h (110210h)	REG110210	7:0	Default: 0x00 Access: R/W
	-	7:6	Reserved.
	MCR_AFCE	5	Auto Flow Control Enable; 1: Enable.
	MCR_LOOPBACK	4	1: Loopback mode. SOUT -> SIN (pad "STX_PAD_O" will be set to "1"). DTR -> DSR. RTS -> CTS. Out1 -> RI. Out2 -> DCD.
	MCR_OUT2	3	In loopback mode, connect to Data Carrier Detect (DCD) signal input.
	MCR_OUT1	2	In loopback mode, connect to Ring Indicator (RI) signal input.
	MCR_RTS	1	Request To Send (RTS) signal control. "0": RTS is "1"; "1": RTS is "0."
	MCR_DTR	0	Data Terminal Ready (DTR) signal control.

FUART Register (Bank = 1102)

Index (Absolute)	Mnemonic	Bit	Description
			"0": DTR is "1"; "1": DTR is "0."
0Ah (110214h)	REG110214	7:0	Default: 0x00 Access: RO
	LSR_ERROR	7	Receiver FIFO Error bit.
	LSR_TX_EMPTY	6	1: Transmitter (TX FIFO and shift registers) Empty indicator. Cleared after writing data into TX FIFO.
	LSR_TXFIFO_EMPTY	5	1: Transmit FIFO is empty. Cleared after writing data into TX FIFO. Gen a Transmitter Holding Register Empty interrupt.
	LSR_BI	4	Break Interrupt bit.
	LSR_FE	3	1: Framing Error indicator. Cleared when reading. Gen a Receiver Line Status interrupt.
	LSR_PE	2	1: Parity Error indicator. Cleared when reading. Gen a Receiver Line Status interrupt.
	LSR_OE	1	1: RX Overrun Error indicator. Cleared when reading. Gen a Receiver Line Status interrupt.
	LSR_DR	0	1: Received Data Ready indicator.
0Ch (110218h)	REG110218	7:0	Default: 0x00 Access: RO
	MSR_DCD_COMP	7	Complement of "DCD" or equal to "OUT2" in loopback.
	MSR_RI_COMP	6	Complement of "RI" or equal to "OUT1" in loopback.
	MSR_DSR_COMP	5	Complement of "DSR" or equal to "DTR" in loopback.
	MSR_CTS_COMP	4	Complement of "CTS" or equal to "RTS" in loopback.
	MSR_DDCD	3	Delta Data Carrier Detect (DDCD) indicator. "1": The "DCD" line has changed its state. Cleared when reading.
	MSR_TERI	2	Trailing Edge of Ring Indicator (TERI) detector. The "RI" line has changed its state from low to high. Cleared when reading.
	MSR_DDSD	1	Delta Data Set Ready (DDSD) indicator. "1": The "DSR" line has changed its state. Cleared when reading.
	MSR_DCTS	0	Delta Clear To Send (DCTS) indicator. "1": The "CTS" line has changed its state. Cleared when reading.

FUART Register (Bank = 1102)

Index (Absolute)	Mnemonic	Bit	Description
0Eh (11021Ch)	REG11021C	7:0	Default: 0x00 Access: RO
	-	7:5	Reserved.
	USR_RFF	4	RX FIFO Full.
	USR_RFNE	3	RX FIFO Not Empty.
	USR_TFE	2	TX FIFO Empty.
	USR_TFNF	1	TX FIFO Not Full.
	USR_BUSY	0	UART busy.
10h (110220h)	REG110220	7:0	Default: 0x00 Access: RO
	-	7:6	Reserved.
	TFL[5:0]	5:0	TX FIFO level.
12h (110224h)	REG110224	7:0	Default: 0x00 Access: RO
	-	7:6	Reserved.
	RFL[5:0]	5:0	RX FIFO level.

URDMA Register (Bank = 1103)

URDMA Register (Bank = 1103)				
Index (Absolute)	Mnemonic	Bit	Description	
60h (1103C0h)	REG1103C0	7:0	Default: 0x00	Access: R/W
	RX_SW_RST	7	RX Engine software reset. 0: De-assert. 1: Assert.	
	TX_SW_RST	6	TX Engine software reset. 0: De-assert. 1: Assert.	
	RX_ENDINA	5	Software select MIU read data, endian definition. 0: MIU_RDATA [63:0] = {B7, B6, B5, B4, B3, B2, B1, B0}. 1: MIU_RDATA [63:0] = {B4, B5, B6, B7, B0, B1, B2, B3}.	
	TX_ENDINA	4	Software select MIU read data, endian definition. 0: MIU_RDATA [63:0] = {B7, B6, B5, B4, B3, B2, B1, B0}. 1: MIU_RDATA [63:0] = {B4, B5, B6, B7, B0, B1, B2, B3}.	
	RX_URDMA_EN	3	URDMA RX Hardware Enable. 0: Disable. 1: Enable.	
	TX_URDMA_EN	2	URDMA TX Hardware Enable. 0: Disable. 1: Enable.	
	URDMA_MODE	1	URDMA Mode Select. When URDMA mode is enabled, URDMA hardware will replace MCU to access UART; otherwise, MCU will still control UART directly. 0: Disable. 1: Enable.	
	SW_RST	0	URDMA software reset. 0: De-assert. 1: Assert.	
60h (1103C1h)	REG1103C1	7:0	Default: 0x00	Access: RO, R/W
	-	7:6	Reserved.	
	RX_BUSY	5	RX controller in BUSY state.	
	TX_BUSY	4	TX controller in BUSY state.	
	RX_OP_MODE	3	RX controller operation mode. 0: Default. 1: Stop RX DMA activity when threshold interrupt occurs, until RX_BUF_BASE is updated.	

URDMA Register (Bank = 1103)				
Index (Absolute)	Mnemonic	Bit	Description	
	-	2:1	Reserved.	
	URDMA2MIU_NS	0	For MIU secure bit.	
61h (1103C2h)	REG1103C2	7:0	Default: 0x40	Access: R/W
	INTR_THRESHOLD[7:0]	7:0	Interrupt threshold, to configure MCU interrupt period. Used to set how much data is to be received from UART, and to alert MCU; byte unit.	
61h (1103C3h)	REG1103C3	7:0	Default: 0x00	Access: R/W
	-	7:4	Reserved.	
	INTR_THRESHOLD[11:8]	3:0	See description of '1103C2h.'	
62h (1103C4h)	REG1103C4	7:0	Default: 0x00	Access: R/W
	TX_BUF_BASE_H[7:0]	7:0	Configure TX buffer base address [26:16] in DRAM, 8-byte unit.	
62h (1103C5h)	REG1103C5	7:0	Default: 0x00	Access: R/W
	TX_BUF_BASE_H[15:8]	7:0	See description of '1103C4h.'	
63h (1103C6h)	REG1103C6	7:0	Default: 0x00	Access: R/W
	TX_BUF_BASE_L[7:0]	7:0	Configure TX buffer base address [15:0] in DRAM, 8-byte unit.	
63h (1103C7h)	REG1103C7	7:0	Default: 0x00	Access: R/W
	TX_BUF_BASE_L[15:8]	7:0	See description of '1103C6h.'	
64h (1103C8h)	REG1103C8	7:0	Default: 0x00	Access: R/W
	TX_BUF_SIZE[7:0]	7:0	Configure TX buffer size in DRAM, 8-byte unit. For example, if BUF_SIZE = 0x400, access region will be 0x000 – 0x3FF.	
64h (1103C9h)	REG1103C9	7:0	Default: 0x00	Access: R/W
	-	7:5	Reserved.	
	TX_BUF_SIZE[12:8]	4:0	See description of '1103C8h.'	
65h (1103CAh)	REG1103CA	7:0	Default: 0x00	Access: RO
	TX_BUF_RPTR[7:0]	7:0	Read TX buffer read pointer from URDMA, byte unit.	
65h (1103CBh)	REG1103CB	7:0	Default: 0x00	Access: RO
	TX_BUF_RPTR[15:8]	7:0	See description of '1103CAh.'	
66h (1103CCh)	REG1103CC	7:0	Default: 0x00	Access: R/W
	TX_BUF_WPTR[7:0]	7:0	Update TX buffer write pointer to URDMA, byte unit.	
66h (1103CDh)	REG1103CD	7:0	Default: 0x00	Access: R/W
	TX_BUF_WPTR[15:8]	7:0	See description of '1103CCh.'	

URDMA Register (Bank = 1103)				
Index (Absolute)	Mnemonic	Bit	Description	
67h (1103CEh)	REG1103CE	7:0	Default: 0x0A	Access: R/W
	-	7:4	Reserved.	
	TX_TIMEOUT[3:0]	3:0	Configure TX timeout value, to clean TX buffer data and send them to UART. 2^(TX_TIMEOUT) cycle unit.	
68h (1103D0h)	REG1103D0	7:0	Default: 0x00	Access: R/W
	RX_BUF_BASE_H[7:0]	7:0	Configure RX buffer base address [26:16] in DRAM, 8-byte unit.	
68h (1103D1h)	REG1103D1	7:0	Default: 0x00	Access: R/W
	RX_BUF_BASE_H[15:8]	7:0	See description of '1103D0h.'	
69h (1103D2h)	REG1103D2	7:0	Default: 0x00	Access: R/W
	RX_BUF_BASE_L[7:0]	7:0	Configure RX buffer base address [15:0] in DRAM, 8-byte unit.	
69h (1103D3h)	REG1103D3	7:0	Default: 0x00	Access: R/W
	RX_BUF_BASE_L[15:8]	7:0	See description of '1103D2h.'	
6Ah (1103D4h)	REG1103D4	7:0	Default: 0x00	Access: R/W
	RX_BUF_SIZE[7:0]	7:0	Configure RX buffer size in DRAM, 8-byte unit. For example, if BUF_SIZE = 0x400, access region will be 0x000 – 0x3FF.	
6Ah (1103D5h)	REG1103D5	7:0	Default: 0x00	Access: R/W
	-	7:5	Reserved.	
	RX_BUF_SIZE[12:8]	4:0	See description of '1103D4h.'	
6Bh (1103D6h)	REG1103D6	7:0	Default: 0x00	Access: RO
	RX_BUF_WPTR[7:0]	7:0	Read RX buffer write pointer from URDMA, byte unit.	
6Bh (1103D7h)	REG1103D7	7:0	Default: 0x00	Access: RO
	RX_BUF_WPTR[15:8]	7:0	See description of '1103D6h.'	
6Ch (1103D8h)	REG1103D8	7:0	Default: 0x0A	Access: R/W
	-	7:4	Reserved.	
	RX_TIMEOUT[3:0]	3:0	Configure RX timeout value, to clean UART RX FIFO and write to DRAM. 2^(RX_TIMEOUT) cycle unit.	
6Dh (1103DAh)	REG1103DA	7:0	Default: 0x00	Access: RO, R/W
	RX_MCU_INTR	7	Identify RX controller to MCU interrupt event. This bit is logical OR of RX_INTR1~2 flags, and is cleared by RX_INTR_CLR.	

URDMA Register (Bank = 1103)				
Index (Absolute)	Mnemonic	Bit	Description	
	-	6	Reserved.	
	RX_INTR2	5	RX interrupt is caused by interrupt threshold function.	
	RX_INTR1	4	RX interrupt is caused by timeout function.	
	-	3	Reserved.	
	RX_INTR2_EN	2	RX interrupt threshold function enable.	
	RX_INTR1_EN	1	RX timeout interrupt function enable.	
	RX_INTR_CLR	0	MCU clear RX control interrupt signal. Write once and auto clear.	
6Dh (1103DBh)	REG1103DB	7:0	Default: 0x00	Access: RO, R/W
	TX_MCU_INTR	7	Identify TX controller to MCU interrupt event.	
	-	6:2	Reserved.	
	TX_INTR_EN	1	TX controller to MCU interrupt enable.	
	TX_INTR_CLR	0	MCU clear TX control interrupt signal. Write once and auto clear.	
6Fh (1103DEh)	REG1103DE	7:0	Default: 0x00	Access: RO
	DEBUG[7:0]	7:0	Reserved.	
6Fh (1103DFh)	REG1103DF	7:0	Default: 0x00	Access: RO
	DEBUG[15:8]	7:0	See description of '1103DEh.'	
70h (1103E0h)	REG1103E0	7:0	Default: 0x00	Access: R/W
	-	7:2	Reserved.	
	CLK_MIU_EN	1	Reserved.	
	CLK_UART_EN	0	Reserved.	
70h (1103E1h)	REG1103E1	7:0	Default: 0x00	Access: R/W
	URDMA2MIU_SEL[15:14]	7:6	UART DMA to MIU. 00: MIU0. 01: MIU1. 10: MIU2. 11: MIU3.	
	-	5:0	Reserved.	
72h (1103E4h)	REG1103E4	7:0	Default: 0x00	Access: R/W
	RES0[7:0]	7:0	Reserved0.	
72h (1103E5h)	REG1103E5	7:0	Default: 0x00	Access: R/W
	RES0[15:8]	7:0	See description of '1103E4h.'	
73h	REG1103E6	7:0	Default: 0x00	Access: R/W



URDMA Register (Bank = 1103)				
Index (Absolute)	Mnemonic	Bit	Description	
(1103E6h)	RES1[7:0]	7:0	Reserved1.	
73h	REG1103E7	7:0	Default: 0x00	Access: R/W
(1103E7h)	RES1[15:8]	7:0	See description of '1103E6h.'	

UART0 Register (Bank = 1108)

UART0 Register (Bank = 1108)				
Index (Absolute)	Mnemonic	Bit	Description	
00h (110800h)	REG110800	7:0	Default: 0x00	Access: R/W
	THR_RBR_DLL[7:0]	7:0	1. When "REG_LCR_DL_ACCESS" = 0: Write: Transmitter Holding Register. Write transmit FIFO; note that writing data to a full FIFO results in the write data being lost. Read: Receiver Buffer. Read receive FIFO; note that any incoming data are lost when FIFO is full and an overrun error occurs. 2. When "REG_LCR_DL_ACCESS" = 1: Divisor Latch LSB.	
02h (110804h)	REG110804	7:0	Default: 0x00	Access: R/W
	IER_DLH[7:0]	7:0	1. When "REG_LCR_DL_ACCESS" = 0: Interrupt Enable Registers (IER); 1: Enabled. Bit [0]: Received Data Available Interrupt and Character Timeout Interrupt. Bit [1]: Transmitter Holding Register Empty Interrupt. Bit [2]: Receiver Line Status Interrupt. Bit [3]: Modem Status interrupt. Bit [7]: Programmable THRE Interrupt. 2. When "REG_LCR_DL_ACCESS" = 1: Divisor Latch MSB. Baud rate = (serial clock freq.) / (16 * divisor).	
04h (110808h)	REG110808	7:0	Default: 0x00	Access: R/W
	FCR_IIR[7:0]	7:0	1. Write: FIFO Control Register (FCR). Bit [0]: FIFO enable. Bit [1]: Write "1" to clear RX FIFO. Bit [2]: Write "1" to clear TX FIFO. Bit [5:4]: Transmit FIFO Empty trigger level. "00": FIFO empty; "01": 2 characters in the FIFO; "10": FIFO 1/4 full; "11": FIFO 1/2 full; Bit [7:6]: Receiver FIFO Interrupt trigger level. "00": 1 character in the FIFO; "01": FIFO 1/4 full; "10": FIFO 1/2 full; "11": FIFO 2 less than full.	

UART0 Register (Bank = 1108)				
Index (Absolute)	Mnemonic	Bit	Description	
			2. Read: Interrupt Identification Registers (IIR). Bit [0]: 1: No interrupt is pending. Bit [3:1]: Interrupt identified. "110": Character timeout. "011": Receiver Line Status. "010": Receiver Data Available. "001": Transmitter Holding Register empty. "000": Modem Status.	
06h (11080Ch)	REG11080C	7:0	Default: 0x03	Access: R/W
	LCR_DL_ACCESS	7	Divisor Latch Access; 1: The divisor latches can be accessed.	
	LCR_BREAK_CTRL	6	Break control bit.	
	-	5	Reserved.	
	LCR_EVEN_PARITY_SEL	4	1: Select even parity.	
	LCR_PARITY_EN	3	1: Generate parity bit on serial out.	
	LCR_STOP_BITS	2	Specify the number of stop bits. "0": 1 stop bit; "1": 1.5 stop bits when 5-bit character length is selected and 2 bits otherwise.	
	LCR_CHAR_BITS[1:0]	1:0	Select number of bits in each character. "00": 5 bits; "01": 6 bits; "10": 7 bits; "11": 8 bits.	
08h (110810h)	REG110810	7:0	Default: 0x00	Access: R/W
	-	7:6	Reserved.	
	MCR_AFCE	5	Auto Flow Control Enable; 1: Enable.	
	MCR_LOOPBACK	4	1: Loopback mode. SOUT -> SIN (pad "STX_PAD_O" will be set to "1"). DTR -> DSR. RTS -> CTS. OUT1 -> RI. OUT2 -> DCD.	
	MCR_OUT2	3	In loopback mode, connect to Data Carrier Detect (DCD) signal input.	
	MCR_OUT1	2	In loopback mode, connect to Ring Indicator (RI) signal input.	
	MCR_RTS	1	Request To Send (RTS) signal control. "0": RTS is "1"; "1": RTS is "0."	
	MCR_DTR	0	Data Terminal Ready (DTR) signal control.	

UART0 Register (Bank = 1108)				
Index (Absolute)	Mnemonic	Bit	Description	
			"0": DTR is "1"; "1": DTR is "0."	
0Ah (110814h)	REG110814	7:0	Default: 0x00	Access: RO
	LSR_ERROR	7	Receiver FIFO Error bit.	
	LSR_TX_EMPTY	6	1: Transmitter (TX FIFO and shift registers) Empty indicator. Cleared after writing data into TX FIFO.	
	LSR_TXFIFO_EMPTY	5	1: Transmit FIFO is empty. Cleared after writing data into TX FIFO. Gen a Transmitter Holding Register Empty interrupt.	
	LSR_BI	4	Break Interrupt bit.	
	LSR_FE	3	1: Framing Error indicator. Cleared when reading. Gen a Receiver Line Status interrupt.	
	LSR_PE	2	1: Parity Error indicator. Cleared when reading. Gen a Receiver Line Status interrupt.	
	LSR_OE	1	1: RX Overrun Error indicator. Cleared when reading. Gen a Receiver Line Status interrupt.	
	LSR_DR	0	1: Received Data Ready indicator.	
0Ch (110818h)	REG110818	7:0	Default: 0x00	Access: RO
	MSR_DCD_COMP	7	Complement of "DCD" or equal to "OUT2" in loopback.	
	MSR_RI_COMP	6	Complement of "RI" or equal to "OUT1" in loopback.	
	MSR_DSR_COMP	5	Complement of "DSR" or equal to "DTR" in loopback.	
	MSR_CTS_COMP	4	Complement of "CTS" or equal to "RTS" in loopback.	
	MSR_DDCD	3	Delta Data Carrier Detect (DDCD) indicator. "1": The "DCD" line has changed its state. Cleared when reading.	
	MSR_TERI	2	Trailing Edge of Ring Indicator (TERI) detector. The "RI" line has changed its state from low to high. Cleared when reading.	
	MSR_DDSD	1	Delta Data Set Ready (DDSD) indicator. "1": The "DSR" line has changed its state. Cleared when reading.	
	MSR_DCTS	0	Delta Clear To Send (DCTS) indicator. "1": The "CTS" line has changed its state. Cleared when reading.	

UART0 Register (Bank = 1108)

Index (Absolute)	Mnemonic	Bit	Description
0Eh (11081Ch)	REG11081C	7:0	Default: 0x00 Access: RO
	-	7:5	Reserved.
	USR_RFF	4	RX FIFO Full.
	USR_RFNE	3	RX FIFO Not Empty.
	USR_TFE	2	TX FIFO Empty.
	USR_TFNF	1	TX FIFO Not Full.
	USR_BUSY	0	UART Busy.
10h (110820h)	REG110820	7:0	Default: 0x00 Access: RO
	-	7:6	Reserved.
	TFL[5:0]	5:0	TX FIFO level.
12h (110824h)	REG110824	7:0	Default: 0x00 Access: RO
	-	7:6	Reserved.
	RFL[5:0]	5:0	RX FIFO level.
14h (110828h)	REG110828	7:0	Default: 0x01 Access: R/W
	-	7:1	Reserved.
	SW_RST	0	SW_RST.

UART1 Register (Bank = 1109)

UART1 Register (Bank = 1109)				
Index (Absolute)	Mnemonic	Bit	Description	
00h (110900h)	REG110900	7:0	Default: 0x00	Access: R/W
	THR_RBR_DLL[7:0]	7:0	1. When "REG_LCR_DL_ACCESS" = 0: Write: Transmitter Holding Register. Write transmit FIFO; note that writing data to a full FIFO results in the write data being lost. Read: Receiver Buffer. Read receive FIFO; note that any incoming data are lost when FIFO is full and an overrun error occurs. 2. When "REG_LCR_DL_ACCESS" = 1: Divisor Latch LSB.	
02h (110904h)	REG110904	7:0	Default: 0x00	Access: R/W
	IER_DLH[7:0]	7:0	1. When "REG_LCR_DL_ACCESS" = 0: Interrupt Enable Registers (IER); 1: Enabled. Bit [0]: Received Data Available Interrupt and Character Timeout Interrupt. Bit [1]: Transmitter Holding Register Empty Interrupt. Bit [2]: Receiver Line Status Interrupt. Bit [3]: Modem Status interrupt. Bit [7]: Programmable THRE Interrupt. 2. When "REG_LCR_DL_ACCESS" = 1: Divisor Latch MSB. Baud rate = (serial clock freq.) / (16 * divisor).	
04h (110908h)	REG110908	7:0	Default: 0x00	Access: R/W
	FCR_IIR[7:0]	7:0	1. Write: FIFO Control Register (FCR). Bit [0]: FIFO enable. Bit [1]: Write "1" to clear RX FIFO. Bit [2]: Write "1" to clear TX FIFO. Bit [5:4]: Transmit FIFO Empty trigger level. "00": FIFO empty; "01": 2 characters in the FIFO; "10": FIFO 1/4 full; "11": FIFO 1/2 full; Bit [7:6]: Receiver FIFO Interrupt trigger level. "00": 1 character in the FIFO; "01": FIFO 1/4 full; "10": FIFO 1/2 full; "11": FIFO 2 less than full.	

UART1 Register (Bank = 1109)

Index (Absolute)	Mnemonic	Bit	Description
			2. Read: Interrupt Identification Registers (IIR). Bit [0]: 1: No interrupt is pending. Bit [3:1]: Interrupt identified. "110": Character timeout. "011": Receiver Line Status. "010": Receiver Data Available. "001": Transmitter Holding Register empty. "000": Modem Status.
06h (11090Ch)	REG11090C	7:0	Default: 0x03 Access: R/W
	LCR_DL_ACCESS	7	Divisor Latch Access; 1: The divisor latches can be accessed.
	LCR_BREAK_CTRL	6	Break control bit.
	-	5	Reserved.
	LCR_EVEN_PARITY_SEL	4	1: Select even parity.
	LCR_PARITY_EN	3	1: Generate parity bit on serial out.
	LCR_STOP_BITS	2	Specify the number of stop bits. "0": 1 stop bit; "1": 1.5 stop bits when 5-bit character length is selected and 2 bits otherwise.
	LCR_CHAR_BITS[1:0]	1:0	Select number of bits in each character. "00": 5 bits; "01": 6 bits; "10": 7 bits; "11": 8 bits.
08h (110910h)	REG110910	7:0	Default: 0x00 Access: R/W
	-	7:6	Reserved.
	MCR_AFCE	5	Auto Flow Control Enable; 1: Enable.
	MCR_LOOPBACK	4	1: Loopback mode. SOUT -> SIN (pad "STX_PAD_O" will be set to "1"). DTR -> DSR. RTS -> CTS. OUT1 -> RI. OUT2 -> DCD.
	MCR_OUT2	3	In loopback mode, connect to Data Carrier Detect (DCD) signal input.
	MCR_OUT1	2	In loopback mode, connect to Ring Indicator (RI) signal input.
	MCR_RTS	1	Request To Send (RTS) signal control. "0": RTS is "1"; "1": RTS is "0."
	MCR_DTR	0	Data Terminal Ready (DTR) signal control.

UART1 Register (Bank = 1109)				
Index (Absolute)	Mnemonic	Bit	Description	
			"0": DTR is "1"; "1": DTR is "0."	
0Ah (110914h)	REG110914	7:0	Default: 0x00	Access: RO
	LSR_ERROR	7	Receiver FIFO Error bit.	
	LSR_TX_EMPTY	6	1: Transmitter (TX FIFO and shift registers) Empty indicator. Cleared after writing data into TX FIFO.	
	LSR_TXFIFO_EMPTY	5	1: Transmit FIFO is empty. Cleared after writing data into TX FIFO. Gen a Transmitter Holding Register Empty interrupt.	
	LSR_BI	4	Break Interrupt bit.	
	LSR_FE	3	1: Framing Error indicator. Cleared when reading. Gen a Receiver Line Status interrupt.	
	LSR_PE	2	1: Parity Error indicator. Cleared when reading. Gen a Receiver Line Status interrupt.	
	LSR_OE	1	1: RX Overrun Error indicator. Cleared when reading. Gen a Receiver Line Status interrupt.	
	LSR_DR	0	1: Received Data Ready indicator.	
0Ch (110918h)	REG110918	7:0	Default: 0x00	Access: RO
	MSR_DCD_COMP	7	Complement of "DCD" or equal to "OUT2" in loopback.	
	MSR_RI_COMP	6	Complement of "RI" or equal to "OUT1" in loopback.	
	MSR_DSR_COMP	5	Complement of "DSR" or equal to "DTR" in loopback.	
	MSR_CTS_COMP	4	Complement of "CTS" or equal to "RTS" in loopback.	
	MSR_DDCD	3	Delta Data Carrier Detect (DDCD) indicator. "1": The "DCD" line has changed its state. Cleared when reading.	
	MSR_TERI	2	Trailing Edge of Ring Indicator (TERI) detector. The "RI" line has changed its state from low to high. Cleared when reading.	
	MSR_DDSD	1	Delta Data Set Ready (DDSD) indicator. "1": The "DSR" line has changed its state. Cleared when reading.	
	MSR_DCTS	0	Delta Clear To Send (DCTS) indicator. "1": The "CTS" line has changed its state. Cleared when reading.	

UART1 Register (Bank = 1109)

Index (Absolute)	Mnemonic	Bit	Description
0Eh (11091Ch)	REG11091C	7:0	Default: 0x00 Access: RO
	-	7:5	Reserved.
	USR_RFF	4	RX FIFO Full.
	USR_RFNE	3	RX FIFO Not Empty.
	USR_TFE	2	TX FIFO Empty.
	USR_TFNF	1	TX FIFO Not Full.
	USR_BUSY	0	UART Busy.
10h (110920h)	REG110920	7:0	Default: 0x00 Access: RO
	-	7:6	Reserved.
	TFL[5:0]	5:0	TX FIFO level.
12h (110924h)	REG110924	7:0	Default: 0x00 Access: RO
	-	7:6	Reserved.
	RFL[5:0]	5:0	RX FIFO level.
14h (110928h)	REG110928	7:0	Default: 0x01 Access: R/W
	-	7:1	Reserved.
	SW_RST	0	SW_RST.

MSPI0 Register (Bank = 1110)

MSPI0 Register (Bank = 1110)				
Index (Absolute)	Mnemonic	Bit	Description	
30h (111060h)	REG111060	7:0	Default: 0x00	Access: R/W
	DATA_LENGTH[7:0]	7:0	DMA mode R/W data length.	
30h (111061h)	REG111061	7:0	Default: 0x00	Access: R/W
	DATA_LENGTH[15:8]	7:0	See description of '111060h.'	
31h (111062h)	REG111062	7:0	Default: 0x00	Access: R/W
	DATA_LENGTH[23:16]	7:0	See description of '111060h.'	
32h (111064h)	REG111064	7:0	Default: 0x00	Access: R/W
	-	7:1	Reserved.	
	DMA_ENABLE	0	DMA mode enable. 1: Enable. 0: Disable.	
33h (111066h)	REG111066	7:0	Default: 0x00	Access: R/W
	-	7:1	Reserved.	
	DMA_RW	0	DMA mode R/W. 0: DMA write-only mode. 1: DMA read-only mode.	
40h (111080h)	REG111080	7:0	Default: 0x00	Access: R/W
	MSPI_WD0[7:0]	7:0	Write buffer0.	
40h (111081h)	REG111081	7:0	Default: 0x00	Access: R/W
	MSPI_WD1[7:0]	7:0	Write buffer1.	
41h (111082h)	REG111082	7:0	Default: 0x00	Access: R/W
	MSPI_WD2[7:0]	7:0	Write buffer2.	
41h (111083h)	REG111083	7:0	Default: 0x00	Access: R/W
	MSPI_WD3[7:0]	7:0	Write buffer3.	
42h (111084h)	REG111084	7:0	Default: 0x00	Access: R/W
	MSPI_WD4[7:0]	7:0	Write buffer4.	
42h (111085h)	REG111085	7:0	Default: 0x00	Access: R/W
	MSPI_WD5[7:0]	7:0	Write buffer5.	
43h (111086h)	REG111086	7:0	Default: 0x00	Access: R/W
	MSPI_WD6[7:0]	7:0	Write buffer6.	
43h (111087h)	REG111087	7:0	Default: 0x00	Access: R/W
	MSPI_WD7[7:0]	7:0	Write buffer7.	
44h	REG111088	7:0	Default: 0x00	Access: RO

MSPIO Register (Bank = 1110)				
Index (Absolute)	Mnemonic	Bit	Description	
(111088h)	MSPI_RD0[7:0]	7:0	Read buffer0.	
44h (111089h)	REG111089 MSPI_RD1[7:0]	7:0 7:0	Default: 0x00	Access: RO
45h (11108Ah)	REG11108A MSPI_RD2[7:0]	7:0 7:0	Default: 0x00	Access: RO
45h (11108Bh)	REG11108B MSPI_RD3[7:0]	7:0 7:0	Default: 0x00	Access: RO
46h (11108Ch)	REG11108C MSPI_RD4[7:0]	7:0 7:0	Default: 0x00	Access: RO
46h (11108Dh)	REG11108D MSPI_RD5[7:0]	7:0 7:0	Default: 0x00	Access: RO
47h (11108Eh)	REG11108E MSPI_RD6[7:0]	7:0 7:0	Default: 0x00	Access: RO
47h (11108Fh)	REG11108F MSPI_RD7[7:0]	7:0 7:0	Default: 0x00	Access: RO
48h (111090h)	REG111090 MSPI_WBF_SIZE[7:0]	7:0 7:0	Default: 0x00	Access: R/W
			Set the number of bytes to be transferred. Max size is 8 bytes. Min size is 0 byte.	
48h (111091h)	REG111091 MSPI_RBF_SIZE[7:0]	7:0 7:0	Default: 0x00	Access: R/W
			Set the number of bytes to be received. Max size is 8 bytes. Min size is 0 byte.	
49h (111092h)	REG111092 MSPI_CTRL[7:0]	7:0 7:0	Default: 0x00	Access: R/W
			Control Register. Bit [7]: Clock Polarity, CPOL. 0: The SCK is set to 0 in idle state. 1: The SCK is set to 1 in idle state. Bit [6]: Clock Phase, CPHA. 0: Data is sampled when the SCK exits the idle state. 1: Data is sampled when the SCK returns to idle state. Bit [5]: Reserved. Bit [4]: 3-wire mode. 0: Disable. 1: Enable. Bit [3]: Reserved.	

MSPI0 Register (Bank = 1110)				
Index (Absolute)	Mnemonic	Bit	Description	
			Bit [2]: Enable MSPI interrupt. 0: Disable. 1: Enable. Bit [1]: Reset. 0: Reset. 1: Not Reset. Bit [0]: Enable MSPI. 0: Disable. 1: Enable.	
49h (111093h)	REG111093	7:0	Default: 0x00	Access: R/W
	MSPI_CLOCK_RATE[7:0]	7:0	Bit [2:0]. 3'b000: CPU_CLOCK/2. 3'b001: CPU_CLOCK/4. 3'b010: CPU_CLOCK/8. 3'b011: CPU_CLOCK/16. 3'b100: CPU_CLOCK/32. 3'b101: CPU_CLOCK/64. 3'b110: CPU_CLOCK/128. 3'b111: CPU_CLOCK/256. Bit [7:3]: Reserved.	
4Ah (111094h)	REG111094	7:0	Default: 0x00	Access: R/W
	TR_START_TIME[7:0]	7:0	The time from "REG_MSPI_TRIGGER" to first SPI clock. 0x00: Delay 1 MSPI clock. 0x01: Delay 2 MSPI clocks. 0x0f: Delay 16 MSPI clocks. 0xff: Delay 256 MSPI clocks.	
4Ah (111095h)	REG111095	7:0	Default: 0x00	Access: R/W
	TR_END_TIME[7:0]	7:0	The time from last SPI clock to "REG_MSPI_DONE_FLAG." 0x00: Delay 1 MSPI clock. 0x01: Delay 2 MSPI clocks. 0x0f: Delay 16 MSPI clocks. 0xff: Delay 256 MSPI clocks.	
4Bh (111096h)	REG111096	7:0	Default: 0x00	Access: R/W
	TBYTE_INTERVAL_TIME[7:0]	7:0	The time between byte to byte transfers. 0x00: No delay. 0x01: Delay 1 MSPI clock. 0x0f: Delay 15 MSPI clocks. 0xff: Delay 255 MSPI clocks.	
4Bh	REG111097	7:0	Default: 0x00	Access: R/W

MSPI0 Register (Bank = 1110)				
Index (Absolute)	Mnemonic	Bit	Description	
(111097h)	RW_TURN_AROUND_TIME[7:0]	7:0	The time between last write and first read. 0x00: No delay. 0x01: Delay 1 MSPI clock. 0x0f: Delay 15 MSPI clocks. 0xff: Delay 255 MSPI clocks.	
4Ch (111098h)	REG111098	7:0	Default: 0xFF	Access: R/W
	MSPI_WD2_BIT_SEL[1:0]	7:6	Bit length selection for write buffer2.	
	MSPI_WD1_BIT_SEL[2:0]	5:3	Bit length selection for write buffer1.	
	MSPI_WD0_BIT_SEL[2:0]	2:0	Bit length selection for write buffer0. The number of bits to be transferred in write buffer0. 3'b111: 8 bits. 3'b110: 7 bits. 3'b101: 6 bits. 3'b100: 5 bits. 3'b011: 4 bits. 3'b010: 3 bits. 3'b001: 2 bits. 3'b000: 1 bit.	
4Ch (111099h)	REG111099	7:0	Default: 0x0F	Access: R/W
	-	7:4	Reserved.	
	MSPI_WD3_BIT_SEL[2:0]	3:1	Bit length selection for write buffer3.	
	MSPI_WD2_BIT_SEL[2]	0	See description of '111098h.'	
4Dh (11109Ah)	REG11109A	7:0	Default: 0xFF	Access: R/W
	MSPI_WD6_BIT_SEL[1:0]	7:6	Bit length selection for write buffer6.	
	MSPI_WD5_BIT_SEL[2:0]	5:3	Bit length selection for write buffer5.	
	MSPI_WD4_BIT_SEL[2:0]	2:0	Bit length selection for write buffer4.	
4Dh (11109Bh)	REG11109B	7:0	Default: 0x0F	Access: R/W
	-	7:4	Reserved.	
	MSPI_WD7_BIT_SEL[2:0]	3:1	Bit length selection for write buffer7.	
	MSPI_WD6_BIT_SEL[2]	0	See description of '11109Ah.'	
4Eh (11109Ch)	REG11109C	7:0	Default: 0xFF	Access: R/W
	MSPI_RD2_BIT_SEL[1:0]	7:6	Bit length selection for read buffer2.	
	MSPI_RD1_BIT_SEL[2:0]	5:3	Bit length selection for read buffer1.	
	MSPI_RD0_BIT_SEL[2:0]	2:0	Bit length selection for read buffer0. The number of bits to be received in read buffer0. 3'b111: 8 bits.	

MSPI0 Register (Bank = 1110)				
Index (Absolute)	Mnemonic	Bit	Description	
			3'b110: 7 bits. 3'b101: 6 bits. 3'b100: 5 bits. 3'b011: 4 bits. 3'b010: 3 bits. 3'b001: 2 bits. 3'b000: 1 bit.	
4Eh (11109Dh)	REG11109D	7:0	Default: 0x0F	Access: R/W
	-	7:4	Reserved.	
	MSPI_RD3_BIT_SEL[2:0]	3:1	Bit length selection for read buffer3.	
	MSPI_RD2_BIT_SEL[2]	0	See description of '11109Ch.'	
4Fh (11109Eh)	REG11109E	7:0	Default: 0xFF	Access: R/W
	MSPI_RD6_BIT_SEL[1:0]	7:6	Bit length selection for read buffer6.	
	MSPI_RD5_BIT_SEL[2:0]	5:3	Bit length selection for read buffer5.	
	MSPI_RD4_BIT_SEL[2:0]	2:0	Bit length selection for read buffer4.	
4Fh (11109Fh)	REG11109F	7:0	Default: 0x0F	Access: R/W
	-	7:4	Reserved.	
	MSPI_RD7_BIT_SEL[2:0]	3:1	Bit length selection for read buffer7.	
	MSPI_RD6_BIT_SEL[2]	0	See description of '11109Eh.'	
50h (1110A0h)	REG1110A0	7:0	Default: 0x00	Access: R/W
	-	7:1	Reserved.	
	LSB_FIRST	0	Data transfer from LSB first.	
5Ah (1110B4h)	REG1110B4	7:0	Default: 0x00	Access: WO
	-	7:1	Reserved.	
	MSPI_TRIGGER	0	Start data transfer.	
5Bh (1110B6h)	REG1110B6	7:0	Default: 0x00	Access: RO
	-	7:1	Reserved.	
	MSPI_DONE_FLAG	0	Busy status, HW sets to one when transfer is completed. 1: Transfer done or interrupt pending. 0: Transfer busy or interrupt not pending.	
5Ch (1110B8h)	REG1110B8	7:0	Default: 0x00	Access: WO
	-	7:1	Reserved.	
	MSPI_CLEAR_DONE_FLAG	0	SW needs to set this bit to clear done flag or interrupt.	
5Dh (1110BAh)	REG1110BA	7:0	Default: 0x00	Access: RO, R/W
	-	7:3	Reserved.	

MSPIO Register (Bank = 1110)				
Index (Absolute)	Mnemonic	Bit	Description	
	SPI_WP_C	2	WRITE_PROTECT input for SPI flash.	
	SPI_WP_OEN	1	WRITE_PROTECT output enable (OEN) for SPI flash.	
	SPI_WP_I	0	WRITE_PROTECT output for SPI flash.	
5Eh (1110BCh)	REG1110BC	7:0	Default: 0x00	Access: RO, R/W
	-	7:3	Reserved.	
	SPI_HOLD_C	2	Hold input for SPI flash.	
	SPI_HOLD_OEN	1	Hold output enable (OEN) for SPI flash.	
	SPI_HOLD_I	0	Hold output for SPI flash.	
5Fh (1110BEh)	REG1110BE	7:0	Default: 0xFF	Access: R/W
	CHIP_SELECT8	7	CHIP_SELECT for SPI Device8. 0: Enable. 1: Disable.	
	CHIP_SELECT7	6	CHIP_SELECT for SPI Device7. 0: Enable. 1: Disable.	
	CHIP_SELECT6	5	CHIP_SELECT for SPI Device6. 0: Enable. 1: Disable.	
	CHIP_SELECT5	4	CHIP_SELECT for SPI Device5. 0: Enable. 1: Disable.	
	CHIP_SELECT4	3	CHIP_SELECT for SPI Device4. 0: Enable. 1: Disable.	
	CHIP_SELECT3	2	CHIP_SELECT for SPI Device3. 0: Enable. 1: Disable.	
	CHIP_SELECT2	1	CHIP_SELECT for SPI Device2. 0: Enable. 1: Disable.	
	CHIP_SELECT1	0	CHIP_SELECT for SPI Device1. 0: Enable. 1: Disable.	

MSPI1 Register (Bank = 1111)

MSPI1 Register (Bank = 1111)				
Index (Absolute)	Mnemonic	Bit	Description	
50h (1111A0h)	REG1111A0	7:0	Default: 0x00	Access: R/W
	DATA_LENGTH[7:0]	7:0	DMA mode R/W data length.	
50h (1111A1h)	REG1111A1	7:0	Default: 0x00	Access: R/W
	DATA_LENGTH[15:8]	7:0	See description of '1111A0h.'	
51h (1111A2h)	REG1111A2	7:0	Default: 0x00	Access: R/W
	DATA_LENGTH[23:16]	7:0	See description of '1111A0h.'	
52h (1111A4h)	REG1111A4	7:0	Default: 0x00	Access: R/W
	-	7:1	Reserved.	
	DMA_ENABLE	0	DMA mode enable. 1: Enable. 0: Disable.	
53h (1111A6h)	REG1111A6	7:0	Default: 0x00	Access: R/W
	-	7:1	Reserved.	
	DMA_RW	0	DMA mode R/W. 0: DMA write-only mode. 1: DMA read-only mode.	
7Dh (1111FAh)	REG1111FA	7:0	Default: 0x00	Access: RO, R/W
	-	7:3	Reserved.	
	SPI_WP_C	2	WRITE_PROTECT input for SPI flash.	
	SPI_WP_OEN	1	WRITE_PROTECT output enable (OEN) for SPI flash.	
	SPI_WP_I	0	WRITE_PROTECT output for SPI flash.	
7Eh (1111FCh)	REG1111FC	7:0	Default: 0x00	Access: RO, R/W
	-	7:3	Reserved.	
	SPI_HOLD_C	2	Hold input for SPI flash.	
	SPI_HOLD_OEN	1	Hold output enable (OEN) for SPI flash.	
	SPI_HOLD_I	0	Hold output for SPI flash.	

MSPI1 Register (Bank = 1111)

MSPI1 Register (Bank = 1111)				
Index (Absolute)	Mnemonic	Bit	Description	
60h (1111C0h)	REG1111C0	7:0	Default: 0x00	Access: R/W
	MSPI_WD0[7:0]	7:0	Write buffer0.	
60h (1111C1h)	REG1111C1	7:0	Default: 0x00	Access: R/W
	MSPI_WD1[7:0]	7:0	Write buffer1.	
61h (1111C2h)	REG1111C2	7:0	Default: 0x00	Access: R/W
	MSPI_WD2[7:0]	7:0	Write buffer2.	
61h (1111C3h)	REG1111C3	7:0	Default: 0x00	Access: R/W
	MSPI_WD3[7:0]	7:0	Write buffer3.	
62h (1111C4h)	REG1111C4	7:0	Default: 0x00	Access: R/W
	MSPI_WD4[7:0]	7:0	Write buffer4.	
62h (1111C5h)	REG1111C5	7:0	Default: 0x00	Access: R/W
	MSPI_WD5[7:0]	7:0	Write buffer5.	
63h (1111C6h)	REG1111C6	7:0	Default: 0x00	Access: R/W
	MSPI_WD6[7:0]	7:0	Write buffer6.	
63h (1111C7h)	REG1111C7	7:0	Default: 0x00	Access: R/W
	MSPI_WD7[7:0]	7:0	Write buffer7.	
64h (1111C8h)	REG1111C8	7:0	Default: 0x00	Access: RO
	MSPI_RD0[7:0]	7:0	Read buffer0.	
64h (1111C9h)	REG1111C9	7:0	Default: 0x00	Access: RO
	MSPI_RD1[7:0]	7:0	Read buffer1.	
65h (1111CAh)	REG1111CA	7:0	Default: 0x00	Access: RO
	MSPI_RD2[7:0]	7:0	Read buffer2.	
65h (1111CBh)	REG1111CB	7:0	Default: 0x00	Access: RO
	MSPI_RD3[7:0]	7:0	Read buffer3.	
66h (1111CCh)	REG1111CC	7:0	Default: 0x00	Access: RO
	MSPI_RD4[7:0]	7:0	Read buffer4.	
66h (1111CDh)	REG1111CD	7:0	Default: 0x00	Access: RO
	MSPI_RD5[7:0]	7:0	Read buffer5.	
67h (1111CEh)	REG1111CE	7:0	Default: 0x00	Access: RO
	MSPI_RD6[7:0]	7:0	Read buffer6.	
67h (1111CFh)	REG1111CF	7:0	Default: 0x00	Access: RO
	MSPI_RD7[7:0]	7:0	Read buffer7.	

MSPI1 Register (Bank = 1111)				
Index (Absolute)	Mnemonic	Bit	Description	
68h (1111D0h)	REG1111D0	7:0	Default: 0x00	Access: R/W
	MSPI_WBF_SIZE[7:0]	7:0	Set the number of bytes to be transferred. Max size is 8 bytes. Min size is 0 byte.	
68h (1111D1h)	REG1111D1	7:0	Default: 0x00	Access: R/W
	MSPI_RBF_SIZE[7:0]	7:0	Set the number of bytes to be received. Max size is 8 bytes. Min size is 0 byte.	
69h (1111D2h)	REG1111D2	7:0	Default: 0x00	Access: R/W
	MSPI_CTRL[7:0]	7:0	Control Register. Bit [7]: Clock Polarity, CPOL. 0: The SCK is set to 0 in idle state. 1: The SCK is set to 1 in idle state. Bit [6]: Clock Phase, CPHA. 0: Data is sampled when the SCK exits the idle state. 1: Data is sampled when the SCK returns to idle state. Bit [5]: Reserved. Bit [4]: 3-wire mode. 0: Disable. 1: Enable. Bit [3]: Reserved. Bit [2]: Enable MSPI interrupt. 0: Disable. 1: Enable. Bit [1]: Reset. 0: Reset. 1: Not Reset. Bit [0]: Enable MSPI. 0: Disable. 1: Enable.	
69h (1111D3h)	REG1111D3	7:0	Default: 0x00	Access: R/W
	MSPI_CLOCK_RATE[7:0]	7:0	Bit [2:0]. 3'b000: CPU_CLOCK/2. 3'b001: CPU_CLOCK/4. 3'b010: CPU_CLOCK/8. 3'b011: CPU_CLOCK/16. 3'b100: CPU_CLOCK/32. 3'b101: CPU_CLOCK/64. 3'b110: CPU_CLOCK/128.	

MSPI1 Register (Bank = 1111)				
Index (Absolute)	Mnemonic	Bit	Description	
			3'b111: CPU_CLOCK/256. Bit [7:3]: Reserved.	
6Ah (1111D4h)	REG1111D4	7:0	Default: 0x00	Access: R/W
	TR_START_TIME[7:0]	7:0	The time from "REG_MSPI_TRIGGER" to first SPI clock. 0x00: Delay 1 MSPI clock. 0x01: Delay 2 MSPI clocks. 0x0f: Delay 16 MSPI clocks. 0xff: Delay 256 MSPI clocks.	
6Ah (1111D5h)	REG1111D5	7:0	Default: 0x00	Access: R/W
	TR_END_TIME[7:0]	7:0	The time from last SPI clock to "REG_MSPI_DONE_FLAG." 0x00: Delay 1 MSPI clock. 0x01: Delay 2 MSPI clocks. 0x0f: Delay 16 MSPI clocks. 0xff: Delay 256 MSPI clocks.	
6Bh (1111D6h)	REG1111D6	7:0	Default: 0x00	Access: R/W
	TBYTE_INTERVAL_TIME[7:0]	7:0	The time between byte to byte transfers. 0x00: No delay. 0x01: Delay 1 MSPI clock. 0x0f: Delay 15 MSPI clocks. 0xff: Delay 255 MSPI clocks.	
6Bh (1111D7h)	REG1111D7	7:0	Default: 0x00	Access: R/W
	RW_TURN_AROUND_TIME[7:0]	7:0	The time between last write and first read. 0x00: No delay. 0x01: Delay 1 MSPI clock. 0x0f: Delay 15 MSPI clocks. 0xff: Delay 255 MSPI clocks.	
6Ch (1111D8h)	REG1111D8	7:0	Default: 0xFF	Access: R/W
	MSPI_WD2_BIT_SEL[1:0]	7:6	Bit length selection for write buffer2.	
	MSPI_WD1_BIT_SEL[2:0]	5:3	Bit length selection for write buffer1.	
	MSPI_WD0_BIT_SEL[2:0]	2:0	Bit length selection for write buffer0. The number of bits to be transferred in write buffer0. 3'b111: 8 bits. 3'b110: 7 bits. 3'b101: 6 bits. 3'b100: 5 bits. 3'b011: 4 bits. 3'b010: 3 bits. 3'b001: 2 bits.	

MSPI1 Register (Bank = 1111)				
Index (Absolute)	Mnemonic	Bit	Description	
			3'b000: 1 bit.	
6Ch (1111D9h)	REG1111D9	7:0	Default: 0x0F	Access: R/W
	-	7:4	Reserved.	
	MSPI_WD3_BIT_SEL[2:0]	3:1	Bit length selection for write buffer3.	
	MSPI_WD2_BIT_SEL[2]	0	See description of '1111D8h.'	
6Dh (1111DAh)	REG1111DA	7:0	Default: 0xFF	Access: R/W
	MSPI_WD6_BIT_SEL[1:0]	7:6	Bit length selection for write buffer6.	
	MSPI_WD5_BIT_SEL[2:0]	5:3	Bit length selection for write buffer5.	
	MSPI_WD4_BIT_SEL[2:0]	2:0	Bit length selection for write buffer4.	
6Dh (1111DBh)	REG1111DB	7:0	Default: 0x0F	Access: R/W
	-	7:4	Reserved.	
	MSPI_WD7_BIT_SEL[2:0]	3:1	Bit length selection for write buffer7.	
	MSPI_WD6_BIT_SEL[2]	0	See description of '1111DAh.'	
6Eh (1111DCh)	REG1111DC	7:0	Default: 0xFF	Access: R/W
	MSPI_RD2_BIT_SEL[1:0]	7:6	Bit length selection for read buffer2.	
	MSPI_RD1_BIT_SEL[2:0]	5:3	Bit length selection for read buffer1.	
	MSPI_RD0_BIT_SEL[2:0]	2:0	Bit length selection for read buffer0. The number of bits to be received in read buffer0. 3'b111: 8 bits. 3'b110: 7 bits. 3'b101: 6 bits. 3'b100: 5 bits. 3'b011: 4 bits. 3'b010: 3 bits. 3'b001: 2 bits. 3'b000: 1 bit.	
6Eh (1111DDh)	REG1111DD	7:0	Default: 0x0F	Access: R/W
	-	7:4	Reserved.	
	MSPI_RD3_BIT_SEL[2:0]	3:1	Bit length selection for read buffer3.	
	MSPI_RD2_BIT_SEL[2]	0	See description of '1111DCh.'	
6Fh (1111DEh)	REG1111DE	7:0	Default: 0xFF	Access: R/W
	MSPI_RD6_BIT_SEL[1:0]	7:6	Bit length selection for read buffer6.	
	MSPI_RD5_BIT_SEL[2:0]	5:3	Bit length selection for read buffer5.	
	MSPI_RD4_BIT_SEL[2:0]	2:0	Bit length selection for read buffer4.	
6Fh	REG1111DF	7:0	Default: 0x0F	Access: R/W

MSPI1 Register (Bank = 1111)				
Index (Absolute)	Mnemonic	Bit	Description	
(1111DFh)	-	7:4	Reserved.	
	MSPI_RD7_BIT_SEL[2:0]	3:1	Bit length selection for read buffer7.	
	MSPI_RD6_BIT_SEL[2]	0	See description of '1111DEh.'	
70h (1111E0h)	REG1111E0	7:0	Default: 0x00	Access: R/W
	-	7:1	Reserved.	
	LSB_FIRST	0	Data transfer from LSB first.	
7Ah (1111F4h)	REG1111F4	7:0	Default: 0x00	Access: WO
	-	7:1	Reserved.	
	MSPI_TRIGGER	0	Start data transfer.	
7Bh (1111F6h)	REG1111F6	7:0	Default: 0x00	Access: RO
	-	7:1	Reserved.	
	MSPI_DONE_FLAG	0	Busy status, HW sets to one when transfer is completed. 1: Transfer done or interrupt pending. 0: Transfer busy or interrupt not pending.	
7Ch (1111F8h)	REG1111F8	7:0	Default: 0x00	Access: WO
	-	7:1	Reserved.	
	MSPI_CLEAR_DONE_FLAG	0	SW needs to set this bit to clear done flag or interrupt.	
7Fh (1111FEh)	REG1111FE	7:0	Default: 0xFF	Access: R/W
	CHIP_SELECT8	7	CHIP_SELECT for SPI Device8. 0: Enable. 1: Disable.	
	CHIP_SELECT7	6	CHIP_SELECT for SPI Device7. 0: Enable. 1: Disable.	
	CHIP_SELECT6	5	CHIP_SELECT for SPI Device6. 0: Enable. 1: Disable.	
	CHIP_SELECT5	4	CHIP_SELECT for SPI Device5. 0: Enable. 1: Disable.	
	CHIP_SELECT4	3	CHIP_SELECT for SPI Device4. 0: Enable. 1: Disable.	
	CHIP_SELECT3	2	CHIP_SELECT for SPI Device3. 0: Enable. 1: Disable.	



MSPI1 Register (Bank = 1111)			
Index (Absolute)	Mnemonic	Bit	Description
	CHIP_SELECT2	1	CHIP_SELECT for SPI Device2. 0: Enable. 1: Disable.
	CHIP_SELECT1	0	CHIP_SELECT for SPI Device1. 0: Enable. 1: Disable.

MIIC0 Register (Bank = 1118)

MIIC0 Register (Bank = 1118)				
Index (Absolute)	Mnemonic	Bit	Description	
00h (111800h)	REG111800	7:0	Default: 0x01	Access: R/W
	MIIC_CFG[7:0]	7:0	MIIC configuration register. Bit [7]: REG_ERROR_DET_EN. 0: Disable. 1: Enable. Bit [6]: REG_OEN_PUSH_EN. 0: Disable. 1: Enable. Bit [5]: Enable filter. 0: Disable. 1: Enable. Bit [4]: Enable timeout interrupt. 0: Disable. 1: Enable. Bit [3]: Enable clock stretching. 0: Disable. 1: Enable. Bit [2]: Interrupt enable. 0: Disable. 1: Enable. Bit [1]: Enable DMA. 0: Disable. 1: Enable. Bit [0]: Reset. 0: Not Reset. 1: Reset.	
01h (111802h)	REG111802	7:0	Default: 0x00	Access: WO
	-	7:1	Reserved.	
	CMD_START	0	MIIC command. [0]: Start.	
01h (111803h)	REG111803	7:0	Default: 0x00	Access: WO
	-	7:1	Reserved.	
	CMD_STOP	0	MIIC command. [1]: Stop.	
02h (111804h)	REG111804	7:0	Default: 0x00	Access: R/W
	WDATA[7:0]	7:0	I2C write data.	
02h	REG111805	7:0	Default: 0x00	Access: RO

MIIC0 Register (Bank = 1118)				
Index (Absolute)	Mnemonic	Bit	Description	
(111805h)	-	7:1	Reserved.	
	WRITE_ACK	0	I2C ACK for write data from slave IIC.	
03h (111806h)	REG111806	7:0	Default: 0x00	Access: RO
	RDATA[7:0]	7:0	I2C read data.	
03h (111807h)	REG111807	7:0	Default: 0x00	Access: R/W
	-	7:2	Reserved.	
	ACK_BIT	1	I2C ACK for read data to slave IIC.	
	RDATA_EN	0	I2C read data trigger.	
04h (111808h)	REG111808	7:0	Default: 0x00	Access: R/W
	-	7:1	Reserved.	
	FLAG	0	MIIC interrupt FLAG.	
05h (11180Ah)	REG11180A	7:0	Default: 0x00	Access: RO
	-	7:5	Reserved.	
	MIIC_STATE[4:0]	4:0	MIIC final state machine (debug only).	
05h (11180Bh)	REG11180B	7:0	Default: 0x00	Access: RO
	-	7	Reserved.	
	MIIC_INT_STATUS[6:0]	6:0	INT_STATUS. [0]: IC_START_DET_INTR. [1]: IC_STOP_DET_INTR. [2]: IC_RX_DONE_INTR. [3]: IC_TX_DONE_INTR. [4]: CLOCK_STRETCHING_INTR. [5]: SCL_ERROR_INTE. [6]: TIME_OUT_INTR.	
06h (11180Ch)	REG11180C	7:0	Default: 0x00	Access: RO
	-	7:5	Reserved.	
	SCLO	4	PAD_SCLO.	
	-	3:2	Reserved.	
	SDAI	1	PAD_SDAI.	
	SCLI	0	PAD_SCLI.	
07h (11180Eh)	REG11180E	7:0	Default: 0x00	Access: R/W
	-	7:2	Reserved.	
	FORCE_DRIVE_OEN	1	Force drive OEN enable.	
	PUSH_MULTI_EN	0	Push multi time enable.	

MIIC0 Register (Bank = 1118)

Index (Absolute)	Mnemonic	Bit	Description
07h (11180Fh)	REG11180F PUSH_COUNT_NUMBER[7:0]	7:0 7:0	Default: 0x00 Access: R/W Push multi time number (number x T).
08h (111810h)	REG111810 STOP_CNT[7:0]	7:0 7:0	Default: 0x00 Access: R/W This register sets the SCL and SDA count for stop.
08h (111811h)	REG111811 STOP_CNT[15:8]	7:0 7:0	Default: 0x00 Access: R/W See description of '111810h.'
09h (111812h)	REG111812 HCNT[7:0]	7:0 7:0	Default: 0x00 Access: R/W This register sets the SCL clock high-period count.
09h (111813h)	REG111813 HCNT[15:8]	7:0 7:0	Default: 0x00 Access: R/W See description of '111812h.'
0Ah (111814h)	REG111814 LCNT[7:0]	7:0 7:0	Default: 0x00 Access: R/W This register sets the SCL clock low-period count.
0Ah (111815h)	REG111815 LCNT[15:8]	7:0 7:0	Default: 0x00 Access: R/W See description of '111814h.'
0Bh (111816h)	REG111816 SDA_CNT[7:0]	7:0 7:0	Default: 0x00 Access: R/W This register sets the clock count between the falling edges of SCL and SDA.
0Bh (111817h)	REG111817 SDA_CNT[15:8]	7:0 7:0	Default: 0x00 Access: R/W See description of '111816h.'
0Ch (111818h)	REG111818 START_CNT[7:0]	7:0 7:0	Default: 0x00 Access: R/W This register sets the SCL and SDA count for start.
0Ch (111819h)	REG111819 START_CNT[15:8]	7:0 7:0	Default: 0x00 Access: R/W See description of '111818h.'
0Dh (11181Ah)	REG11181A DATA_LAT_CNT[7:0]	7:0 7:0	Default: 0x00 Access: R/W This register sets the data latch timing.
0Dh (11181Bh)	REG11181B DATA_LAT_CNT[15:8]	7:0 7:0	Default: 0x00 Access: R/W See description of '11181Ah.'
0Eh (11181Ch)	REG11181C TIMEOUT_CNT[7:0]	7:0 7:0	Default: 0x00 Access: R/W This register sets timing delay if timeout interrupt occurs.
0Eh (11181Dh)	REG11181D TIMEOUT_CNT[15:8]	7:0 7:0	Default: 0x00 Access: R/W See description of '11181Ch.'
0Fh	REG11181E	7:0	Default: 0x00 Access: R/W

MIIC0 Register (Bank = 1118)				
Index (Absolute)	Mnemonic	Bit	Description	
(11181Eh)	-	7:3	Reserved.	
	SCLI_DELAY[2:0]	2:0	Reserved.	
20h (111840h)	REG111840	7:0	Default: 0x1A	Access: RO, R/W
	-	7:6	Reserved.	
	MIU_NS	5	MIU secure bit.	
	MIU_PRIORITY	4	Set MIU priority.	
	MIU_RST	3	MIU software reset.	
	DMA_CFG[2:0]	2:0	DMA configuration register. Bit [2]: Interrupt enable. 0: Disable. 1: Enable. Bit [1]: DMA software reset. 0: Reset. 1: Not Reset.	
21h (111842h)	REG111842	7:0	Default: 0x00	Access: R/W
	MIU_ADDR[7:0]	7:0	Get TX data or put RX data address in DRAM.	
21h (111843h)	REG111843	7:0	Default: 0x00	Access: R/W
	MIU_ADDR[15:8]	7:0	See description of '111842h.'	
22h (111844h)	REG111844	7:0	Default: 0x00	Access: R/W
	MIU_ADDR[23:16]	7:0	See description of '111842h.'	
22h (111845h)	REG111845	7:0	Default: 0x00	Access: R/W
	MIU_ADDR[31:24]	7:0	See description of '111842h.'	
23h (111846h)	REG111846	7:0	Default: 0x00	Access: R/W
	MIU_SEL	7	MIIC channel select.	
	READ_CMD	6	MIIC transfer format. 1: Read. 0: Write.	
	STOP_DISABLE	5	MIIC transfer format. 1: S + data ... 0: S + data ... + P.	
	-	4:0	Reserved.	
24h (111848h)	REG111848	7:0	Default: 0x00	Access: R/W
	-	7:1	Reserved.	
	DMA_TRANSFER_DONE	0	SW needs to set this bit to clear TRANSFER_DONE flag or interrupt in order to receive the subsequent DMA	

MIIC0 Register (Bank = 1118)				
Index (Absolute)	Mnemonic	Bit	Description	
			TRANSFER_DONE flag or interrupt.	
25h (11184Ah)	REG11184A	7:0	Default: 0x00	Access: R/W
	CMD_DATA[7:0]	7:0	I2C TX Data Buffer and Command.	
25h (11184Bh)	REG11184B	7:0	Default: 0x00	Access: R/W
	CMD_DATA[15:8]	7:0	See description of '11184Ah.'	
26h (11184Ch)	REG11184C	7:0	Default: 0x00	Access: R/W
	CMD_DATA[23:16]	7:0	See description of '11184Ah.'	
26h (11184Dh)	REG11184D	7:0	Default: 0x00	Access: R/W
	CMD_DATA[31:24]	7:0	See description of '11184Ah.'	
27h (11184Eh)	REG11184E	7:0	Default: 0x00	Access: R/W
	CMD_DATA[39:32]	7:0	See description of '11184Ah.'	
27h (11184Fh)	REG11184F	7:0	Default: 0x00	Access: R/W
	CMD_DATA[47:40]	7:0	See description of '11184Ah.'	
28h (111850h)	REG111850	7:0	Default: 0x00	Access: R/W
	CMD_DATA[55:48]	7:0	See description of '11184Ah.'	
28h (111851h)	REG111851	7:0	Default: 0x00	Access: R/W
	CMD_DATA[63:56]	7:0	See description of '11184Ah.'	
29h (111852h)	REG111852	7:0	Default: 0x00	Access: R/W
	-	7:4	Reserved.	
	CMD_LEN[3:0]	3:0	Transfer command register length (0 – 8).	
2Ah (111854h)	REG111854	7:0	Default: 0x00	Access: R/W
	DATA_LEN[7:0]	7:0	Transfer command register length.	
2Ah (111855h)	REG111855	7:0	Default: 0x00	Access: R/W
	DATA_LEN[15:8]	7:0	See description of '111854h.'	
2Bh (111856h)	REG111856	7:0	Default: 0x00	Access: R/W
	DATA_LEN[23:16]	7:0	See description of '111854h.'	
2Bh (111857h)	REG111857	7:0	Default: 0x00	Access: R/W
	DATA_LEN[31:24]	7:0	See description of '111854h.'	
2Ch (111858h)	REG111858	7:0	Default: 0x00	Access: RO
	DMA_TC[7:0]	7:0	DMA transfer count register for MIIC0 (debug only).	
2Ch (111859h)	REG111859	7:0	Default: 0x00	Access: RO
	DMA_TC[15:8]	7:0	See description of '111858h.'	
2Dh	REG11185A	7:0	Default: 0x00	Access: RO

MIIC0 Register (Bank = 1118)				
Index (Absolute)	Mnemonic	Bit	Description	
(11185Ah)	DMA_TC[23:16]	7:0	See description of '111858h.'	
2Dh (11185Bh)	REG11185B	7:0	Default: 0x00	Access: RO
	DMA_TC[31:24]	7:0	See description of '111858h.'	
2Eh (11185Ch)	REG11185C	7:0	Default: 0x00	Access: R/W
	SAR[7:0]	7:0	I2C Slave Address. [9:0]: 10-bit mode slave address. [6:0]: Normal mode slave address.	
2Eh (11185Dh)	REG11185D	7:0	Default: 0x00	Access: R/W
	-	7:3	Reserved.	
	10BIT_MODE	2	I2C Slave Address mode setting. 1: 10-bit mode slave address. 0: Normal mode slave address.	
	SAR[9:8]	1:0	See description of '11185Ch.'	
2Fh (11185Eh)	REG11185E	7:0	Default: 0x00	Access: R/W
	-	7:1	Reserved.	
	DMA_TRIGGER	0	DMA transfer trigger.	
2Fh (11185Fh)	REG11185F	7:0	Default: 0x00	Access: R/W
	-	7:1	Reserved.	
	RE_TRIGGER	0	Re-trigger DMA transfer for data transfer not completed.	
31h (111862h)	REG111862	7:0	Default: 0x00	Access: RO
	STATE[7:0]	7:0	DMA FSM (debug only).	
31h (111863h)	REG111863	7:0	Default: 0x00	Access: RO
	-	7:1	Reserved.	
	MIU_LAST_DONE_Z	0	MIU last done z (debug only).	

MIIC1 Register (Bank = 1119)

MIIC1 Register (Bank = 1119)				
Index (Absolute)	Mnemonic	Bit	Description	
00h (111900h)	REG111900	7:0	Default: 0x01	Access: R/W
	MIIC_CFG[7:0]	7:0	MIIC configuration register. Bit [7]: REG_ERROR_DET_EN. 0: Disable. 1: Enable. Bit [6]: REG_OEN_PUSH_EN. 0: Disable. 1: Enable. Bit [5]: Enable filter. 0: Disable. 1: Enable. Bit [4]: Enable timeout interrupt. 0: Disable. 1: Enable. Bit [3]: Enable clock stretching. 0: Disable. 1: Enable. Bit [2]: Interrupt enable. 0: Disable. 1: Enable. Bit [1]: Enable DMA. 0: Disable. 1: Enable. Bit [0]: Reset. 0: Not Reset. 1: Reset.	
01h (111902h)	REG111902	7:0	Default: 0x00	Access: WO
	-	7:1	Reserved.	
	CMD_START	0	MIIC command. [0]: Start.	
01h (111903h)	REG111903	7:0	Default: 0x00	Access: WO
	-	7:1	Reserved.	
	CMD_STOP	0	MIIC command. [1]: Stop.	
02h (111904h)	REG111904	7:0	Default: 0x00	Access: R/W
	WDATA[7:0]	7:0	I2C write data.	
02h	REG111905	7:0	Default: 0x00	Access: RO

MIIC1 Register (Bank = 1119)				
Index (Absolute)	Mnemonic	Bit	Description	
(111905h)	-	7:1	Reserved.	
	WRITE_ACK	0	I2C ACK for write data from slave IIC.	
03h (111906h)	REG111906	7:0	Default: 0x00	Access: RO
	RDATA[7:0]	7:0	I2C read data.	
03h (111907h)	REG111907	7:0	Default: 0x00	Access: R/W
	-	7:2	Reserved.	
	ACK_BIT	1	I2C ACK for read data to slave IIC.	
	RDATA_EN	0	I2C read data trigger.	
04h (111908h)	REG111908	7:0	Default: 0x00	Access: R/W
	-	7:1	Reserved.	
	FLAG	0	MIIC interrupt FLAG.	
05h (11190Ah)	REG11190A	7:0	Default: 0x00	Access: RO
	-	7:5	Reserved.	
	MIIC_STATE[4:0]	4:0	MIIC final state machine (debug only).	
05h (11190Bh)	REG11190B	7:0	Default: 0x00	Access: RO
	-	7	Reserved.	
	MIIC_INT_STATUS[6:0]	6:0	INT_STATUS. [0]: IC_START_DET_INTR. [1]: IC_STOP_DET_INTR. [2]: IC_RX_DONE_INTR. [3]: IC_TX_DONE_INTR. [4]: CLOCK_STRETCHING_INTR. [5]: SCL_ERROR_INTE. [6]: TIME_OUT_INTR.	
06h (11190Ch)	REG11190C	7:0	Default: 0x00	Access: RO
	-	7:5	Reserved.	
	SCLO	4	PAD_SCLO.	
	-	3:2	Reserved.	
	SDAI	1	PAD_SDAI.	
	SCLI	0	PAD_SCLI.	
07h (11190Eh)	REG11190E	7:0	Default: 0x00	Access: R/W
	-	7:2	Reserved.	
	FORCE_DRIVE_OEN	1	FORCE_DRIVE_OEN for pad.	
	PUSH_MULTI_EN	0	PUSH_MULTI enable.	

MIIC1 Register (Bank = 1119)				
Index (Absolute)	Mnemonic	Bit	Description	
07h (11190Fh)	REG11190F PUSH_COUNT_NUMBER[7:0]	7:0	Default: 0x00	Access: R/W
		7:0	Count number for push multi T.	
08h (111910h)	REG111910 STOP_CNT[7:0]	7:0	Default: 0x00	Access: R/W
		7:0	This register sets the SCL and SDA count for stop.	
08h (111911h)	REG111911 STOP_CNT[15:8]	7:0	Default: 0x00	Access: R/W
		7:0	See description of '111910h.'	
09h (111912h)	REG111912 HCNT[7:0]	7:0	Default: 0x00	Access: R/W
		7:0	This register sets the SCL clock high-period count.	
09h (111913h)	REG111913 HCNT[15:8]	7:0	Default: 0x00	Access: R/W
		7:0	See description of '111912h.'	
0Ah (111914h)	REG111914 LCNT[7:0]	7:0	Default: 0x00	Access: R/W
		7:0	This register sets the SCL clock low-period count.	
0Ah (111915h)	REG111915 LCNT[15:8]	7:0	Default: 0x00	Access: R/W
		7:0	See description of '111914h.'	
0Bh (111916h)	REG111916 SDA_CNT[7:0]	7:0	Default: 0x00	Access: R/W
		7:0	This register sets the clock count between the falling edges of SCL and SDA.	
0Bh (111917h)	REG111917 SDA_CNT[15:8]	7:0	Default: 0x00	Access: R/W
		7:0	See description of '111916h.'	
0Ch (111918h)	REG111918 START_CNT[7:0]	7:0	Default: 0x00	Access: R/W
		7:0	This register sets the SCL and SDA count for start.	
0Ch (111919h)	REG111919 START_CNT[15:8]	7:0	Default: 0x00	Access: R/W
		7:0	See description of '111918h.'	
0Dh (11191Ah)	REG11191A DATA_LAT_CNT[7:0]	7:0	Default: 0x00	Access: R/W
		7:0	This register sets the data latch timing.	
0Dh (11191Bh)	REG11191B DATA_LAT_CNT[15:8]	7:0	Default: 0x00	Access: R/W
		7:0	See description of '11191Ah.'	
0Eh (11191Ch)	REG11191C TIMEOUT_CNT[7:0]	7:0	Default: 0x00	Access: R/W
		7:0	This register sets timing delay if timeout interrupt occurs.	
0Eh (11191Dh)	REG11191D TIMEOUT_CNT[15:8]	7:0	Default: 0x00	Access: R/W
		7:0	See description of '11191Ch.'	
0Fh (11191Eh)	REG11191E -	7:3	Default: 0x00	Access: R/W
		7:3	Reserved.	

MIIC1 Register (Bank = 1119)				
Index (Absolute)	Mnemonic	Bit	Description	
	SCLI_DELAY[2:0]	2:0	Reserved.	
20h (111940h)	REG111940	7:0	Default: 0x1A	Access: RO, R/W
	-	7:6	Reserved.	
	MIU_NS	5	MIU secure bit.	
	MIU_PRIORITY	4	Set MIU priority.	
	MIU_RST	3	MIU software reset.	
	DMA_CFG[2:0]	2:0	DMA configuration register. Bit [2]: Interrupt enable. 0: Disable. 1: Enable. Bit [1]: DMA software reset. 0: Reset. 1: Not Reset.	
21h (111942h)	REG111942	7:0	Default: 0x00	Access: R/W
	MIU_ADDR[7:0]	7:0	Get TX data or put RX data address in DRAM.	
21h (111943h)	REG111943	7:0	Default: 0x00	Access: R/W
	MIU_ADDR[15:8]	7:0	See description of '111942h.'	
22h (111944h)	REG111944	7:0	Default: 0x00	Access: R/W
	MIU_ADDR[23:16]	7:0	See description of '111942h.'	
22h (111945h)	REG111945	7:0	Default: 0x00	Access: R/W
	MIU_ADDR[31:24]	7:0	See description of '111942h.'	
23h (111946h)	REG111946	7:0	Default: 0x00	Access: R/W
	MIU_SEL	7	MIIC channel select.	
	READ_CMD	6	MIIC transfer format. 1: Read. 0: Write.	
	STOP_DISABLE	5	MIIC transfer format. 1: S + data ... 0: S + data ... + P.	
	-	4:0	Reserved.	
24h (111948h)	REG111948	7:0	Default: 0x00	Access: R/W
	-	7:1	Reserved.	
	DMA_TRANSFER_DONE	0	SW needs to set this bit to clear TRANSFER_DONE flag or interrupt in order to receive the subsequent DMA TRANSFER_DONE flag or interrupt.	

MIIC1 Register (Bank = 1119)				
Index (Absolute)	Mnemonic	Bit	Description	
25h (11194Ah)	REG11194A	7:0	Default: 0x00	Access: R/W
	CMD_DATA[7:0]	7:0	I2C TX data buffer and command.	
25h (11194Bh)	REG11194B	7:0	Default: 0x00	Access: R/W
	CMD_DATA[15:8]	7:0	See description of '11194Ah.'	
26h (11194Ch)	REG11194C	7:0	Default: 0x00	Access: R/W
	CMD_DATA[23:16]	7:0	See description of '11194Ah.'	
26h (11194Dh)	REG11194D	7:0	Default: 0x00	Access: R/W
	CMD_DATA[31:24]	7:0	See description of '11194Ah.'	
27h (11194Eh)	REG11194E	7:0	Default: 0x00	Access: R/W
	CMD_DATA[39:32]	7:0	See description of '11194Ah.'	
27h (11194Fh)	REG11194F	7:0	Default: 0x00	Access: R/W
	CMD_DATA[47:40]	7:0	See description of '11194Ah.'	
28h (111950h)	REG111950	7:0	Default: 0x00	Access: R/W
	CMD_DATA[55:48]	7:0	See description of '11194Ah.'	
28h (111951h)	REG111951	7:0	Default: 0x00	Access: R/W
	CMD_DATA[63:56]	7:0	See description of '11194Ah.'	
29h (111952h)	REG111952	7:0	Default: 0x00	Access: R/W
	-	7:4	Reserved.	
	CMD_LEN[3:0]	3:0	Transfer command register length (0 – 8).	
2Ah (111954h)	REG111954	7:0	Default: 0x00	Access: R/W
	DATA_LEN[7:0]	7:0	Transfer command register length.	
2Ah (111955h)	REG111955	7:0	Default: 0x00	Access: R/W
	DATA_LEN[15:8]	7:0	See description of '111954h.'	
2Bh (111956h)	REG111956	7:0	Default: 0x00	Access: R/W
	DATA_LEN[23:16]	7:0	See description of '111954h.'	
2Bh (111957h)	REG111957	7:0	Default: 0x00	Access: R/W
	DATA_LEN[31:24]	7:0	See description of '111954h.'	
2Ch (111958h)	REG111958	7:0	Default: 0x00	Access: RO
	DMA_TC[7:0]	7:0	DMA transfer count register for MIIC0 (debug only).	
2Ch (111959h)	REG111959	7:0	Default: 0x00	Access: RO
	DMA_TC[15:8]	7:0	See description of '111958h.'	
2Dh (11195Ah)	REG11195A	7:0	Default: 0x00	Access: RO
	DMA_TC[23:16]	7:0	See description of '111958h.'	

MIIC1 Register (Bank = 1119)				
Index (Absolute)	Mnemonic	Bit	Description	
2Dh (11195Bh)	REG11195B	7:0	Default: 0x00	Access: RO
	DMA_TC[31:24]	7:0	See description of '111958h.'	
2Eh (11195Ch)	REG11195C	7:0	Default: 0x00	Access: R/W
	SAR[7:0]	7:0	I2C Slave Address. [9:0]: 10-bit mode slave address. [6:0]: Normal mode slave address.	
2Eh (11195Dh)	REG11195D	7:0	Default: 0x00	Access: R/W
	-	7:3	Reserved.	
	10BIT_MODE	2	I2C Slave Address mode setting. 1: 10-bit mode slave address. 0: Normal mode slave address.	
	SAR[9:8]	1:0	See description of '11195Ch.'	
2Fh (11195Eh)	REG11195E	7:0	Default: 0x00	Access: R/W
	-	7:1	Reserved.	
	DMA_TRIGGER	0	DMA transfer trigger.	
2Fh (11195Fh)	REG11195F	7:0	Default: 0x00	Access: R/W
	-	7:1	Reserved.	
	RE_TRIGGER	0	Re-trigger DMA transfer for data transfer not completed.	
31h (111962h)	REG111962	7:0	Default: 0x00	Access: RO
	STATE[7:0]	7:0	DMA FSM (debug only).	
31h (111963h)	REG111963	7:0	Default: 0x00	Access: RO
	-	7:1	Reserved.	
	MIU_LAST_DONE_Z	0	MIU last done z (debug only).	

SDIO0 Register (Bank = 1410)

SDIO0 Register (Bank = 1410)				
Index (Absolute)	Mnemonic	Bit	Description	
00h (141000h)	REG141000	7:0	Default: 0x00	Access: R/W
	CARD2_CHANGE	7	Card2 inserted or removed event.	
	CARD_CHANGE	6	Card inserted or removed event.	
	R2N_RDY_INT	5	R2N FIFO Ready interrupt event. REG_JOB_DIR = 1: Write to eMMC is completed (CIFD RFIFO is empty). REG_JOB_DIR = 0: read from eMMC is completed (CIFD WFIFO is full).	
	BUSYEND_INT	4	Busy end interrupt.	
	SDIO_INT	3	SDIO interrupt event.	
	ERR_STS	2	Error status event for REG_0x0D [5:0].	
	CMD_END	1	SD/MMC card command and response transaction complete event.	
	DMA_END	0	Data transaction complete event.	
00h (141001h)	REG141001	7:0	Default: 0x00	Access: R/W
	DUMMY_INT[7:0]	7:0	Dummy interrupt event.	
01h (141002h)	REG141002	7:0	Default: 0x00	Access: R/W
	CARD2_CHANGE_EN	7	Card2 inserted or removed event interrupt enable.	
	CARD_CHANGE_EN	6	Card inserted or removed event interrupt enable.	
	R2N_RDY_INT_EN	5	R2N_RDY_INT interrupt enable.	
	BUSYEND_INT_EN	4	BUSYEND_INT interrupt enable.	
	SDIO_INT_EN	3	SDIO_INT interrupt enable.	
	ERR_STS_EN	2	Error status event enable.	
	CMD_END_EN	1	CMD_END interrupt enable.	
	DMA_END_EN	0	DMA_END interrupt enable.	
01h (141003h)	REG141003	7:0	Default: 0x00	Access: R/W
	DUMMY_INT_EN[7:0]	7:0	Dummy interrupt enable.	
02h (141004h)	REG141004	7:0	Default: 0x00	Access: R/W
	-	7:6	Reserved.	
	MIU_BUS_EN[1:0]	5:4	MIU bus burst length. 0: Burst 1. 1: Burst 2. 2: Burst 4. 3: Burst 8.	

SDIO0 Register (Bank = 1410)				
Index (Absolute)	Mnemonic	Bit	Description	
	MIU_SELECT[1:0]	3:2	MIU0/MIU1/MIU2/MIU3 selection, default 0 = MIU0.	
	MIU_W_PRIORITY	1	MIU write request priority. 0: Low priority. 1: High priority.	
	MIU_R_PRIORITY	0	MIU read request priority. 0: Low priority. 1: High priority.	
03h (141006h)	REG141006	7:0	Default: 0x00	Access: R/W
	DMA_ADDR_15_0[7:0]	7:0	DMA Address [15:0].	
03h (141007h)	REG141007	7:0	Default: 0x00	Access: R/W
	DMA_ADDR_15_0[15:8]	7:0	See description of '141006h.'	
04h (141008h)	REG141008	7:0	Default: 0x00	Access: R/W
	DMA_ADDR_31_16[7:0]	7:0	DMA Address [31:16].	
04h (141009h)	REG141009	7:0	Default: 0x00	Access: R/W
	DMA_ADDR_31_16[15:8]	7:0	See description of '141008h.'	
05h (14100Ah)	REG14100A	7:0	Default: 0x00	Access: R/W
	DMA_LEN_15_0[7:0]	7:0	DMA Length [15:0].	
05h (14100Bh)	REG14100B	7:0	Default: 0x00	Access: R/W
	DMA_LEN_15_0[15:8]	7:0	See description of '14100Ah.'	
06h (14100Ch)	REG14100C	7:0	Default: 0x00	Access: R/W
	DMA_LEN_31_16[7:0]	7:0	DMA Length [31:16].	
06h (14100Dh)	REG14100D	7:0	Default: 0x00	Access: R/W
	DMA_LEN_31_16[15:8]	7:0	See description of '14100Ch.'	
07h (14100Eh)	REG14100E	7:0	Default: 0x00	Access: R/W
	FUNC_DUMMY[4:0]	7:3	Dummy function enable.	
	SDIO_MOD	2	SDIO card function enable.	
	SD_EN	1	SD card function enable.	
	EMMC_EN	0	MMC function enable.	
07h (14100Fh)	REG14100F	7:0	Default: 0x00	Access: R/W
	FUNC_DUMMY[12:5]	7:0	See description of '14100Eh.'	
08h (141010h)	REG141010	7:0	Default: 0x00	Access: R/W
	BLK_CNT[7:0]	7:0	Total block count for this job.	
08h (141011h)	REG141011	7:0	Default: 0x00	Access: R/W
	BLK_CNT[15:8]	7:0	See description of '141010h.'	

SDIO0 Register (Bank = 1410)				
Index (Absolute)	Mnemonic	Bit	Description	
09h (141012h)	REG141012	7:0	Default: 0x00	Access: R/W
	BLK_SIZE[7:0]	7:0	Block Size (0 – 32768 bytes).	
09h (141013h)	REG141013	7:0	Default: 0x00	Access: R/W
	BLK_SIZE[15:8]	7:0	See description of '141012h.'	
0Ah (141014h)	REG141014	7:0	Default: 0x00	Access: R/W
	RSP_SIZE_DUMMY[1:0]	7:6	Response size dummy.	
	RSP_SIZE[5:0]	5:0	Expected response size (byte count) for SD/MMC card.	
0Ah (141015h)	REG141015	7:0	Default: 0x00	Access: R/W
	CMD_SIZE_DUMMY[1:0]	7:6	Command size dummy.	
	CMD_SIZE[5:0]	5:0	Command transfer size (byte count) for SD/MMC.	
0Bh (141016h)	REG141016	7:0	Default: 0x00	Access: R/W
	DMA_RD_CLK_STOP	7	SD reads DMA stop clock when DMA ends.	
	-	6	Reserved.	
	DATSYNC	5	Synchronize data bus, for SD1.1 specification.	
	DEST	4	SD/MMC data transfer destination. 0: DMA mode. 1: R2N mode.	
	-	3	Reserved.	
	BUSWIDTH[1:0]	2:1	00: 1-bit. 01: 4-bit. 10: 8-bit. 11: Reserved.	
	CLK_EN	0	SD/MMC output clock enable.	
0Bh (141017h)	REG141017	7:0	Default: 0x00	Access: R/W
	SD_MOD_DUMMY[6:0]	7:1	SD Mode dummy.	
	DIS_WR_BUSY_CHK	0	Disable write busy check after last block write.	
0Ch (141018h)	REG141018	7:0	Default: 0x00	Access: R/W
	CHK_CMD	7	Enable command checking.	
	JOB_START	6	Job start.	
	ADMA_EN	5	Advanced DMA mode enable.	
	JOB_DIR	4	SD/MMC data transfer direction. 0: Read from card. 1: Write to card.	
	DTRX_EN	3	SD/MMC data transmit/receive enable (job finish auto clear).	

SDIO0 Register (Bank = 1410)

Index (Absolute)	Mnemonic	Bit	Description
	CMD_EN	2	SD/MMC transmit command enable (job finish auto clear).
	RSP_EN	1	SD/MMC receive command response enable.
	RSPR2_EN	0	SD/MMC receive command response for R2 type.
0Ch (141019h)	REG141019	7:0	Default: 0x00 Access: R/W
	SD_CTL_DUMMY[5:0]	7:2	SD Control dummy.
	ERR_DET_ON	1	Enable error detection for REG_ERR_STS interrupt.
	BUSY_DET_ON	0	Enable D0 high detection.
0Dh (14101Ah)	REG14101A	7:0	Default: 0x00 Access: RO, R/W
	-	7	Reserved.
	CARD_BUSY	6	SD/MMC card busy status, 1 indicates SD card busy.
	DAT_RD_TOUT	5	Received data phase timeout event (Wait start-bit timeout).
	CMDRSP_CERR	4	Received command phase: Response CRC error event.
	CMD_NORSP	3	Transmitted command phase: Response timeout event (time out = 64 clocks), which means there is no response on CMD line.
	DAT_WR_TOUT	2	Transmitted data phase timeout event.
	DAT_WR_CERR	1	Transmitted data phase: "CRC status = error" from SD/MMC card, which means SD/MMC card has encountered a flash program error.
	DAT_RD_CERR	0	Received data phase CRC error event.
0Dh (14101Bh)	REG14101B	7:0	Default: 0x00 Access: RO
	DATALINE[7:0]	7:0	SD/MMC DATA Line 7 – 0.
0Eh (14101Ch)	REG14101C	7:0	Default: 0x00 Access: R/W
	BOOT_DUMMY[4:0]	7:3	Reserved for eMMC boot mode.
	BOOT_MODE	2	Enable boot mode.
	EMMC_RSTZ_EN	1	EMMC Reset function enable.
	EMMC_RSTZ	0	EMMC Reset value.
0Eh (14101Dh)	REG14101D	7:0	Default: 0x00 Access: R/W
	BOOT_DUMMY[12:5]	7:0	See description of '14101Ch.'
0Fh (14101Eh)	REG14101E	7:0	Default: 0x00 Access: R/W
	MACRO_EN	7	Macro enable.

SDIO0 Register (Bank = 1410)

Index (Absolute)	Mnemonic	Bit	Description
	DDR_MOD_DUMMY[2:0]	6:4	DDR mode dummy register.
	PRE_FULL_SEL[1:0]	3:2	Adjust the pre-full signal for R2N mode.
	PAD_IN_RDY_SEL	1	Enable pad in RDY selection.
	PAD_IN_BYPASS	0	Bypass the pad input (without latching 1-T).
0Fh (14101Fh)	REG14101F	7:0	Default: 0x00 Access: R/W
	PAD_IN_MASK	7	Mask for GPIO pad input (response, write CRC status, read data input).
	FALL_LATCH	6	Latch 1T with CLK_FCIE_N for pad-in path (FCIE_SDR_MACRO). 1: Enable. 0: Disable.
	PAD_IN_SEL	5	Latch 1T with feedback clock for pad-in path (FCIE_SDR_MACRO). 1: Enable. 0: Disable.
	MACRO32B_EN	4	32-bit macro enable.
	PAD_IN_SEL_IP	3	Select the clock source for pad-in signals of SD macro. 0: From feedback clock. 1: From IP/IP_N clock.
	PAD_CLK_SEL	2	Select feedback clock for SD macro.
	-	1	Reserved.
	DDR_EN	0	Enable DDR mode.
10h (141020h)	REG141020	7:0	Default: 0x00 Access: R/W
	DDR_TOGGLE_CNT[7:0]	7:0	DDR data toggle count for flash macro.
10h (141021h)	REG141021	7:0	Default: 0x00 Access: R/W
	DDR_TOGGLE_CNT[15:8]	7:0	See description of '141020h.'
11h (141022h)	REG141022	7:0	Default: 0x00 Access: R/W
	SDIO_INT_TUNE_CLR[0]	7	Tune timing for SDIO interrupt period clear.
	SDIO_INT_TUNE[2:0]	6:4	Tune timing for SDIO interrupt period set.
	SDIO_DET_INT_SRC	3	SDIO interrupt source selection in SDIO_INT_MOD = 0 (edge trigger) or 1 (level trigger).
	SDIO_INT_MOD_SW_EN	2	Select SDIO interrupt mode by SW (REG_SDIO_INT_MOD).
	SDIO_INT_MOD[1:0]	1:0	SDIO_INT_MOD = 00: Continuous interrupt detect. SDIO_INT_MOD = 01: CMD12 or IO Abort command interrupt detect.

SDIO0 Register (Bank = 1410)				
Index (Absolute)	Mnemonic	Bit	Description	
			SDIO_INT_MOD = 10: Single block read/write interrupt detect. SDIO_INT_MOD = 11: Multi-block read/write interrupt detect.	
11h (141023h)	REG141023	7:0	Default: 0x00	Access: R/W
	SDIO_INT_ASYNC_EN	7	Enable SDIO INT ASYNC mode.	
	SDIO_INT_TUNE_SW	6	Select SDIO INT tuning by SW.	
	SDIO_INT_STOP_DMA	5	Stop SDIO INT period when DMA is done.	
	SDIO_BLK_GAP_DIS	4	Disable SDIO interrupt detect during block gap.	
	SDIO_RDWAIT_EN	3	Enable read wait function.	
	-	2	Reserved.	
	SDIO_INT_TUNE_CLR[2:1]	1:0	See description of '141022h.'	
12h (141024h)	REG141024	7:0	Default: 0x00	Access: R/W
	RSP_SHIFT_DUMMY[2:0]	7:5	RSP shift count dummy.	
	RSP_SHIFT_SEL	4	Select RSP shift count tuning by HW or SW. 1: SW. 0: HW.	
	RSP_SHIFT_TUNE[3:0]	3:0	RSP shift counter.	
12h (141025h)	REG141025	7:0	Default: 0x00	Access: R/W
	RSP_SHIFT_DUMMY[10:3]	7:0	See description of '141024h.'	
13h (141026h)	REG141026	7:0	Default: 0x00	Access: R/W
	RSTOP_SHIFT_DUMMY[2:0]	7:5	Read stop clock shift count dummy.	
	RSTOP_SHIFT_SEL	4	Select read stop clock shift count tuning by HW or SW. 1: SW. 0: HW.	
	RSTOP_SHIFT_TUNE[3:0]	3:0	Read stop clock shift counter.	
13h (141027h)	REG141027	7:0	Default: 0x00	Access: R/W
	WRSTS_SHIFT_DUMMY[2:0]	7:5	CRC status shift count dummy.	
	WRSTS_SHIFT_SEL	4	Select CRC status shift count tuning by HW or SW. 1: SW. 0: HW.	
	WRSTS_SHIFT_TUNE[3:0]	3:0	CRC status shift counter.	
14h (141028h)	REG141028	7:0	Default: 0x00	Access: R/W
	ZDEC_DUMMY[5:0]	7:2	ZDEC dummy.	
	SD2ZDEC_PTR_CLR	1	Clear job count and script ID for ZDEC.	

SDIO0 Register (Bank = 1410)				
Index (Absolute)	Mnemonic	Bit	Description	
	ZDEC_EN	0	Enable ZDEC function.	
14h (141029h)	REG141029	7:0	Default: 0x00	Access: R/W
	ZDEC_DUMMY[13:6]	7:0	See description of '141028h.'	
15h (14102Ah)	REG14102A	7:0	Default: 0x00	Access: R/W
	TEST_DUMMY[2:0]	7:5	Test mode dummy.	
	BIST_MODE	4	Test clock FCIE status.	
	SD_DEBUG_MOD[2:0]	3:1	DEBUG_MOD [2:0] definition. 0: State Machine. 1: SD Control. 2: MMA Control. 3: SDIO Control. 4: DDR/HS200 Control. 5: Reserved. 6: Reserved. 7: Reserved.	
	SDDR_L	0	Firmware writes 1 to drive SD interface, data bus and command line low.	
15h (14102Bh)	REG14102B	7:0	Default: 0x00	Access: R/W
	TEST_DUMMY[10:3]	7:0	See description of '14102Ah.'	
16h (14102Ch)	REG14102C	7:0	Default: 0xFF	Access: R/W
	MIU_BANK_SIZE[7:0]	7:0	MMA Bank Size.	
16h (14102Dh)	REG14102D	7:0	Default: 0x03	Access: R/W
	-	7:2	Reserved.	
	MIU_BANK_SIZE[9:8]	1:0	See description of '14102Ch.'	
17h (14102Eh)	REG14102E	7:0	Default: 0x00	Access: R/W
	WR_SBIT_TIMER[7:0]	7:0	Start-bit timer for eMMC block write.	
17h (14102Fh)	REG14102F	7:0	Default: 0x00	Access: R/W
	WR_SBIT_TIMER_EN	7	Enable write start-bit timeout detect.	
	WR_SBIT_TIMER[14:8]	6:0	See description of '14102Eh.'	
18h (141030h)	REG141030	7:0	Default: 0x00	Access: R/W
	RD_SBIT_TIMER[7:0]	7:0	Start-bit timer for eMMC block read.	
18h (141031h)	REG141031	7:0	Default: 0x00	Access: R/W
	RD_SBIT_TIMER_EN	7	Enable start-bit timeout detect.	
	RD_SBIT_TIMER[14:8]	6:0	See description of '141030h.'	
20h	REG141040	7:0	Default: 0x00	Access: R/W

SDIO0 Register (Bank = 1410)

Index (Absolute)	Mnemonic	Bit	Description
(141040h)	CMD_IDX[7:0]	7:0	Command Index/Response Index.
20h (141041h)	REG141041 CMD_ARG0[7:0]	7:0 7:0	Default: 0x00 Command Argument0/Response0. Access: R/W
21h (141042h)	REG141042 CMD_ARG1[7:0]	7:0 7:0	Default: 0x00 Command Argument1/Response1. Access: R/W
21h (141043h)	REG141043 CMD_ARG2[7:0]	7:0 7:0	Default: 0x00 Command Argument2/Response2. Access: R/W
22h (141044h)	REG141044 CMD_ARG3[7:0]	7:0 7:0	Default: 0x00 Command Argument3/Response3. Access: R/W
22h (141045h)	REG141045 CMD_ARG4[7:0]	7:0 7:0	Default: 0x00 Command Argument4/Response4. Access: R/W
23h (141046h)	REG141046 CMD_ARG5[7:0]	7:0 7:0	Default: 0x00 Command Argument5/Response5. Access: R/W
23h (141047h)	REG141047 CMD_ARG6[7:0]	7:0 7:0	Default: 0x00 Command Argument6/Response6. Access: R/W
24h (141048h)	REG141048 CMD_ARG7[7:0]	7:0 7:0	Default: 0x00 Command Argument7/Response7. Access: R/W
24h (141049h)	REG141049 CMD_ARG8[7:0]	7:0 7:0	Default: 0x00 Command Argument8/Response8. Access: R/W
25h (14104Ah)	REG14104A CMD_ARG9[7:0]	7:0 7:0	Default: 0x00 Command Argument9/Response9. Access: R/W
25h (14104Bh)	REG14104B CMD_ARG10[7:0]	7:0 7:0	Default: 0x00 Command Argument10/Response10. Access: R/W
26h (14104Ch)	REG14104C CMD_ARG11[7:0]	7:0 7:0	Default: 0x00 Command Argument11/Response11. Access: R/W
26h (14104Dh)	REG14104D CMD_ARG12[7:0]	7:0 7:0	Default: 0x00 Command Argument12/Response12. Access: R/W
27h (14104Eh)	REG14104E CMD_ARG13[7:0]	7:0 7:0	Default: 0x00 Command Argument13/Response13. Access: R/W
27h (14104Fh)	REG14104F CMD_ARG14[7:0]	7:0 7:0	Default: 0x00 Command Argument14/Response14. Access: R/W
28h (141050h)	REG141050 CMD_ARG15[7:0]	7:0 7:0	Default: 0x00 Command Argument15/Response15. Access: R/W

SDIO0 Register (Bank = 1410)				
Index (Absolute)	Mnemonic	Bit	Description	
28h (141051h)	REG141051	7:0	Default: 0x00	Access: R/W
	CMD_ARG16[7:0]	7:0	Command Argument16/Response16.	
2Fh (14105Eh)	REG14105E	7:0	Default: 0x00	Access: R/W
	-	7:1	Reserved.	
	SDIO_DET_ON	0	SDIO Interrupt detect on enable.	
30h (141060h)	REG141060	7:0	Default: 0x00	Access: RO, R/W
	-	7:4	Reserved.	
	RBUF_EMPTY	3	CIFD read buffer empty.	
	RBUF_FULL_TRIG	2	Trigger CIFD read buffer full.	
	WBUF_EMPTY_TRIG	1	Trigger CIFD write buffer empty.	
	WBUF_FULL	0	CIFD write buffer full.	
31h (141062h)	REG141062	7:0	Default: 0x00	Access: R/W
	-	7:2	Reserved.	
	RBUF_EMPTY_INT_EN	1	RBUF_EMPTY interrupt enable.	
31h (141063h)	REG141063	7:0	Default: 0x00	Access: R/W
	-	7:2	Reserved.	
	F_RBUF_EMPTY_INT	1	Force RBUF_EMPTY interrupt active, for test.	
34h (141068h)	REG141068	7:0	Default: 0xFF	Access: R/W
	SD_POWER_RD_MASK[7:0]	7:0	Power save mode, read data mask bits. 0: Mask. 1: Valid.	
34h (141069h)	REG141069	7:0	Default: 0xFF	Access: R/W
	SD_POWER_RD_MASK[15:8]	7:0	See description of '141068h.'	
35h (14106Ah)	REG14106A	7:0	Default: 0x08	Access: RO, R/W
	BAT_SAVE_EVENT	7	Power Saving Mode when Battery lost event occurs, can be cleared by REG_SD_POWER_SAVE_RST = 0.	
	RST_SAVE_EVENT	6	Power Saving Mode when Reset event occurs, can be cleared by REG_SD_POWER_SAVE_RST = 0.	
	RIU_SAVE_EVENT	5	Power Saving Mode when RIU emulation event occurs, can be cleared by REG_SD_POWER_SAVE_RST = 0.	
	POWER_SAVE_INT_FORCE	4	Force Power Save interrupt active, for test.	
	SD_POWER_SAVE_RST	3	Software reset Power Save HW, default is 1, set 0 to reset	

SDIO0 Register (Bank = 1410)

Index (Absolute)	Mnemonic	Bit	Description
			HW.
	POWER_SAVE_MODE_INT_EN	2	Power Save interrupt enable, active high.
	SD_POWER_SAVE_RIU	1	SW sets register to emulate power lost event, active high.
	POWER_SAVE_MODE	0	Power Save HW enable, active high.
35h (14106Bh)	REG14106B	7:0	Default: 0x00 Access: R/W
	POWER_SAVE_MODE_INT	7	Power Save interrupt event.
	-	6:2	Reserved.
	RST_SD_POWER_SAVE_MASK	1	Mask RST_SD_POWER_SAVE signal.
	BAT_SD_POWER_SAVE_MASK	0	Mask BAT_SD_POWER_SAVE signal.
36h (14106Ch)	REG14106C	7:0	Default: 0x00 Access: RO
	-	7:1	Reserved.
	BIST_ERR_FLAG	0	BIST mode status.
36h (14106Dh)	REG14106D	7:0	Default: 0x00 Access: R/W
	SEL_BIST_PAT[3:0]	7:4	Select BIST pattern format. 0: 32'h00000000. 1: 32'hffffff. 2: 32'hff00ff00. 3: 32'h00ff00ff. 4: 32'h55555555. 5: 32'haaaaaaaa. 6: 32'h55aa55aa. 7: 32'h55aa55aa. 8: 32'ha55aa55a. 9: 32'h5aa55aa5. 10: Increasing. 11: Decreasing. 12 – 15: Reserved.
	-	3:0	Reserved.
37h (14106Eh)	REG14106E	7:0	Default: 0x00 Access: R/W
	-	7:3	Reserved.
	IMI_SEL	2	Select EMI or IMI. 1: IMI. 0: EMI.
	BOOTSRAM_ACCESS_SEL	1	Select the owner to 12K boot SRAM access when NAND boot is enabled. 1: FCIE4.

SDIO0 Register (Bank = 1410)				
Index (Absolute)	Mnemonic	Bit	Description	
			0: CA-9.	
	NAND_BOOT_EN	0	Enable FCIE Boot operation.	
38h (141070h)	REG141070	7:0	Default: 0x00	Access: RO
	FCIE_DBUS_15_0[7:0]	7:0	Debug bus [15:0].	
38h (141071h)	REG141071	7:0	Default: 0x00	Access: RO
	FCIE_DBUS_15_0[15:8]	7:0	See description of '141070h.'	
39h (141072h)	REG141072	7:0	Default: 0x00	Access: RO
	FCIE_DBUS_23_16[7:0]	7:0	Debug bus [23:16].	
39h (141073h)	REG141073	7:0	Default: 0x00	Access: RO, R/W
	SRAM_SD_EN	7	SRAM power down enable.	
	-	6	Reserved.	
	MMA1_BISTFAIL	5	SRAM mma1 512-byte BIST Test Fail.	
	MMA0_BISTFAIL	4	SRAM mma0 512-byte BIST Test Fail.	
	DEBUG_MOD[3:0]	3:0	DEBUG_MOD selection.	
3Eh (14107Ch)	REG14107C	7:0	Default: 0x01	Access: RO, R/W
	-	7:6	Reserved.	
	TEST_ECC_STS	5	Status of ECC clock test.	
	TEST_MCU_STS	4	Status of MCU clock test.	
	TEST_MIE_STS	3	Status of FCIE clock test.	
	TEST_MIU_STS	2	Status of MIU clock test.	
	TEST_CLK	1	Test clock source enable.	
	FCIE_CLK_EN	0	FCIE clock enable.	
3Fh (14107Eh)	REG14107E	7:0	Default: 0x01	Access: RO, R/W
	-	7:5	Reserved.	
	RST_ECC_STS	4	Software reset status of ECC clock domain.	
	RST_MCU_STS	3	Software reset status of MCU clock domain.	
	RST_MIE_STS	2	Software reset status of MIE clock domain.	
	RST_MIU_STS	1	Software reset status of MIU clock domain.	
	FCIE_SOFT_RST	0	FCIE module software reset, active low.	

BACH0 Register (Bank = 1502)

BACH0 Register (Bank = 1502)				
Index (Absolute)	Mnemonic	Bit	Description	
40h (150280h)	REG150280	7:0	Default: 0x00	Access: R/W
	DMA1_CTRL_0[7:0]	7:0	0x80 DMA1_CTRL_0. [15] REG_WR_UNDERRUN_INT_EN. DMA write underrun threshold interrupt enable. 0 = disable. 1 = enable. [14] REG_WR_OVERRUN_INT_EN. DMA write overrun threshold interrupt enable. 0 = disable. 1 = enable. [13] REG_RD_UNDERRUN_INT_EN. DMA read underrun threshold interrupt enable. 0 = disable. 1 = enable. [12] REG_RD_OVERRUN_INT_EN. DMA read overrun threshold interrupt enable. 0 = disable. 1 = enable. [11] REG_WR_FULL_INT_EN. DMA write full interrupt enable. 0 = disable. 1 = enable. [10] REG_RD_EMPTY_INT_EN. DMA read empty interrupt enable. 0 = disable. 1 = enable. [9] REG_WR_FULL_FLAG_CLR / REG_WR_LOCALBUF_FULL_CLR. DMA write full flag clear / DMA write local buffer full flag clear. 0 = normal. 1 = clear. [8] REG_RD_EMPTY_FLAG_CLR / REG_RD_LOCALBUF_EMPTY_CLR. DMA read full flag clear / DMA read local buffer full flag clear. 0 = normal. 1 = clear. [6] REG_SEL_TES_BUS.	

BACH0 Register (Bank = 1502)

Index (Absolute)	Mnemonic	Bit	Description
			DMA test bus selection. 0: Select flag. 1: Select test bus. [5] REG_RD_LR_SWAP_EN. DMA read left/right channel swap enable. 0 = normal. 1 = swap. [4] REG_PRIORITY_KEEP_HIGH. DMA read priority keep high. 0 = normal. 1 = keep high priority. [3] REG_RD_BYTE_SWAP_EN. DMA read byte swap enable. 0 = normal. 1 = swap. [2] REG_RD_LEVEL_CNT_LIVE_MASK. DMA read level counter live mask. 0 = level counter free run mode. 1 = level counter will stop while buffer is empty. [1] REG_ENABLE. DMA enable. 0 = disable. 1 = enable. [0] REG_SW_RST_DMA. DMA software reset. 0 = normal. 1 = reset.
40h (150281h)	REG150281	7:0	Default: 0x00 Access: R/W
	DMA1_CTRL_0[15:8]	7:0	See description of '150280h.'
41h (150282h)	REG150282	7:0	Default: 0x00 Access: R/W
	DMA1_CTRL_1[7:0]	7:0	0x82 DMA1_CTRL_1. [15] REG_RD_ENABLE. DMA read enable. 0 = disable. 1 = enable. [14] REG_RD_INIT. DMA read initialization. 0 = normal. 1 = initialization. [13] REG_RD_TRIG.

BACH0 Register (Bank = 1502)				
Index (Absolute)	Mnemonic	Bit	Description	
			DMA read trigger. 0 = normal. 1 = trigger. [12] REG_RD_LEVEL_CNT_MASK. DMA read level counter mask. 0 = normal. 1 = mask. [11:0] REG_RD_BASE_ADDR [11:0]. DMA read base address [11:0].	
41h (150283h)	REG150283	7:0	Default: 0x00	Access: R/W
	DMA1_CTRL_1[15:8]	7:0	See description of '150282h.'	
42h (150284h)	REG150284	7:0	Default: 0x00	Access: R/W
	DMA1_CTRL_2[7:0]	7:0	0x84 DMA1_CTRL_2. [15] Reserved. [14:0] REG_RD_BASE_ADDR [14:0]. DMA read base address [26:12].	
42h (150285h)	REG150285	7:0	Default: 0x00	Access: R/W
	DMA1_CTRL_2[15:8]	7:0	See description of '150284h.'	
43h (150286h)	REG150286	7:0	Default: 0x00	Access: R/W
	DMA1_CTRL_3[7:0]	7:0	0x86 DMA1_CTRL_3. [15:0] REG_RD_BUFF_SIZE. DMA read buffer size.	
43h (150287h)	REG150287	7:0	Default: 0x00	Access: R/W
	DMA1_CTRL_3[15:8]	7:0	See description of '150286h.'	
44h (150288h)	REG150288	7:0	Default: 0x00	Access: R/W
	DMA1_CTRL_4[7:0]	7:0	0x88 DMA1_CTRL_4. [15:0] REG_RD_SIZE. DMA read size.	
44h (150289h)	REG150289	7:0	Default: 0x00	Access: R/W
	DMA1_CTRL_4[15:8]	7:0	See description of '150288h.'	
45h (15028Ah)	REG15028A	7:0	Default: 0x00	Access: R/W
	DMA1_CTRL_5[7:0]	7:0	0x8A DMA1_CTRL_5. [15:0] REG_RD_OVERRUN_TH. DMA read overrun threshold.	
45h (15028Bh)	REG15028B	7:0	Default: 0x00	Access: R/W
	DMA1_CTRL_5[15:8]	7:0	See description of '15028Ah.'	

BACH0 Register (Bank = 1502)				
Index (Absolute)	Mnemonic	Bit	Description	
46h (15028Ch)	REG15028C	7:0	Default: 0x00	Access: R/W
	DMA1_CTRL_6[7:0]	7:0	0x8C DMA1_CTRL_6. [15:0] REG_RD_UNDERRUN_TH. DMA read underrun threshold.	
46h (15028Dh)	REG15028D	7:0	Default: 0x00	Access: R/W
	DMA1_CTRL_6[15:8]	7:0	See description of '15028Ch.'	
47h (15028Eh)	REG15028E	7:0	Default: 0x00	Access: RO
	DMA1_CTRL_7[7:0]	7:0	0x8E DMA1_CTRL_7. [15:0] REG_RD_LEVEL_CNT. DMA read level counter.	
47h (15028Fh)	REG15028F	7:0	Default: 0x00	Access: RO
	DMA1_CTRL_7[15:8]	7:0	See description of '15028Eh.'	
48h (150290h)	REG150290	7:0	Default: 0x00	Access: RO
	DMA1_CTRL_8[7:0]	7:0	0x90 DMA1_CTRL_8. [7] REG_RD_LOCALBUF_EMPTY. DMA read local buffer empty flag. [6] REG_WR_LOCALBUF_FULL. DMA write local buffer full flag. [5] REG_WR_FULL_FLAG. DMA write full flag. [4] REG_RD_EMPTY_FLAG. DMA read empty flag. [3] REG_RD_OVERRUN_FLAG. DMA read overrun flag. [2] REG_RD_UNDERRUN_FLAG. DMA read underrun flag. [1] REG_WR_OVERRUN_FLAG. DMA write overrun flag. [0] REG_WR_UNDERRUN_FLAG. DMA write underrun flag.	
48h (150291h)	REG150291	7:0	Default: 0x00	Access: RO
	DMA1_CTRL_8[15:8]	7:0	See description of '150290h.'	
49h (150292h)	REG150292	7:0	Default: 0x00	Access: R/W
	DMA1_CTRL_9[7:0]	7:0	0x92 DMA1_CTRL_9. [15] REG_WR_ENABLE. DMA write enable. 0 = disable. 1 = enable.	

BACH0 Register (Bank = 1502)

Index (Absolute)	Mnemonic	Bit	Description
			[14] REG_WR_INIT. DMA write initialization. 0 = normal. 1 = initialization. [13] REG_WR_TRIG. DMA write trigger. 0 = normal. 1 = trigger. [12] REG_WR_LEVEL_CNT_MASK. DMA write level counter mask. 0 = normal. 1 = mask. [11:0] REG_WR_BASE_ADDR [11:0]. DMA write base address [11:0].
49h (150293h)	REG150293	7:0	Default: 0x00 Access: R/W
	DMA1_CTRL_9[15:8]	7:0	See description of '150292h.'
4Ah (150294h)	REG150294	7:0	Default: 0x00 Access: R/W
	DMA1_CTRL_10[7:0]	7:0	0x94 DMA1_CTRL_10. [15] Reserved. [14:0] REG_WR_BASE_ADDR [14:0]. DMA write base address [26:12].
4Ah (150295h)	REG150295	7:0	Default: 0x00 Access: R/W
	DMA1_CTRL_10[15:8]	7:0	See description of '150294h.'
4Bh (150296h)	REG150296	7:0	Default: 0x00 Access: R/W
	DMA1_CTRL_11[7:0]	7:0	0x96 DMA1_CTRL_11. [15:0] REG_WR_BUFF_SIZE. DMA write buffer size.
4Bh (150297h)	REG150297	7:0	Default: 0x00 Access: R/W
	DMA1_CTRL_11[15:8]	7:0	See description of '150296h.'
4Ch (150298h)	REG150298	7:0	Default: 0x00 Access: R/W
	DMA1_CTRL_12[7:0]	7:0	0x98 DMA1_CTRL_12. [15:0] REG_WR_SIZE. DMA write size.
4Ch (150299h)	REG150299	7:0	Default: 0x00 Access: R/W
	DMA1_CTRL_12[15:8]	7:0	See description of '150298h.'
4Dh (15029Ah)	REG15029A	7:0	Default: 0x00 Access: R/W
	DMA1_CTRL_13[7:0]	7:0	0x9A DMA1_CTRL_13.

BACH0 Register (Bank = 1502)				
Index (Absolute)	Mnemonic	Bit	Description	
			[15:0] REG_WR_OVERRUN_TH. DMA write overrun threshold.	
4Dh (15029Bh)	REG15029B	7:0	Default: 0x00	Access: R/W
	DMA1_CTRL_13[15:8]	7:0	See description of '15029Ah.'	
4Eh (15029Ch)	REG15029C	7:0	Default: 0x00	Access: R/W
	DMA1_CTRL_14[7:0]	7:0	0x9C DMA1_CTRL_14. [15:0] REG_WR_UNDERRUN_TH. DMA write underrun threshold.	
4Eh (15029Dh)	REG15029D	7:0	Default: 0x00	Access: R/W
	DMA1_CTRL_14[15:8]	7:0	See description of '15029Ch.'	
4Fh (15029Eh)	REG15029E	7:0	Default: 0x00	Access: RO
	DMA1_CTRL_15[7:0]	7:0	0x9E DMA1_CTRL_15. [15:0] REG_WR_LEVEL_CNT. DMA write level counter.	
4Fh (15029Fh)	REG15029F	7:0	Default: 0x00	Access: RO
	DMA1_CTRL_15[15:8]	7:0	See description of '15029Eh.'	
50h (1502A0h)	REG1502A0	7:0	Default: 0x00	Access: R/W
	DMA2_CTRL_0[7:0]	7:0	0xA0 DMA2_CTRL_0. [15] REG_WR_UNDERRUN_INT_EN. DMA write underrun threshold interrupt enable. 0 = disable. 1 = enable. [14] REG_WR_OVERRUN_INT_EN. DMA write overrun threshold interrupt enable. 0 = disable. 1 = enable. [13] REG_RD_UNDERRUN_INT_EN. DMA read underrun threshold interrupt enable. 0 = disable. 1 = enable. [12] REG_RD_OVERRUN_INT_EN. DMA read overrun threshold interrupt enable. 0 = disable. 1 = enable. [11] REG_WR_FULL_INT_EN. DMA write full interrupt enable. 0 = disable. 1 = enable.	

BACH0 Register (Bank = 1502)			
Index (Absolute)	Mnemonic	Bit	Description
			[10] REG_RD_EMPTY_INT_EN. DMA read empty interrupt enable. 0 = disable. 1 = enable. [9] REG_WR_FULL_FLAG_CLR / REG_WR_LOCALBUF_FULL_CLR. DMA write full flag clear / DMA write local buffer full flag clear. 0 = normal. 1 = clear. [8] REG_RD_EMPTY_FLAG_CLR / REG_RD_LOCALBUF_EMPTY_CLR. DMA read full flag clear / DMA read local buffer full flag clear. 0 = normal. 1 = clear. [6] REG_SEL_TES_BUS. DMA test bus selection. 0: Select flag. 1: Select test bus. [5] REG_RD_LR_SWAP_EN. DMA read left/right channel swap enable. 0 = normal. 1 = swap. [4] REG_PRIORITY_KEEP_HIGH. DMA read priority keep high. 0 = normal. 1 = keep high priority. [3] REG_RD_BYTE_SWAP_EN. DMA read byte swap enable. 0 = normal. 1 = swap. [2] REG_RD_LEVEL_CNT_LIVE_MASK. DMA read level counter live mask. 0 = level counter free run mode. 1 = level counter will stop while buffer is empty. [1] REG_ENABLE. DMA enable. 0 = disable. 1 = enable. [0] REG_SW_RST_DMA.

BACH0 Register (Bank = 1502)				
Index (Absolute)	Mnemonic	Bit	Description	
			DMA software reset. 0 = normal. 1 = reset.	
50h (1502A1h)	REG1502A1	7:0	Default: 0x00	Access: R/W
	DMA2_CTRL_0[15:8]	7:0	See description of '1502A0h.'	
51h (1502A2h)	REG1502A2	7:0	Default: 0x00	Access: R/W
	DMA2_CTRL_1[7:0]	7:0	0xA2 DMA2_CTRL_1. [15] REG_RD_ENABLE. DMA read enable. 0 = disable. 1 = enable. [14] REG_RD_INIT. DMA read initialization. 0 = normal. 1 = initialization. [13] REG_RD_TRIG. DMA read trigger. 0 = normal. 1 = trigger. [12] REG_RD_LEVEL_CNT_MASK. DMA read level counter mask. 0 = normal. 1 = mask. [11:0] REG_RD_BASE_ADDR [11:0]. DMA read base address [11:0].	
51h (1502A3h)	REG1502A3	7:0	Default: 0x00	Access: R/W
	DMA2_CTRL_1[15:8]	7:0	See description of '1502A2h.'	
52h (1502A4h)	REG1502A4	7:0	Default: 0x00	Access: R/W
	DMA2_CTRL_2[7:0]	7:0	0xA4 DMA2_CTRL_2. [15] Reserved. [14:0] REG_RD_BASE_ADDR [14:0]. DMA read base address [26:12].	
52h (1502A5h)	REG1502A5	7:0	Default: 0x00	Access: R/W
	DMA2_CTRL_2[15:8]	7:0	See description of '1502A4h.'	
53h (1502A6h)	REG1502A6	7:0	Default: 0x00	Access: R/W
	DMA2_CTRL_3[7:0]	7:0	0xA6 DMA2_CTRL_3. [15:0] REG_RD_BUFF_SIZE. DMA read buffer size.	

BACH0 Register (Bank = 1502)				
Index (Absolute)	Mnemonic	Bit	Description	
53h (1502A7h)	REG1502A7	7:0	Default: 0x00	Access: R/W
	DMA2_CTRL_3[15:8]	7:0	See description of '1502A6h.'	
54h (1502A8h)	REG1502A8	7:0	Default: 0x00	Access: R/W
	DMA2_CTRL_4[7:0]	7:0	0xA8 DMA2_CTRL_4. [15:0] REG_RD_SIZE. DMA read size.	
54h (1502A9h)	REG1502A9	7:0	Default: 0x00	Access: R/W
	DMA2_CTRL_4[15:8]	7:0	See description of '1502A8h.'	
55h (1502AAh)	REG1502AA	7:0	Default: 0x00	Access: R/W
	DMA2_CTRL_5[7:0]	7:0	0xAA DMA2_CTRL_5. [15:0] REG_RD_OVERRUN_TH. DMA read overrun threshold.	
55h (1502ABh)	REG1502AB	7:0	Default: 0x00	Access: R/W
	DMA2_CTRL_5[15:8]	7:0	See description of '1502AAh.'	
56h (1502ACh)	REG1502AC	7:0	Default: 0x00	Access: R/W
	DMA2_CTRL_6[7:0]	7:0	0xAC DMA2_CTRL_6. [15:0] REG_RD_UNDERRUN_TH. DMA read underrun threshold.	
56h (1502ADh)	REG1502AD	7:0	Default: 0x00	Access: R/W
	DMA2_CTRL_6[15:8]	7:0	See description of '1502ACh.'	
57h (1502AEh)	REG1502AE	7:0	Default: 0x00	Access: RO
	DMA2_CTRL_7[7:0]	7:0	0xAE DMA2_CTRL_7. [15:0] REG_RD_LEVEL_CNT. DMA read level counter.	
57h (1502AFh)	REG1502AF	7:0	Default: 0x00	Access: RO
	DMA2_CTRL_7[15:8]	7:0	See description of '1502AEh.'	
58h (1502B0h)	REG1502B0	7:0	Default: 0x00	Access: RO
	DMA2_CTRL_8[7:0]	7:0	0xB0 DMA2_CTRL_8. [7] REG_RD_LOCALBUF_EMPTY. DMA read local buffer empty flag. [6] REG_WR_LOCALBUF_FULL. DMA write local buffer full flag. [5] REG_WR_FULL_FLAG. DMA write full flag. [4] REG_RD_EMPTY_FLAG. DMA read empty flag.	

BACH0 Register (Bank = 1502)				
Index (Absolute)	Mnemonic	Bit	Description	
			[3] REG_RD_OVERRUN_FLAG. DMA read overrun flag. [2] REG_RD_UNDERRUN_FLAG. DMA read underrun flag. [1] REG_WR_OVERRUN_FLAG. DMA write overrun flag. [0] REG_WR_UNDERRUN_FLAG. DMA write underrun flag.	
58h (1502B1h)	REG1502B1	7:0	Default: 0x00	Access: RO
	DMA2_CTRL_8[15:8]	7:0	See description of '1502B0h.'	
59h (1502B2h)	REG1502B2	7:0	Default: 0x00	Access: R/W
	DMA2_CTRL_9[7:0]	7:0	0xB2 DMA2_CTRL_9. [15] REG_WR_ENABLE. DMA write enable. 0 = disable. 1 = enable. [14] REG_WR_INIT. DMA write initialization. 0 = normal. 1 = initialization. [13] REG_WR_TRIG. DMA write trigger. 0 = normal. 1 = trigger. [12] REG_WR_LEVEL_CNT_MASK. DMA write level counter mask. 0 = normal. 1 = mask. [11:0] REG_WR_BASE_ADDR [11:0]. DMA write base address [11:0].	
59h (1502B3h)	REG1502B3	7:0	Default: 0x00	Access: R/W
	DMA2_CTRL_9[15:8]	7:0	See description of '1502B2h.'	
5Ah (1502B4h)	REG1502B4	7:0	Default: 0x00	Access: R/W
	DMA2_CTRL_10[7:0]	7:0	0xB4 DMA2_CTRL_10. [15] Reserved. [14:0] REG_WR_BASE_ADDR [14:0]. DMA write base address [24:12].	
5Ah	REG1502B5	7:0	Default: 0x00	Access: R/W

BACH0 Register (Bank = 1502)				
Index (Absolute)	Mnemonic	Bit	Description	
(1502B5h)	DMA2_CTRL_10[15:8]	7:0	See description of '1502B4h.'	
5Bh (1502B6h)	REG1502B6	7:0	Default: 0x00	Access: R/W
	DMA2_CTRL_11[7:0]	7:0	0xB6 DMA2_CTRL_11. [15:0] REG_WR_BUFF_SIZE. DMA write buffer size.	
5Bh (1502B7h)	REG1502B7	7:0	Default: 0x00	Access: R/W
	DMA2_CTRL_11[15:8]	7:0	See description of '1502B6h.'	
5Ch (1502B8h)	REG1502B8	7:0	Default: 0x00	Access: R/W
	DMA2_CTRL_12[7:0]	7:0	0xB8 DMA2_CTRL_12. [15:0] REG_WR_SIZE. DMA write size.	
5Ch (1502B9h)	REG1502B9	7:0	Default: 0x00	Access: R/W
	DMA2_CTRL_12[15:8]	7:0	See description of '1502B8h.'	
5Dh (1502BAh)	REG1502BA	7:0	Default: 0x00	Access: R/W
	DMA2_CTRL_13[7:0]	7:0	0xBA DMA2_CTRL_13. [15:0] REG_WR_OVERRUN_TH. DMA write overrun threshold.	
5Dh (1502BBh)	REG1502BB	7:0	Default: 0x00	Access: R/W
	DMA2_CTRL_13[15:8]	7:0	See description of '1502BAh.'	
5Eh (1502BCh)	REG1502BC	7:0	Default: 0x00	Access: R/W
	DMA2_CTRL_14[7:0]	7:0	0xBC DMA2_CTRL_14. [15:0] REG_WR_UNDERRUN_TH. DMA write underrun threshold.	
5Eh (1502BDh)	REG1502BD	7:0	Default: 0x00	Access: R/W
	DMA2_CTRL_14[15:8]	7:0	See description of '1502BCh.'	
5Fh (1502BEh)	REG1502BE	7:0	Default: 0x00	Access: RO
	DMA2_CTRL_15[7:0]	7:0	0xBE DMA2_CTRL_15. [15:0] REG_WR_LEVEL_CNT. DMA write level counter.	
5Fh (1502BFh)	REG1502BF	7:0	Default: 0x00	Access: RO
	DMA2_CTRL_15[15:8]	7:0	See description of '1502BEh.'	
70h (1502E0h)	REG1502E0	7:0	Default: 0x00	Access: R/W
	DMA_TEST_CTRL0[7:0]	7:0	0xE0 DMA_TEST_CTRL0. [0] DMA_TEST_WR_EN. DMA test write enable. [1] DMA_TEST_WR.	

BACH0 Register (Bank = 1502)				
Index (Absolute)	Mnemonic	Bit	Description	
			DMA test write. [3:2] DMA_TEST_WR_SEL. DMA test write selection. [4] DMA_TEST_RD_EN. DMA test read enable. [5] DMA_TEST_RD. DMA test read. [7:6] DMA_TEST_RD_SEL. DMA test read selection.	
70h (1502E1h)	REG1502E1	7:0	Default: 0x00	Access: R/W
	DMA_TEST_CTRL0[15:8]	7:0	See description of '1502E0h.'	
71h (1502E2h)	REG1502E2	7:0	Default: 0x00	Access: R/W
	DMA_TEST_CTRL1[7:0]	7:0	0xE2 DMA_TEST_CTRL1. [15:0] DMA_WR_DATA_L_MUX. DMA write data left channel value.	
71h (1502E3h)	REG1502E3	7:0	Default: 0x00	Access: R/W
	DMA_TEST_CTRL1[15:8]	7:0	See description of '1502E2h.'	
72h (1502E4h)	REG1502E4	7:0	Default: 0x00	Access: R/W
	DMA_TEST_CTRL2[7:0]	7:0	0xE4 DMA_TEST_CTRL2. [15:0] DMA_WR_DATA_R_MUX. DMA write data right channel value.	
72h (1502E5h)	REG1502E5	7:0	Default: 0x00	Access: R/W
	DMA_TEST_CTRL2[15:8]	7:0	See description of '1502E4h.'	
73h (1502E6h)	REG1502E6	7:0	Default: 0x00	Access: RO
	DMA_TEST_CTRL3[7:0]	7:0	0xE6 DMA_TEST_CTRL3. [15:0] DMA_RD_DATA_L. DMA read data left channel.	
73h (1502E7h)	REG1502E7	7:0	Default: 0x00	Access: RO
	DMA_TEST_CTRL3[15:8]	7:0	See description of '1502E6h.'	
74h (1502E8h)	REG1502E8	7:0	Default: 0x00	Access: RO
	DMA_TEST_CTRL4[7:0]	7:0	0xE8 DMA_TEST_CTRL4. [15:0] DMA_RD_DATA_R. DMA read data right channel.	
74h (1502E9h)	REG1502E9	7:0	Default: 0x00	Access: RO
	DMA_TEST_CTRL4[15:8]	7:0	See description of '1502E8h.'	
75h	REG1502EA	7:0	Default: 0x00	Access: R/W

BACH0 Register (Bank = 1502)

Index (Absolute)	Mnemonic	Bit	Description
(1502EAh)	DMA_TEST_CTRL5[7:0]	7:0	0xEA DMA_TEST_CTRL5. DMA1 sine generator setting. [15] REG_SINE_GEN_EN. Sine generator enable. [14] REG_SINE_GEN_L. [13] REG_SINE_GEN_R. Sine generator left/right channel enable. [12] REG_SINE_GEN_RD_WR. Sine generator read/write select. 0 = read. 1 = write. [11] REG_DMA_TEST_SEL_H. DMA test bus low/high word select. 0 = low word. 1 = high word. [10:8] REG_DMA_TEST_SEL. DMA test bus selection. [3:0] REG_SINE_GEN_FREQ. Sine generator frequency selection. [7:4] REG_SINE_GEN_GAIN. Sine generator gain selection.
75h (1502EBh)	REG1502EB	7:0	Default: 0x00 Access: R/W
	DMA_TEST_CTRL5[15:8]	7:0	See description of '1502EAh.'
76h (1502ECh)	REG1502EC	7:0	Default: 0x00 Access: R/W
	DMA_TEST_CTRL6[7:0]	7:0	0xEC DMA_TEST_CTRL6. DMA2 sine generator setting. [15] REG_SINE_GEN_EN. Sine generator enable. [14] REG_SINE_GEN_L. [13] REG_SINE_GEN_R. Sine generator left/right channel enable. [12] REG_SINE_GEN_RD_WR. Sine generator read/write select. 0 = read. 1 = write. [11] REG_DMA_TEST_SEL_H. DMA test bus low/high word select. 0 = low word. 1 = high word. [10:8] REG_DMA_TEST_SEL.

BACH0 Register (Bank = 1502)				
Index (Absolute)	Mnemonic	Bit	Description	
			DMA test bus selection. [3:0] REG_SINE_GEN_FREQ. Sine generator frequency selection. [7:4] REG_SINE_GEN_GAIN. Sine generator gain selection.	
76h	REG1502ED	7:0	Default: 0x00	Access: R/W
(1502EDh)	DMA_TEST_CTRL6[15:8]	7:0	See description of '1502ECh.'	
77h	REG1502EE	7:0	Default: 0x00	Access: R/W
(1502EEh)	DMA_TEST_CTRL7[7:0]	7:0	0xEE DMA_TEST_CTRL7. [15] REG_DMA1_RD_MONO. 0 = disable. 1 = enable. [14] REG_DMA1_WR_MONO. 0 = disable. 1 = enable. [13] REG_DMA1_RD_MONO_COPY. 0 = R channel outputs zero. 1 = R channel copies L channel. [12] REG_DMA2_RD_MONO. 0 = disable. 1 = enable. [11] REG_DMA2_WR_MONO. 0 = disable. 1 = enable. [10] REG_DMA2_RD_MONO_COPY. 0 = R channel outputs zero. 1 = R channel copies L channel. [7:6] REG_DMA2_MIU_SEL. [5:4] REG_DMA1_MIU_SEL. [3] REG_DMA_2_SINE_GEN_GAIN_EXTEND. SINE_GEN_GAIN [4:0] = {SINE_GEN_GAIN_EXTEND, SINE_GEN_GAIN [3:0]}. [2] REG_DMA_1_SINE_GEN_GAIN_EXTEND. SINE_GEN_GAIN [4:0] = {SINE_GEN_GAIN_EXTEND, SINE_GEN_GAIN [3:0]}. [1] REG_DMA2_RD_WR_ENABLE. 0 = disable. 1 = enable. [0] REG_DMA1_RD_WR_ENABLE. 0 = disable.	



BACH0 Register (Bank = 1502)				
Index (Absolute)	Mnemonic	Bit	Description	
			1 = enable.	
77h (1502EFh)	REG1502EF	7:0	Default: 0x00	Access: R/W
	DMA_TEST_CTRL7[15:8]	7:0	See description of '1502EEh.'	

BACH0 Register (Bank = 1502)

BACH0 Register (Bank = 1502)				
Index (Absolute)	Mnemonic	Bit	Description	
20h (150240h)	REG150240	7:0	Default: 0x07	Access: R/W
	MMC1_DPGA_CFG1[7:0]	7:0	0x40 MMC1_DPGA_CFG1. [15:11] REG_OFFSET_R. Gain Offset for the right channel DPGA. Gain Offset = (signed) DPGA_GAIN_OFFSET_R * (-0.5dB). [10:6] REG_OFFSET_L. Gain Offset for the left channel DPGA. Gain Offset = (signed) DPGA_GAIN_OFFSET_L * (-0.5dB). [5:3] STEP. Step Size for the fading function in the right channel. 000 = 1 sample. 001 = 2 samples. 010 = 4 samples. 011 = 8 samples. 100 = 16 samples. 101 = 32 samples. 110 = 64 samples. 111 = reserved. [2] MUTE_2_ZERO. DPGA mute to zero. 0 = disable. 1 = enable. [1] FADING_EN. Fading enable. 0 = disable. 1 = enable. [0] DPGA_EN. DPGA Enable. 0 = disable. 1 = enable.	
20h (150241h)	REG150241	7:0	Default: 0x00	Access: R/W
	MMC1_DPGA_CFG1[15:8]	7:0	See description of '150240h.'	
21h (150242h)	REG150242	7:0	Default: 0x00	Access: R/W
	MMC1_DPGA_CFG2[7:0]	7:0	0x42 MMC1_DPGA_CFG2. [15:8] REG_GAIN_R. Gain for the right channel DPGA. Gain = (signed) DPGA_GAIN_R * (-0.5dB). [7:0] REG_GAIN_L.	

BACH0 Register (Bank = 1502)				
Index (Absolute)	Mnemonic	Bit	Description	
			Gain for the left channel DPGA. Gain = (signed) DPGA_GAIN_L * (-0.5dB).	
21h (150243h)	REG150243	7:0	Default: 0x00	Access: R/W
	MMC1_DPGA_CFG2[15:8]	7:0	See description of '150242h.'	
22h (150244h)	REG150244	7:0	Default: 0x07	Access: R/W
	MMC2_DPGA_CFG1[7:0]	7:0	0x44 MMC2_DPGA_CFG1. [15:11] REG_OFFSET_R. Gain Offset for the right channel DPGA. Gain Offset = (signed) DPGA_GAIN_OFFSET_R * (-0.5dB). [10:6] REG_OFFSET_L. Gain Offset for the left channel DPGA. Gain Offset = (signed) DPGA_GAIN_OFFSET_L * (-0.5dB). [5:3] STEP. Step Size for the fading function in the right channel. 000 = 1 sample. 001 = 2 samples. 010 = 4 samples. 011 = 8 samples. 100 = 16 samples. 101 = 32 samples. 110 = 64 samples. 111 = reserved. [2] MUTE_2_ZERO. DPGA mute to zero. 0 = disable. 1 = enable. [1] FADING_EN. Fading enable. 0 = disable. 1 = enable. [0] DPGA_EN. DPGA Enable. 0 = disable. 1 = enable.	
22h (150245h)	REG150245	7:0	Default: 0x00	Access: R/W
	MMC2_DPGA_CFG1[15:8]	7:0	See description of '150244h.'	
23h (150246h)	REG150246	7:0	Default: 0x00	Access: R/W
	MMC2_DPGA_CFG2[7:0]	7:0	0x46 MMC2_DPGA_CFG2. [15:8] REG_GAIN_R.	

BACH0 Register (Bank = 1502)				
Index (Absolute)	Mnemonic	Bit	Description	
			Gain for the right channel DPGA. Gain = (signed) DPGA_GAIN_R * (-0.5dB). [7:0] REG_GAIN_L. Gain for the left channel DPGA. Gain = (signed) DPGA_GAIN_L * (-0.5dB).	
23h (150247h)	REG150247	7:0	Default: 0x00	Access: R/W
	MMC2_DPGA_CFG2[15:8]	7:0	See description of '150246h.'	
24h (150248h)	REG150248	7:0	Default: 0x07	Access: R/W
	ADC_DPGA_CFG1[7:0]	7:0	0x48 ADC_DPGA_CFG1. Gain Offset for the right channel DPGA. Gain Offset = (signed) DPGA_GAIN_OFFSET_R * (-0.5dB). [10:6] REG_OFFSET_L. Gain Offset for the left channel DPGA. Gain Offset = (signed) DPGA_GAIN_OFFSET_L * (-0.5dB). [5:3] STEP. Step Size for the fading function in the right channel. 000 = 1 sample. 001 = 2 samples. 010 = 4 samples. 011 = 8 samples. 100 = 16 samples. 101 = 32 samples. 110 = 64 samples. 111 = reserved. [2] MUTE_2_ZERO. DPGA mute to zero. 0 = disable. 1 = enable. [1] FADING_EN. Fading enable. 0 = disable. 1 = enable. [0] DPGA_EN. DPGA Enable. 0 = disable. 1 = enable.	
24h (150249h)	REG150249	7:0	Default: 0x00	Access: R/W
	ADC_DPGA_CFG1[15:8]	7:0	See description of '150248h.'	
25h	REG15024A	7:0	Default: 0x00	Access: R/W

BACH0 Register (Bank = 1502)				
Index (Absolute)	Mnemonic	Bit	Description	
(15024Ah)	ADC_DPGA_CFG2[7:0]	7:0	0x4A ADC_DPGA_CFG2. [15:8] REG_GAIN_R. Gain for the right channel DPGA. Gain = (signed) DPGA_GAIN_R * (-0.5dB). [7:0] REG_GAIN_L. Gain for the left channel DPGA. Gain = (signed) DPGA_GAIN_L * (-0.5dB).	
25h	REG15024B	7:0	Default: 0x00	Access: R/W
(15024Bh)	ADC_DPGA_CFG2[15:8]	7:0	See description of '15024Ah.'	
26h	REG15024C	7:0	Default: 0x07	Access: R/W
(15024Ch)	STM_DPGA_CFG1[7:0]	7:0	0x4C STM_DPGA_CFG1. Gain Offset for the right channel DPGA. Gain Offset = (signed) DPGA_GAIN_OFFSET_R * (-0.5dB). [10:6] REG_OFFSET_L. Gain Offset for the left channel DPGA. Gain Offset = (signed) DPGA_GAIN_OFFSET_L * (-0.5dB). [5:3] STEP. Step Size for the fading function in the right channel. 000 = 1 sample. 001 = 2 samples. 010 = 4 samples. 011 = 8 samples. 100 = 16 samples. 101 = 32 samples. 110 = 64 samples. 111 = reserved. [2] MUTE_2_ZERO. DPGA mute to zero. 0 = disable. 1 = enable. [1] FADING_EN. Fading enable. 0 = disable. 1 = enable. [0] DPGA_EN. DPGA Enable. 0 = disable. 1 = enable.	
26h	REG15024D	7:0	Default: 0x00	Access: R/W

BACH0 Register (Bank = 1502)				
Index (Absolute)	Mnemonic	Bit	Description	
(15024Dh)	STM_DPGA_CFG1[15:8]	7:0	See description of '15024Ch.'	
27h (15024Eh)	REG15024E	7:0	Default: 0x00	Access: R/W
	STM_DPGA_CFG2[7:0]	7:0	0x4E STM_DPGA_CFG2. [15:8] REG_GAIN_R. Gain for the right channel DPGA. Gain = (signed) DPGA_GAIN_R * (-0.5dB). [7:0] REG_GAIN_L. Gain for the left channel DPGA. Gain = (signed) DPGA_GAIN_L * (-0.5dB).	
27h (15024Fh)	REG15024F	7:0	Default: 0x00	Access: R/W
	STM_DPGA_CFG2[15:8]	7:0	See description of '15024Eh.'	
28h (150250h)	REG150250	7:0	Default: 0x00	Access: R/W
	AEC1_DPGA_CFG1[7:0]	7:0	0x50 Reserved.	
28h (150251h)	REG150251	7:0	Default: 0x00	Access: R/W
	AEC1_DPGA_CFG1[15:8]	7:0	See description of '150250h.'	
29h (150252h)	REG150252	7:0	Default: 0x00	Access: R/W
	AEC1_DPGA_CFG2[7:0]	7:0	0x52 Reserved.	
29h (150253h)	REG150253	7:0	Default: 0x00	Access: R/W
	AEC1_DPGA_CFG2[15:8]	7:0	See description of '150252h.'	
2Ah (150254h)	REG150254	7:0	Default: 0x07	Access: R/W
	AEC2_DPGA_CFG1[7:0]	7:0	0x54 AEC2_DPGA_CFG1. Gain Offset for the right channel DPGA. Gain Offset = (signed) DPGA_GAIN_OFFSET_R * (-0.5dB). [10:6] REG_OFFSET_L. Gain Offset for the left channel DPGA. Gain Offset = (signed) DPGA_GAIN_OFFSET_L * (-0.5dB). [5:3] STEP. Step Size for the fading function in the right channel. 000 = 1 sample. 001 = 2 samples. 010 = 4 samples. 011 = 8 samples. 100 = 16 samples. 101 = 32 samples. 110 = 64 samples. 111 = reserved. [2] MUTE_2_ZERO.	

BACH0 Register (Bank = 1502)				
Index (Absolute)	Mnemonic	Bit	Description	
			DPGA mute to zero. 0 = disable. 1 = enable. [1] FADING_EN. Fading enable. 0 = disable. 1 = enable. [0] DPGA_EN. DPGA Enable. 0 = disable. 1 = enable.	
2Ah (150255h)	REG150255	7:0	Default: 0x00	Access: R/W
	AEC2_DPGA_CFG1[15:8]	7:0	See description of '150254h.'	
2Bh (150256h)	REG150256	7:0	Default: 0x00	Access: R/W
	AEC2_DPGA_CFG2[7:0]	7:0	0x56 AEC2_DPGA_CFG2. [15:8] REG_GAIN_R. Gain for the right channel DPGA. Gain = (signed) DPGA_GAIN_R * (-0.5dB). [7:0] REG_GAIN_L. Gain for the left channel DPGA. Gain = (signed) DPGA_GAIN_L * (-0.5dB).	
2Bh (150257h)	REG150257	7:0	Default: 0x00	Access: R/W
	AEC2_DPGA_CFG2[15:8]	7:0	See description of '150256h.'	
2Ch (150258h)	REG150258	7:0	Default: 0x07	Access: R/W
	AEC3_DPGA_CFG1[7:0]	7:0	0x58 AEC3_DPGA_CFG1. Gain Offset for the right channel DPGA. Gain Offset = (signed) DPGA_GAIN_OFFSET_R * (-0.5dB). [10:6] REG_OFFSET_L. Gain Offset for the left channel DPGA. Gain Offset = (signed) DPGA_GAIN_OFFSET_L * (-0.5dB). [5:3] STEP. Step Size for the fading function in the right channel. 000 = 1 sample. 001 = 2 samples. 010 = 4 samples. 011 = 8 samples. 100 = 16 samples. 101 = 32 samples. 110 = 64 samples.	

BACH0 Register (Bank = 1502)				
Index (Absolute)	Mnemonic	Bit	Description	
			111 = reserved. [2] MUTE_2_ZERO. DPGA mute to zero. 0 = disable. 1 = enable. [1] FADING_EN. Fading enable. 0 = disable. 1 = enable. [0] DPGA_EN. DPGA Enable. 0 = disable. 1 = enable.	
2Ch (150259h)	REG150259	7:0	Default: 0x00	Access: R/W
	AEC3_DPGA_CFG1[15:8]	7:0	See description of '150258h.'	
2Dh (15025Ah)	REG15025A	7:0	Default: 0x00	Access: R/W
	AEC3_DPGA_CFG2[7:0]	7:0	0x5A AEC3_DPGA_CFG2. [15:8] REG_GAIN_R. Gain for the right channel DPGA. Gain = (signed) DPGA_GAIN_R * (-0.5dB). [7:0] REG_GAIN_L. Gain for the left channel DPGA. Gain = (signed) DPGA_GAIN_L * (-0.5dB).	
2Dh (15025Bh)	REG15025B	7:0	Default: 0x00	Access: R/W
	AEC3_DPGA_CFG2[15:8]	7:0	See description of '15025Ah.'	
2Eh (15025Ch)	REG15025C	7:0	Default: 0x07	Access: R/W
	GSMDL_DPGA_CFG1[7:0]	7:0	0x5C GSMDL_DPGA_CFG1. Gain Offset for the right channel DPGA. Gain Offset = (signed) DPGA_GAIN_OFFSET_R * (-0.5dB). [10:6] REG_OFFSET_L. Gain Offset for the left channel DPGA. Gain Offset = (signed) DPGA_GAIN_OFFSET_L * (-0.5dB). [5:3] STEP. Step Size for the fading function in the right channel. 000 = 1 sample. 001 = 2 samples. 010 = 4 samples. 011 = 8 samples. 100 = 16 samples.	

BACH0 Register (Bank = 1502)				
Index (Absolute)	Mnemonic	Bit	Description	
			101 = 32 samples. 110 = 64 samples. 111 = reserved. [2] MUTE_2_ZERO. DPGA mute to zero. 0 = disable. 1 = enable. [1] FADING_EN. Fading enable. 0 = disable. 1 = enable. [0] DPGA_EN. DPGA Enable. 0 = disable. 1 = enable.	
2Eh (15025Dh)	REG15025D	7:0	Default: 0x00	Access: R/W
	GSMDL_DPGA_CFG1[15:8]	7:0	See description of '15025Ch.'	
2Fh (15025Eh)	REG15025E	7:0	Default: 0x00	Access: R/W
	GSMDL_DPGA_CFG2[7:0]	7:0	0x5E GSMDL_DPGA_CFG2. [15:8] REG_GAIN_R. Gain for the right channel DPGA. Gain = (signed) DPGA_GAIN_R * (-0.5dB). [7:0] REG_GAIN_L. Gain for the left channel DPGA. Gain = (signed) DPGA_GAIN_L * (-0.5dB).	
2Fh (15025Fh)	REG15025F	7:0	Default: 0x00	Access: R/W
	GSMDL_DPGA_CFG2[15:8]	7:0	See description of '15025Eh.'	
30h (150260h)	REG150260	7:0	Default: 0x07	Access: R/W
	MMCDEC1_DPGA_CFG1[7:0]	7:0	0x60 MMCDEC1_DPGA_CFG1. Gain Offset for the right channel DPGA. Gain Offset = (signed) DPGA_GAIN_OFFSET_R * (-0.5dB). [10:6] REG_OFFSET_L. Gain Offset for the left channel DPGA. Gain Offset = (signed) DPGA_GAIN_OFFSET_L * (-0.5dB). [5:3] STEP. Step Size for the fading function in the right channel. 000 = 1 sample. 001 = 2 samples. 010 = 4 samples.	

BACH0 Register (Bank = 1502)				
Index (Absolute)	Mnemonic	Bit	Description	
			011 = 8 samples. 100 = 16 samples. 101 = 32 samples. 110 = 64 samples. 111 = reserved. [2] MUTE_2_ZERO. DPGA mute to zero. 0 = disable. 1 = enable. [1] FADING_EN. Fading enable. 0 = disable. 1 = enable. [0] DPGA_EN. DPGA Enable. 0 = disable. 1 = enable.	
30h (150261h)	REG150261	7:0	Default: 0x00	Access: R/W
	MMCDEC1_DPGA_CFG1[15:8]	7:0	See description of '150260h.'	
31h (150262h)	REG150262	7:0	Default: 0x00	Access: R/W
	MMCDEC1_DPGA_CFG2[7:0]	7:0	0x62 MMCDEC1_DPGA_CFG2. [15:8] REG_GAIN_R. Gain for the right channel DPGA. Gain = (signed) DPGA_GAIN_R * (-0.5dB). [7:0] REG_GAIN_L. Gain for the left channel DPGA. Gain = (signed) DPGA_GAIN_L * (-0.5dB).	
31h (150263h)	REG150263	7:0	Default: 0x00	Access: R/W
	MMCDEC1_DPGA_CFG2[15:8]	7:0	See description of '150262h.'	
32h (150264h)	REG150264	7:0	Default: 0x07	Access: R/W
	MMCDEC2_DPGA_CFG1[7:0]	7:0	0x64 MMCDEC2_DPGA_CFG1. Gain Offset for the right channel DPGA. Gain Offset = (signed) DPGA_GAIN_OFFSET_R * (-0.5dB). [10:6] REG_OFFSET_L. Gain Offset for the left channel DPGA. Gain Offset = (signed) DPGA_GAIN_OFFSET_L * (-0.5dB). [5:3] STEP. Step Size for the fading function in the right channel. 000 = 1 sample.	

BACH0 Register (Bank = 1502)				
Index (Absolute)	Mnemonic	Bit	Description	
			001 = 2 samples. 010 = 4 samples. 011 = 8 samples. 100 = 16 samples. 101 = 32 samples. 110 = 64 samples. 111 = reserved. [2] MUTE_2_ZERO. DPGA mute to zero. 0 = disable. 1 = enable. [1] FADING_EN. Fading enable. 0 = disable. 1 = enable. [0] DPGA_EN. DPGA Enable. 0 = disable. 1 = enable.	
32h (150265h)	REG150265	7:0	Default: 0x00	Access: R/W
	MMCDEC2_DPGA_CFG1[15:8]	7:0	See description of '150264h.'	
33h (150266h)	REG150266	7:0	Default: 0x00	Access: R/W
	MMCDEC2_DPGA_CFG2[7:0]	7:0	0x66 MMCDEC2_DPGA_CFG2. [15:8] REG_GAIN_R. Gain for the right channel DPGA. Gain = (signed) DPGA_GAIN_R * (-0.5dB). [7:0] REG_GAIN_L. Gain for the left channel DPGA. Gain = (signed) DPGA_GAIN_L * (-0.5dB).	
33h (150267h)	REG150267	7:0	Default: 0x00	Access: R/W
	MMCDEC2_DPGA_CFG2[15:8]	7:0	See description of '150266h.'	
34h (150268h)	REG150268	7:0	Default: 0x00	Access: R/W
	DPGA_RESERVED_1[7:0]	7:0	0x68 DPGA_RESERVED_1. [15] DPGA_STATUS_CLR. 0 = normal. 1 = clear.	
34h (150269h)	REG150269	7:0	Default: 0x00	Access: R/W
	DPGA_RESERVED_1[15:8]	7:0	See description of '150268h.'	

BACH0 Register (Bank = 1502)				
Index (Absolute)	Mnemonic	Bit	Description	
35h (15026Ah)	REG15026A	7:0	Default: 0x00	Access: R/W
	DPGA_RESERVED_2[7:0]	7:0	0x6A DPGA_RESERVED_2.	
35h (15026Bh)	REG15026B	7:0	Default: 0x00	Access: R/W
	DPGA_RESERVED_2[15:8]	7:0	See description of '15026Ah.'	
36h (15026Ch)	REG15026C	7:0	Default: 0x00	Access: R/W
	DPGA_MUTE_CFG[7:0]	7:0	0x6C DPGA_MUTE_CFG. [15] DPGA mute done selection. 0: REG_DPGA_MUTE_STS[9] = MMCDEC2_MUTE_DONE_L. REG_DPGA_MUTE_STS[8] = MMCDEC1_MUTE_DONE_L. REG_DPGA_MUTE_STS[7] = GSMDL_MUTE_DONE_L. REG_DPGA_MUTE_STS[6] = AEC3_MUTE_DONE_L. REG_DPGA_MUTE_STS[5] = AEC2_MUTE_DONE_L. REG_DPGA_MUTE_STS[4] = AEC1_MUTE_DONE_L. REG_DPGA_MUTE_STS[3] = STM_MUTE_DONE_L. REG_DPGA_MUTE_STS[2] = ADC_MUTE_DONE_L. REG_DPGA_MUTE_STS[1] = MMC2_MUTE_DONE_L. REG_DPGA_MUTE_STS[0] = MMC1_MUTE_DONE_L. 1: REG_DPGA_MUTE_STS[9] = MMCDEC2_MUTE_DONE_R. REG_DPGA_MUTE_STS[8] = MMCDEC1_MUTE_DONE_R. REG_DPGA_MUTE_STS[7] = GSMDL_MUTE_DONE_R. REG_DPGA_MUTE_STS[6] = AEC3_MUTE_DONE_R. REG_DPGA_MUTE_STS[5] = AEC2_MUTE_DONE_R. REG_DPGA_MUTE_STS[4] = AEC1_MUTE_DONE_R. REG_DPGA_MUTE_STS[3] = STM_MUTE_DONE_R. REG_DPGA_MUTE_STS[2] = ADC_MUTE_DONE_R. REG_DPGA_MUTE_STS[1] = MMC2_MUTE_DONE_R. REG_DPGA_MUTE_STS[0] = MMC1_MUTE_DONE_R. [14] DPGA mute done interrupt enable. 0 = disable. 1 = enable. [9:0] DPGA mute done bypass. 0 = normal. 1 = bypass. [9] MMCDEC2_MUTE_DONE. [8] MMCDEC1_MUTE_DONE. [7] GSMDL_MUTE_DONE. [6] AEC3_MUTE_DONE. [5] AEC2_MUTE_DONE.	

BACH0 Register (Bank = 1502)

Index (Absolute)	Mnemonic	Bit	Description
			[4] AEC1_MUTE_DONE. [3] STM_MUTE_DONE. [2] ADC_MUTE_DONE. [1] MMC2_MUTE_DONE. [0] MMC1_MUTE_DONE.
36h (15026Dh)	REG15026D	7:0	Default: 0x00 Access: R/W
	DPGA_MUTE_CFG[15:8]	7:0	See description of '15026Ch.'
37h (15026Eh)	REG15026E	7:0	Default: 0x00 Access: RO
	DPGA_MUTE_STS[7:0]	7:0	0x6E DPGA_MUTE_STS. [15:10] DPGA_STATUS. [9:0] refer to 0x6C [15].
37h (15026Fh)	REG15026F	7:0	Default: 0x00 Access: RO
	DPGA_MUTE_STS[15:8]	7:0	See description of '15026Eh.'

BACH0 Register (Bank = 1502)

BACH0 Register (Bank = 1502)				
Index (Absolute)	Mnemonic	Bit	Description	
00h (150200h)	REG150200	7:0	Default: 0xFF	Access: R/W
	ENABLE_CTRL[7:0]	7:0	0x00. [15] REG_TIME_GEN_EN. Time GEN enable. 0 = disable. 1 = enable. [14] REG_INIT_SRAM_EN. Initialize SRAM. 0 = disable. 1 = enable. [13] REG_BP_GSM_HPF_UL. Bypass GSM HPF up-link. 0 = normal. 1 = bypass. [12] REG_BP_GSM_HPF_DL. Bypass GSM HPF down-link. 0 = normal. 1 = bypass. [11] REG_BP_ADC_HPF2. Bypass ADC HPF2. 0 = normal. 1 = bypass. [10] REG_BP_ADC_HPF1. Bypass ADC HPF1. 0 = normal. 1 = bypass. [9] Reserved. [8] REG_EN_INT_1. Enable interpolation 1. 0 = disable. 1 = enable. [7] REG_EN_INT_2. Enable interpolation 2. 0 = disable. 1 = enable. [6] REG_EN_DEC_1. Enable decimation 1. 0 = disable. 1 = enable.	

BACH0 Register (Bank = 1502)

Index (Absolute)	Mnemonic	Bit	Description	
			[5] REG_EN_DEC_2. Enable decimation 2. 0 = disable. 1 = enable. [4] REG_EN_DEC_3. Enable decimation 3. 0 = disable. 1 = enable. [3] REG_EN_FIR_1. Enable 9-tap FIR. 0 = disable. 1 = enable. [2] REG_EN_ASRC. Enable ASRC. 0 = disable. 1 = enable. [1] REG_EN_SDM. Enable SDM. 0 = disable. 1 = enable. [0] REG_EN_CODEC. Enable CODEC. 0 = disable. 1 = enable.	
00h (150201h)	REG150201	7:0	Default: 0xC1	Access: R/W
	ENABLE_CTRL[15:8]	7:0	See description of `150200h.`	
01h (150202h)	REG150202	7:0	Default: 0x80	Access: R/W
	SRO_SEL[7:0]	7:0	0x02. [15:14] REG_CIC_1_SEL. Decimation 1 sampling rate selection. 0: 8KHz. 1: 16KHz. 2: 32KHz. 3: 48KHz. [13:12] REG_CIC_2_SEL. Decimation 2 sampling rate selection. 0: 8KHz. 1: 16KHz. 2: 32KHz. 3: 48KHz.	

BACH0 Register (Bank = 1502)				
Index (Absolute)	Mnemonic	Bit	Description	
			[11:10] REG_CIC_3_SEL. Decimation 3 sampling rate selection. 0: 8KHz. 1: 16KHz. 2: 32KHz. 3: 48KHz. [9:8] REG_CODEC_SEL. CODEC 1 sampling rate selection. 0: 8KHz. 1: 16KHz. 2: 32KHz. 3: 48KHz. [7:4] REG_SRC1_SEL. SRC 1 sampling rate selection. 0: 8KHz. 1: 11KHz. 2: 12KHz. 3: 16KHz. 4: 22KHz. 5: 24KHz. 6: 32KHz. 7: 44KHz. 8: 48KHz. [3:0] REG_SRC2_SEL. SRC 2 sampling rate selection. 0: 8KHz. 1: 11KHz. 2: 12KHz. 3: 16KHz. 4: 22KHz. 5: 24KHz. 6: 32KHz. 7: 44KHz. 8: 48KHz.	
01h (150203h)	REG150203	7:0	Default: 0x00	Access: R/W
	SR0_SEL[15:8]	7:0	See description of '150202h.'	
02h (150204h)	REG150204	7:0	Default: 0x03	Access: R/W
	SR1_SEL[7:0]	7:0	0x04. [15] REG_RESET_AU. Audio reset.	

BACH0 Register (Bank = 1502)			
Index (Absolute)	Mnemonic	Bit	Description
			0 = normal. 1 = reset. [14] REG_RESET_IMC. IMC reset. 0 = normal. 1 = reset. [13] REG_RESET_DPGA. DPGA reset. 0 = normal. 1 = reset. [12] REG_RESET_AB. Audioband reset. 0 = normal. 1 = reset. [11] REG_RESET_ASRC. ASRC reset. 0 = normal. 1 = reset. [10] REG_RESET_SDM. SDM reset. 0 = normal. 1 = reset. [9] REG_RESET_GPEQ. GPEQ reset. 0 = normal. 1 = reset. [8] REG_RESET_SYNTH. Synthesizer reset. 0 = normal. 1 = reset. [6] REG_ICG_FORCE_ENABLE_ON. 1 = MIU domain clock will be gated. 0 = MIU domain clock will always be on. [5] REG_PCM2AU_SEL_1. 0 = a-law. 1 = u-law. [4] REG_PCM2AU_SEL_2. 0 = a-law. 1 = u-law. [3] REG_ADC_RSEL. ADC sampling rate selection.

BACH0 Register (Bank = 1502)

Index (Absolute)	Mnemonic	Bit	Description	
			0: 24MHz. 1: 12MHz. [2] REG_DAC_RSEL. DAC sampling rate selection. 0: 24MHz. 1: 12MHz. [1:0] REG_ASRC1_SEL. ASRC sampling rate selection. 0: 8KHz. 1: 16KHz. 2: 32KHz. 3: 48KHz.	
02h (150205h)	REG150205	7:0	Default: 0x00	Access: R/W
	SR1_SEL[15:8]	7:0	See description of '150204h.'	
03h (150206h)	REG150206	7:0	Default: 0x80	Access: R/W
	MUX0_SEL[7:0]	7:0	0x06. [15] REG_DMA4_WR_SEL. 0 = select AEC_REF. 1 = select AEC_REF2. [14:13] REG_DBB_MUX_SEL. 0 = select AEC_REF. 1 = select ADC2_PATH. 2 = select {ADC2_PATH, AEC_REF}. [12] REG_ADDA_SEL. 0 = select DMA2 read output. 1 = select ADC1_path & ADC2_path. [11:10] REG_AGC1_MX_SEL. AGC matrix selection. 0 = select GPEQ output. 1 = select MMC1 DPGA output. 2 = select CODEC mux output. 3 = reserved. [9] REG_GSM_RCH_SEL. DBB I2S TX right channel source selection. 0 = select GSM uplink HPF. 1 = select MMC1 decimation. [8] REG_DMA2_WR_SEL. DMA 2 write source selection. 0 = select MMC1 decimation. 1 = select AMR record.	

BACH0 Register (Bank = 1502)

Index (Absolute)	Mnemonic	Bit	Description
			[7] REG_AEC_BP_SEL. AEC bypass. 0 = normal. 1 = bypass. [6] REG_GSM_DL_SEL. GSM downlink source selection. 0 = DBB I2S RX. 1 = AMR decode. [5] REG_MMC1_SRC_SEL. MMC1 SRC source selection. 0 = ASRC. 1 = DMA 1 read. [4] REG_MMC2_SRC_SEL. MMC2 SRC source selection. 0 = GSM downlink HPF. 1 = DMA2 read. [3:2] REG_ASRC_MUX1_SEL. ASRC source selection. 0 = DMA1 read. 1 = FM RX. 2 = External I2S RX. 3 = SPDIF RX. [1:0] REG_CODECS_MUX_SEL. CODEC source selection. 0 = ADC. 1 = CODEC I2S. 2 = Bluetooth I2S RX.
03h (150207h)	REG150207	7:0	Default: 0x00 Access: R/W
	MUX0_SEL[15:8]	7:0	See description of '150206h.'
04h (150208h)	REG150208	7:0	Default: 0x00 Access: R/W
	MUX1_SEL[7:0]	7:0	0x08. [15:12] REG_ADC_HPF_N. ADC HPF Number. [11] REG_SAT_SEL. ADC HPF saturation select. 0 = normal mode. 1 = old mode. [10:8] REG_CPO_0_SEL. Cut point 0 selection. [7] REG_HDMI_MUX_SEL.

BACH0 Register (Bank = 1502)				
Index (Absolute)	Mnemonic	Bit	Description	
			0 = data rate follows 3404 [13:12]. 1 = data rate follows HDMI MN synthesizer. [6:4] REG_CPO_1_SEL. Cut point 1 selection. [3] REG_GPEQ_DRC_5P19_SEL. Change GPEQ format from 1.23 to 5.19. [2:0] REG_CPO_2_SEL. Cut point 2 selection.	
04h (150209h)	REG150209	7:0	Default: 0x00	Access: R/W
	MUX1_SEL[15:8]	7:0	See description of '150208h.'	
05h (15020Ah)	REG15020A	7:0	Default: 0x00	Access: R/W
	MUX2_SEL[7:0]	7:0	0x0A. [15] REG_SRAM_CG_EN. SRAM clock gating. 0 = normal. 1 = gated. [14] REG_DBB_PCM_MODE. 0: I2S mode. 1: PCM mode. [13] REG_BT_PCM_MODE. 0: I2S mode. 1: PCM mode. [12] REG_CODEC_PCM_MODE. 0: I2S mode. 1: PCM mode. [11:10] REG_DRC_RT_FAST_MODE. Increase DRC's attack and release time. 0: Normal. 1: 2 times faster. 2: 4 times faster. 3: 4 times faster. [9] REG_TEST_BUS_SEL. DWA 6-bit test data input selection. 0 = from CHIPTOP audio test pad. 1 = from BACH pad. [8] REG_CPO_TEST. Cut point out test. [7] REG_CPI_0_EN. Cut point input 0 enable. 0 = disable.	

BACH0 Register (Bank = 1502)

Index (Absolute)	Mnemonic	Bit	Description
			1 = enable. [6] REG_CPI_1_EN. Cut point input 1 enable. 0 = disable. 1 = enable. [5] REG_CPI_2_EN. Cut point input 2 enable. 0 = disable. 1 = enable. [4] REG_CPI_3_EN. Cut point input 3 enable. 0 = disable. 1 = enable. [3] REG_CPI_4_EN. Cut point input 4 enable. 0 = disable. 1 = enable. [2] REG_CPO_0_EN. Cut point output 0 enable. 0 = disable. 1 = enable. [1] REG_CPO_1_EN. Cut point output 1 enable. 0 = disable. 1 = enable. [0] REG_CPO_2_EN. Cut point output 2 enable. 0 = disable. 1 = enable.
05h (15020Bh)	REG15020B	7:0	Default: 0x80 Access: R/W
	MUX2_SEL[15:8]	7:0	See description of '15020Ah.'
06h (15020Ch)	REG15020C	7:0	Default: 0x00 Access: R/W
	MIX0_SEL[7:0]	7:0	0x0C. [15:14] REG_S2M_ADC_SEL. MMC Decimation 1 stereo to mono selection. 0 = (left channel + right channel)/2. 1 = (left channel + right channel). 2 = left channel. 3 = right channel. [13:12] REG_DTMF_MIXER_L.

BACH0 Register (Bank = 1502)

Index (Absolute)	Mnemonic	Bit	Description
			0 = mute. 1 = HPF. 2 = DTMF. 3 = mix. [11:10] REG_S2M_MMCDDEC1_SEL. MMC Decimation 1 stereo to mono selection. 0 = (left channel + right channel)/2. 1 = (left channel + right channel). 2 = left channel. 3 = right channel. [9:8] REG_S2M_MMCDDEC2_SEL. MMC Decimation 2 stereo to mono selection. 0 = (left channel + right channel)/2. 1 = (left channel + right channel). 2 = left channel. 3 = right channel. [7:6] REG_ARM_MIXER_L. AMR mixer. 0 = mute. 1 = AEC2 DPGA. 2 = GSM down-link. 3 = mix. [5:4] REG_GSM_UL_MIXER_L. GSM up-link mixer. 0 = mute. 1 = AEC 3 DPGA. 2 = MMC decimation 1. 3 = mix. [3:2] REG_ST_MIXER_L. Side tone mixer left channel. 0 = mute. 1 = MMC 2 DPGA. 2 = Side tone mixer DPGA. 3 = mix. [1:0] REG_ST_MIXER_R. Side tone mixer right channel. 0 = mute. 1 = MMC 2 DPGA. 2 = Side tone mixer DPGA. 3 = mix.
06h	REG15020D	7:0	Default: 0x00 Access: R/W

BACH0 Register (Bank = 1502)

Index (Absolute)	Mnemonic	Bit	Description
(15020Dh)	MIX0_SEL[15:8]	7:0	See description of '15020Ch.'
07h (15020Eh)	REG15020E	7:0	Default: 0x00 Access: R/W
	MIX1_SEL[7:0]	7:0	0x0E. [15:14] REG_DAC_DIN_L_SEL. DAC left channel selection. 0 = mute. 1 = normal. 2 = left/right swap. 3 = (left + right)/2. [13:12] REG_DAC_DIN_R_SEL. DAC right channel selection. 0 = mute. 1 = normal. 2 = left/right swap. 3 = (left + right)/2. [11:10] REG_HD_DAC_DIN_L_SEL. HD DAC left channel selection. 0 = mute. 1 = normal. 2 = left/right swap. 3 = (left + right)/2. [9:8] REG_HD_DAC_DIN_L_SEL. DAC right channel selection. 0 = mute. 1 = normal. 2 = left/right swap. 3 = (left + right)/2. [7:6] REG_SRC_MIXER1_L. SRC left channel mixer 1. 0 = mute. 1 = MMC1 SRC. 2 = MMC2 SRC. 3 = mix. [5:4] REG_SRC_MIXER1_R. SRC right channel mixer 1. 0 = mute. 1 = MMC1 SRC. 2 = MMC2 SRC. 3 = mix. [3:2] REG_SRC_MIXER2_L.

BACH0 Register (Bank = 1502)

Index (Absolute)	Mnemonic	Bit	Description
			SRC left channel mixer 2. 0 = mute. 1 = MMC1 SRC. 2 = MMC2 SRC. 3 = mix. [1:0] REG_SRC_MIXER2_R. SRC left channel mixer 2. 0 = mute. 1 = MMC1 SRC. 2 = MMC2 SRC. 3 = mix.
07h (15020Fh)	REG15020F	7:0	Default: 0x00 Access: R/W
	MIX1_SEL[15:8]	7:0	See description of '15020Eh.'
08h (150210h)	REG150210	7:0	Default: 0x00 Access: R/W
	SDM_OFFSET[7:0]	7:0	0x10. SDM_OFFSET [15:0].
08h (150211h)	REG150211	7:0	Default: 0x00 Access: R/W
	SDM_OFFSET[15:8]	7:0	See description of '150210h.'
09h (150212h)	REG150212	7:0	Default: 0x00 Access: R/W
	SDM_CTRL[7:0]	7:0	0x12. [15:14] SDM_OFFSET [17:16]. [12] REG_SDM_TEST_EN. SDM test enable. 0 = disable. 1 = enable. [11] REG_SDM_SFT_DIS. SDM shift disable. 0 = enable. 1 = disable. [10] REG_DAC_TEST_EN. DAC test enable. 0 = disable. 1 = enable. [9] REG_FIX_MSB_EN. DWA fixed MSB enable. 0 = disable. 1 = enable. [8:4] REG_FIX_MSB_SEL. DWA fixed MSN selection.

BACH0 Register (Bank = 1502)				
Index (Absolute)	Mnemonic	Bit	Description	
			[3] REG_DITHER_EN. SDM dither enable. 0 = disable. 1 = enable. [2:0] REG_DITHER_SEL. SDM dither selection.	
09h (150213h)	REG150213	7:0	Default: 0x02	Access: R/W
	SDM_CTRL[15:8]	7:0	See description of '150212h.'	
0Ah (150214h)	REG150214	7:0	Default: 0x00	Access: R/W
	NF_SYNTH_L[7:0]	7:0	0x14. ASRC N.F synthesizer value low word. Dec2hex (round $[48000000/48000 \cdot 2^{20/128}]$).	
0Ah (150215h)	REG150215	7:0	Default: 0x00	Access: R/W
	NF_SYNTH_L[15:8]	7:0	See description of '150214h.'	
0Bh (150216h)	REG150216	7:0	Default: 0x7D	Access: R/W
	NF_SYNTH_H[7:0]	7:0	0x16. ASRC N.F synthesizer value high word. Dec2hex (round $[48000000/48000 \cdot 2^{20/128}]$).	
0Bh (150217h)	REG150217	7:0	Default: 0x00	Access: R/W
	NF_SYNTH_H[15:8]	7:0	See description of '150216h.'	
0Ch (150218h)	REG150218	7:0	Default: 0x00	Access: RO
	FS_SYNTH_L[7:0]	7:0	0x18. ASRC FS synthesizer low word. $48000000/\text{hex2dec}(\text{REG_FS_SYNTH_H}, \text{FS_SYNTH_L}) \cdot 2^{16}$.	
0Ch (150219h)	REG150219	7:0	Default: 0x00	Access: RO
	FS_SYNTH_L[15:8]	7:0	See description of '150218h.'	
0Dh (15021Ah)	REG15021A	7:0	Default: 0x00	Access: RO
	FS_SYNTH_H[7:0]	7:0	0x1A. ASRC FS synthesizer high word. $48000000/\text{hex2dec}(\text{FS_SYNTH_H}, \text{REG_FS_SYNTH_L}) \cdot 2^{16}$.	
0Dh (15021Bh)	REG15021B	7:0	Default: 0x00	Access: RO
	FS_SYNTH_H[15:8]	7:0	See description of '15021Ah.'	
0Eh (15021Ch)	REG15021C	7:0	Default: 0x17	Access: R/W
	SPDIF_IN_CTRL0[7:0]	7:0	0x1C.	

BACH0 Register (Bank = 1502)

Index (Absolute)	Mnemonic	Bit	Description
			[15:12] COARSE_K. Audio SPDIF CDR coarse frequency detection threshold K value (locked range detection offset value). 0000: N.A. 0001: Minimum range. ... 1111: Maximum range. [11:8] PHASE_OFFSET. Audio SPDIF CDR output clock phase selection. (1 sampling phase = 5 ns). 0111: Delay 7 sampling phase offset. ... 0000: Delay 0 sampling phase offset. 1000: Early 0 sampling phase offset. 1001: Early 1 sampling phase offset. ... 1111: Early 7 sampling phase offset. [7:4] FDC_KD_REG. Audio SPDIF clock data recovery (CDR) frequency detect coarse KD (Gain) select (protect CDR from harmonic locking). 0000: Special use for S/W. 0001: Minimum gain. ... 1111: Maximum gain. [3:1] PD_RESOLUTION. Audio SPDIF CDR Phase tracking band width control. 000: Phase tracking fastest. ... 111: Phase tracking slowest. [0] ENABLE_CDR. Audio SPDIF CDR enable. 0 = Disable. 1 = Enable.
0Eh (15021Dh)	REG15021D	7:0	Default: 0x30 Access: R/W
	SPDIF_IN_CTRL0[15:8]	7:0	See description of '15021Ch.'
0Fh (15021Eh)	REG15021E	7:0	Default: 0x02 Access: R/W
	SPDIF_IN_CTRL1[7:0]	7:0	0x1E. [15:0] REG_SPDIF_RX_CTRL1. [3] REG_RX_FIFO_EX.

BACH0 Register (Bank = 1502)

Index (Absolute)	Mnemonic	Bit	Description
			SPDIF RX FIFO extension. 0 = normal. 1 = extension. [2] REG_RX_FIFO_CLR. SPDIF RX FIFO clear. 0 = normal. 1 = clear. [1] REG_SPDIF_IN_BLOCK_NO_CHECK. SPDIF in block no check. 0 = normal. 1 = no check. [0] PHASE_TRACKER. Phase tracker.
0Fh (15021Fh)	REG15021F	7:0	Default: 0x00 Access: R/W
	SPDIF_IN_CTRL1[15:8]	7:0	See description of '15021Eh.'
10h (150220h)	REG150220	7:0	Default: 0x00 Access: R/W
	BT_I2S_RX_CTRL[7:0]	7:0	0x20. [15] RESETB_RX. I2S RX reset. 0 = reset. 1 = normal. [14] REG_I2S_RX_FMT. I2S RX format. 0 = I2S justified. 1 = left justified. [13] REG_I2S_RX_FIFO_CLR. I2S RX FIFO clear. 0 = normal. 1 = clear. [12] Reserved. [11] REG_I2S_RX_BCK_INV. I2S RX BCK invert. 0 = normal. 1 = invert. [10] REG_I2S_RX_FIFO_EX. I2S RX FIFO extension. 0 = normal. 1 = extension. [9:8] REG_I2S_RX_BWH. I2S RX bit width.

BACH0 Register (Bank = 1502)

Index (Absolute)	Mnemonic	Bit	Description
			00 = 16-bit. 01 = reserved. 10 = reserved. 11 = reserved. [7:6] I2S_RX_MUX_SEL_R. I2S RX right channel selection. 00 = normal. 01 = left/right swap. 10 = (left + right)/2. 11 = (left + right). [5:4] I2S_RX_MUX_SEL_L. I2S RX left channel selection. 00 = normal. 01 = left/right swap. 10 = (left + right)/2. 11 = (left + right).
10h (150221h)	REG150221 BT_I2S_RX_CTRL[15:8]	7:0 7:0	Default: 0x94 Access: R/W See description of '150220h.'
11h (150222h)	REG150222 BT_I2S_TX_CTRL[7:0]	7:0 7:0	Default: 0x00 Access: R/W 0x22. 2S TX format. 0 = I2S justified. 1 = left justified. [13] REG_I2S_TX_FIFO_CLR. I2S TX FIFO clear. 0 = normal. 1 = clear. [12] Reserved. [11] REG_I2S_TX_BCK_INV. I2S TX BCK invert. 0 = normal. 1 = invert. [10] REG_I2S_TX_FIFO_EX. I2S TX FIFO extension. 0 = normal. 1 = extension. [9:8] REG_I2S_TX_BWH. I2S TX bit width. 00 = 16 bit. 01 = reserved.

BACH0 Register (Bank = 1502)

Index (Absolute)	Mnemonic	Bit	Description	
			10 = reserved. 11 = reserved. [7:6] I2S_TX_MUX_SEL_R. I2S TX right channel selection. 00 = normal. 01 = left/right swap. 10 = (left + right)/2. 11 = (left + right). [5:4] I2S_TX_MUX_SEL_L. I2S TX left channel selection. 00 = normal. 01 = left/right swap. 10 = (left + right)/2. 11 = (left + right).	
11h (150223h)	REG150223	7:0	Default: 0x94	Access: R/W
	BT_I2S_TX_CTRL[15:8]	7:0	See description of '150222h.'	
12h (150224h)	REG150224	7:0	Default: 0x00	Access: R/W
	CODEC_I2S_RX_CTRL[7:0]	7:0	0x24. [15] RESETB_RX. I2S RX reset. 0 = reset. 1 = normal. [14] REG_I2S_RX_FMT. I2S RX format. 0 = I2S justified. 1 = left justified. [13] REG_I2S_RX_FIFO_CLR. I2S RX FIFO clear. 0 = normal. 1 = clear. [12] REG_I2S_RX_MS. I2S RX master/slave mode. 0 = slave mode. 1 = master mode. [11] REG_I2S_RX_BCK_INV. I2S RX BCK invert. 0 = normal. 1 = invert. [10] REG_I2S_RX_FIFO_EX. I2S RX FIFO extension.	

BACH0 Register (Bank = 1502)

Index (Absolute)	Mnemonic	Bit	Description
			0 = normal. 1 = extension. [9:8] REG_I2S_RX_BWH. I2S RX bit width. 00 = 16 bit. 01 = reserved. 10 = reserved. 11 = reserved. [7:6] I2S_RX_MUX_SEL_R. I2S RX right channel selection. 00 = normal. 01 = left/right swap. 10 = (left + right)/2. 11 = (left + right). [5:4] I2S_RX_MUX_SEL_L. I2S RX left channel selection. 00 = normal. 01 = left/right swap. 10 = (left + right)/2. 11 = (left + right).
12h (150225h)	REG150225	7:0	Default: 0x94 Access: R/W
	CODEC_I2S_RX_CTRL[15:8]	7:0	See description of '150224h.'
13h (150226h)	REG150226	7:0	Default: 0x00 Access: R/W
	CODEC_I2S_TX_CTRL[7:0]	7:0	0x26. [15] RESETB_TX. I2S TX reset. 0 = reset. 1 = normal. [14] REG_I2S_TX_FMT. I2S TX format. 0 = I2S justified. 1 = left justified. [13] REG_I2S_TX_FIFO_CLR. I2S TX FIFO clear. 0 = normal. 1 = clear. [12] REG_I2S_TX_MS. I2S TX master/slave mode. 0 = slave mode.

BACH0 Register (Bank = 1502)

Index (Absolute)	Mnemonic	Bit	Description
			1 = master mode. [11] REG_I2S_TX_BCK_INV. I2S TX BCK invert. 0 = normal. 1 = invert. [10] REG_I2S_TX_FIFO_EX. I2S TX FIFO extension. 0 = normal. 1 = extension. [9:8] REG_I2S_TX_BWH. I2S TX bit width. 00 = 16 bit. 01 = reserved. 10 = reserved. 11 = reserved. [7:6] I2S_TX_MUX_SEL_R. I2S TX right channel selection. 00 = normal. 01 = left/right swap. 10 = (left + right)/2. 11 = (left + right). [5:4] I2S_TX_MUX_SEL_L. I2S TX left channel selection. 00 = normal. 01 = left/right swap. 10 = (left + right)/2. 11 = (left + right).
13h (150227h)	REG150227	7:0	Default: 0x94 Access: R/W
	CODEC_I2S_TX_CTRL[15:8]	7:0	See description of '150226h.'
14h (150228h)	REG150228	7:0	Default: 0x00 Access: R/W
	EXT_I2S_RX_CTRL[7:0]	7:0	0x28. [15] RESETB_RX. I2S RX reset. 0 = reset. 1 = normal. [14] REG_I2S_RX_FMT. I2S RX format. 0 = I2S justified. 1 = left justified.

BACH0 Register (Bank = 1502)

Index (Absolute)	Mnemonic	Bit	Description	
			[13] REG_I2S_RX_FIFO_CLR. I2S RX FIFO clear. 0 = normal. 1 = clear. [12] REG_I2S_RX_MS. I2S RX master/slave mode. 0 = slave mode. 1 = master mode. [11] REG_I2S_RX_BCK_INV. I2S RX BCK invert. 0 = normal. 1 = invert. [10] REG_I2S_RX_FIFO_EX. I2S RX FIFO extension. 0 = normal. 1 = extension. [9:8] REG_I2S_RX_BWH. I2S RX bit width. 00 = 16 bit. 01 = reserved. 10 = reserved. 11 = reserved. [7:6] I2S_RX_MUX_SEL_R. I2S RX right channel selection. 00 = normal. 01 = left/right swap. 10 = (left + right)/2. 11 = (left + right). [5:4] I2S_RX_MUX_SEL_L. I2S RX left channel selection. 00 = normal. 01 = left/right swap. 10 = (left + right)/2. 11 = (left + right).	
14h (150229h)	REG150229	7:0	Default: 0x84	Access: R/W
	EXT_I2S_RX_CTRL[15:8]	7:0	See description of '150228h.'	
15h (15022Ah)	REG15022A	7:0	Default: 0x00	Access: R/W
	EXT_I2S_TX_CTRL[7:0]	7:0	0x2A. [15] RSTB_AU_HDMI_I2S_TX. 0 = reset.	

BACH0 Register (Bank = 1502)

Index (Absolute)	Mnemonic	Bit	Description	
			1 = enable. [14] REG_HDMI_I2S_FMT. 0 = I2S justified. 1 = left justified. [12] REG_HDMI_MS_MODE. I2S TX master/slave mode. 0 = slave mode. 1 = master mode. [11] REG_HDMI_I2S_TX_BCK_INV. I2S TX BCK invert. 0 = normal. 1 = invert. [9:8] REG_HDMI_I2S_WDTH. I2S TX bit width. 00 = 16 bit. 01 = 24 bit. 10 = 32 bit. 11 = reserved. [1] PAD_HDMI_TEST. 0 = normal. 1 = select HDMI I2S related signals to PAD_BT. [0] REG_RESET_HDMI. 0 = normal. 1 = reset HDMI I2S related HW.	
15h (15022Bh)	REG15022B	7:0	Default: 0xD0	Access: R/W
	EXT_I2S_TX_CTRL[15:8]	7:0	See description of '15022Ah.'	
16h (15022Ch)	REG15022C	7:0	Default: 0x00	Access: R/W
	GSM_I2S_RX_CTRL[7:0]	7:0	0x2C. [15] RESETB_RX. I2S RX reset. 0 = reset. 1 = normal. [14] REG_I2S_RX_FMT. I2S RX format. 0 = I2S justified. 1 = left justified. [13] REG_I2S_RX_FIFO_CLR. I2S RX FIFO clear. 0 = normal. 1 = clear.	

BACH0 Register (Bank = 1502)

Index (Absolute)	Mnemonic	Bit	Description
			[12] Reserved. [11] REG_I2S_RX_BCK_INV. I2S RX BCK invert. 0 = normal. 1 = invert. [10] REG_I2S_RX_FIFO_EX. I2S RX FIFO extension. 0 = normal. 1 = extension. [9:8] REG_I2S_RX_BWH. I2S RX bit width. 00 = 16 bit. 01 = reserved. 10 = reserved. 11 = reserved. [7:6] I2S_RX_MUX_SEL_R. I2S RX right channel selection. 00 = normal. 01 = left/right swap. 10 = (left + right)/2. 11 = (left + right). [5:4] I2S_RX_MUX_SEL_L. I2S RX left channel selection. 00 = normal. 01 = left/right swap. 10 = (left + right)/2. 11 = (left + right).
16h (15022Dh)	REG15022D	7:0	Default: 0x94 Access: R/W
	GSM_I2S_RX_CTRL[15:8]	7:0	See description of '15022Ch.'
17h (15022Eh)	REG15022E	7:0	Default: 0x00 Access: R/W
	GSM_I2S_TX_CTRL[7:0]	7:0	0x2E. [15] RESETB_TX. I2S TX reset. 0 = reset. 1 = normal. [14] REG_I2S_TX_FMT. I2S TX format. 0 = I2S justified. 1 = left justified. [13] REG_I2S_TX_FIFO_CLR.

BACH0 Register (Bank = 1502)

Index (Absolute)	Mnemonic	Bit	Description	
			I2S TX FIFO clear. 0 = normal. 1 = clear. [12] REG_I2S_TX_MS. I2S TX master/slave mode. 0 = slave mode. 1 = master mode. [11] REG_I2S_TX_BCK_INV. I2S TX BCK invert. 0 = normal. 1 = invert. [10] REG_I2S_TX_FIFO_EX. I2S TX FIFO extension. 0 = normal. 1 = extension. [9:8] REG_I2S_TX_BWH. I2S TX bit width. 00 = 16 bit. 01 = reserved. 10 = reserved. 11 = reserved. [7:6] I2S_TX_MUX_SEL_R. I2S TX right channel selection. 00 = normal. 01 = left/right swap. 10 = (left + right)/2. 11 = (left + right). [5:4] I2S_TX_MUX_SEL_L. I2S TX left channel selection. 00 = normal. 01 = left/right swap. 10 = (left + right)/2. 11 = (left + right).	
17h (15022Fh)	REG15022F	7:0	Default: 0x94	Access: R/W
	GSM_I2S_TX_CTRL[15:8]	7:0	See description of '15022Eh.'	
18h (150230h)	REG150230	7:0	Default: 0x00	Access: R/W
	FM_RX_CTRL[7:0]	7:0	0x30. [15] RESETB_RX. FM RX reset. 0 = reset.	

BACH0 Register (Bank = 1502)

Index (Absolute)	Mnemonic	Bit	Description
			1 = normal. [13] REG_FM_RX_FIFO_CLR. FM RX FIFO clear. 0 = normal. 1 = clear. [10] REG_FM_RX_FIFO_EX. FM RX FIFO extension. 0 = normal. 1 = extension. [7:6] FM_RX_MUX_SEL_R. FM RX right channel selection. 00 = normal. 01 = left/right swap. 10 = (left + right)/2. 11 = (left + right). [5:4] FM_RX_MUX_SEL_L. FM RX left channel selection. 00 = normal. 01 = left/right swap. 10 = (left + right)/2. 11 = (left + right).
18h (150231h)	REG150231 FM_RX_CTRL[15:8]	7:0	Default: 0x84 Access: R/W
19h (150232h)	REG150232 SPDIF_OUT_CTRL0[7:0]	7:0	Default: 0x00 Access: R/W
19h (150233h)	REG150233 SPDIF_OUT_CTRL0[15:8]	7:0	Default: 0x00 Access: R/W
1Ah (150234h)	REG150234 SPDIF_OUT_CTRL1[7:0]	7:0	Default: 0x00 Access: R/W
1Ah (150235h)	REG150235 SPDIF_OUT_CTRL1[15:8]	7:0	Default: 0x00 Access: R/W
1Bh (150236h)	REG150236 SPDIF_OUT_CTRL2[7:0]	7:0	Default: 0x00 Access: R/W

BACH0 Register (Bank = 1502)

Index (Absolute)	Mnemonic	Bit	Description
			SPDIF TX FIFO extension. 0 = normal. 1 = extension. [14] REG_TX_FIFO_CLR. SPDIF TX FIFO clear. 0 = normal. 1 = clear. [13] REG_CNT_RST. Counter reset. 0 = normal. 1 = reset. [12] REG_SPDIF_FIX. SPDIF fixed value mode. 0 = normal. 1 = fixed. [11] REG_SPDIF_FIX_VAL. SPDIF fixed value. [10] REG_SPDIF_CFG. SPDIF encoder configuration. 0: Normal. 1: Invert. [9] REG_SPDIF_VALIDITY. SPDIF validity. [8] REG_SPDIF_EN_ABS. Channel status control by RIU enable. 0: Disable. 1: Enable. [7:0] REG_SPDIF_OUT_CS4 [7:0].
1Bh (150237h)	REG150237	7:0	Default: 0x00 Access: R/W
	SPDIF_OUT_CTRL2[15:8]	7:0	See description of '150236h.'
1Ch (150238h)	REG150238	7:0	Default: 0x00 Access: R/W
	PAD_TEST[7:0]	7:0	0x38. [15] enable INT test mode. [14] INT test mode value to MMP. [13] INT test mode value to DBB.
1Ch (150239h)	REG150239	7:0	Default: 0x00 Access: R/W
	PAD_TEST[15:8]	7:0	See description of '150238h.'
1Dh (15023Ah)	REG15023A	7:0	Default: 0x00 Access: R/W
	SYNTH_CTRL[7:0]	7:0	0x3A.

BACH0 Register (Bank = 1502)				
Index (Absolute)	Mnemonic	Bit	Description	
			[15:14] REG_SYNC_CLK1_SEL. DAC clock sync phase selection. [13:12] REG_SYNC_CLK2_SEL. ADC clock sync phase selection. [11] REG_ASRC_FIFO_CLR. 0 = normal. 1 = clear. [10] REG_AUTO_UPD_DIS. Auto update disable. 0 = enable. 1 = disable. [9] REG_FS_SYNTH_TRIG. ASRC FS synthesizer trigger. [8] REG_FS_SYNTH_FORCE. ASRC FS synthesizer force lock current frequency. [7:5] REG_FS_SYNTH_BW. ASRC FS synthesizer bandwidth selection. [4:0] REG_CMP_CIC. CIC compensation value.	
1Dh	REG15023B	7:0	Default: 0x00	Access: R/W
(15023Bh)	SYNTH_CTRL[15:8]	7:0	See description of '15023Ah.'	
1Eh	REG15023C	7:0	Default: 0x00	Access: R/W
(15023Ch)	PAD0_CFG[7:0]	7:0	0x3C. [14] PAD_DBB_TEST. DBB Pad output test. 0 = Disable. 1 = Enable. [13] PAD_OUTPUT_TEST. Pad output test. 0 = normal. 1 = test mode. [12] EXT_I2S_DG_EN. External I2S de-glitch enable. 0 = disable. 1 = enable. [11] DBB_SOURCE_SEL. [11] CODEC_DAAD2_LOOP. CODEC DAAD2 loop. 0 = normal mode. 1 = loop DAC to ADC2.	

BACH0 Register (Bank = 1502)

Index (Absolute)	Mnemonic	Bit	Description
			[10] DBB_I2S_LOOP. DBB I2S loop. 0 = normal mode. 1 = loop TX to RX. [9] BT_I2S_LOOP. Bluetooth I2S loop. 0 = normal mode. 1 = loop TX to RX. [8] CODEC_I2S_LOOP. CODEC I2S loop. 0 = normal mode. 1 = loop TX to RX. [7] CODEC_DAAD_LOOP. CODEC DAAD loop. 0 = normal mode. 1 = loop DAC to ADC. [6] CODEC_TEST_SEL. CODEC test selection. 0 = normal mode. 1 = ADC clock/data, DAC clock output to BACH pad. [5] BT_SOURCE_SEL. Bluetooth source selection. 0 = internal Bluetooth. 1 = external Bluetooth. [4] CODEC_I2S_TRX_MCK_INV. CODEC I2S TRX MCLK invert. 0 = normal mode. 1 = invert mode. [3:2] SPDIF_IN_SEL. SPDIF input pad selection. 0 = CODEC I2S MCK. 1 = CODEC I2S SDI. 2 = Bluetooth I2S SDI. 3 = External I2S SDI. [1:0] SPDIF_OUT_SEL. 0 = reserved. 1 = CODEC I2S SDO. 2 = Bluetooth I2S SDO. 3 = CODEC I2S MCK.
1Eh	REG15023D	7:0	Default: 0x00
			Access: R/W

BACH0 Register (Bank = 1502)				
Index (Absolute)	Mnemonic	Bit	Description	
(15023Dh)	PAD0_CFG[15:8]	7:0	See description of '15023Ch.'	
1Fh (15023Eh)	REG15023E	7:0	Default: 0x00	Access: R/W
	PAD1_CFG[7:0]	7:0	0x3E. [15] REG_PAD_OEN_SEL. [14] REG_ADC_SYNC_SEL. [13] REG_PAD_BT_WCK_S_INV. [12] REG_PAD_CODEC_WCK_S_INV. [11] REG_EXT_I2S_RX_WCK_OEN. [10] REG_EXT_I2S_RX_BCK_OEN. [9] REG_EXT_I2S_RX_SDI_OEN. [8] REG_CODEC_I2S_TRX_WCK_OEN. [7] REG_CODEC_I2S_TRX_BCK_OEN. [6] REG_CODEC_I2S_TRX_SDI_OEN. [5] REG_CODEC_I2S_TRX_SDO_OEN. [4] REG_CODEC_I2S_TRX_MCK_OEN. [3] REG_BT_I2S_TRX_WCK_OEN. [2] REG_BT_I2S_TRX_BCK_OEN. [1] REG_BT_I2S_TRX_SDI_OEN. [0] REG_BT_I2S_TRX_SDO_OEN.	
1Fh (15023Fh)	REG15023F	7:0	Default: 0x00	Access: R/W
	PAD1_CFG[15:8]	7:0	See description of '15023Eh.'	
60h (1502C0h)	REG1502C0	7:0	Default: 0x00	Access: R/W
	COUNTER1_CTRL[7:0]	7:0	0xC0. [27:0] REG_COUNTER1_LEVEL. [30:28] REG_COUNTER1_SEL. Select counter source. 000: I2S TDM RX WCK. 001: I2S TDM TX WCK. 010: USB2.0 SOF. 011: USB3.0 SOF. 1xx: Reserved. [31] REG_COUNTER1_TRIG. Clear counter value and trigger counter whenever write 1.	
60h (1502C1h)	REG1502C1	7:0	Default: 0x00	Access: R/W
	COUNTER1_CTRL[15:8]	7:0	See description of '1502C0h.'	
61h (1502C2h)	REG1502C2	7:0	Default: 0x00	Access: R/W
	COUNTER1_CTRL[23:16]	7:0	See description of '1502C0h.'	
61h	REG1502C3	7:0	Default: 0x00	Access: R/W

BACH0 Register (Bank = 1502)				
Index (Absolute)	Mnemonic	Bit	Description	
(1502C3h)	COUNTER1_CTRL[31:24]	7:0	See description of '1502C0h.'	
62h (1502C4h)	REG1502C4	7:0	Default: 0x00	Access: RO
	COUNTER1_STATUS[7:0]	7:0	0xC2. [27:0] REG_COUNTER1_VALUE. DMA write level counter. [30] REG_COUNTER1_BUSY. 1: Busy. 0: Idle. [31] REG_COUNTER1_FLAG. Counter1 value reaches REG_COUNTER1_LEVEL.	
62h (1502C5h)	REG1502C5	7:0	Default: 0x00	Access: RO
	COUNTER1_STATUS[15:8]	7:0	See description of '1502C4h.'	
63h (1502C6h)	REG1502C6	7:0	Default: 0x00	Access: RO
	COUNTER1_STATUS[23:16]	7:0	See description of '1502C4h.'	
63h (1502C7h)	REG1502C7	7:0	Default: 0x00	Access: RO
	COUNTER1_STATUS[31:24]	7:0	See description of '1502C4h.'	
64h (1502C8h)	REG1502C8	7:0	Default: 0xF9	Access: R/W
	ENABLE_CTRL_AU2[7:0]	7:0	0xC8. [11] REG_BP_ADC2_HPF2. Bypass ADC2 HPF2. 0 = normal. 1 = bypass. [10] REG_BP_ADC2_HPF1. Bypass ADC2 HPF1. 0 = normal. 1 = bypass. [9] Reserved. [8] REG_EN_INT_1_AU2. Enable interpolation 1. 0 = disable. 1 = enable. [7] REG_EN_INT_2_AU2. Enable interpolation 2. 0 = disable. 1 = enable. [6] REG_EN_DEC_1_AU2. Enable decimation 1. 0 = disable.	

BACH0 Register (Bank = 1502)

Index (Absolute)	Mnemonic	Bit	Description
			1 = enable. [5] REG_EN_DEC_2_AU2. Enable decimation 2. 0 = disable. 1 = enable. [4] REG_EN_DEC_3_AU2. Enable decimation 3. 0 = disable. 1 = enable. [3] REG_EN_FIR_1_AU2. Enable 9-tap FIR. 0 = disable. 1 = enable. [2] REG_EN_ASRC_AU2. Enable ASRC. 0 = disable. 1 = enable. [0] REG_EN_CODEEC_AU2. Enable CODEC. 0 = disable. 1 = enable.
64h (1502C9h)	REG1502C9	7:0	Default: 0x01 Access: R/W
	ENABLE_CTRL_AU2[15:8]	7:0	See description of '1502C8h.'
65h (1502CAh)	REG1502CA	7:0	Default: 0x00 Access: R/W
	SRO_SEL_AU2[7:0]	7:0	0xCA. [15:14] REG_CIC_1_SEL_AU2. Decimation 1 sampling rate selection. 0: 8KHz. 1: 16KHz. 2: 32KHz. 3: 48KHz. [13:12] REG_CIC_2_SEL_AU2. Decimation 2 sampling rate selection. 0: 8KHz. 1: 16KHz. 2: 32KHz. 3: 48KHz. [11:10] REG_CIC_3_SEL_AU2. Decimation 3 sampling rate selection. 0: 8KHz.

BACH0 Register (Bank = 1502)

Index (Absolute)	Mnemonic	Bit	Description
			1: 16KHz. 2: 32KHz. 3: 48KHz. [9:8] REG_CODECS_SEL_AU2. CODEC 1 sampling rate selection. 0: 8KHz. 1: 16KHz. 2: 32KHz. 3: 48KHz.
65h (1502CBh)	REG1502CB	7:0	Default: 0x00 Access: R/W
	SR0_SEL_AU2[15:8]	7:0	See description of '1502CAh.'
66h (1502CCh)	REG1502CC	7:0	Default: 0x00 Access: R/W
	MUX4_SEL[7:0]	7:0	0xCC. [15:12] REG_ADC2_HPF_N. ADC HPF Number. [11] REG_ADC2_SAT_SEL. ADC HPF saturation select. 0 = normal mode. 1 = old mode. [10:8] REG_SDM_DINL_SEL. 000: AB1_L. 001: AB1_R. 010: (AB1_L+AB1_R)/2. 011: (AB1_L+AB1_R). 100: AB2_L. 101: AB2_R. 110: (AB2_L+AB2_R)/2. 111: (AB2_L+AB2_R). [7] REG_SDM_DINL_INV. 0: SDM_DINL. 1: - SDM_DINL. [6:4] REG_SDM_DINR_SEL. 000: AB1_R. 001: AB1_L. 010: (AB1_L+AB1_R)/2. 011: (AB1_L+AB1_R). 100: AB2_R. 101: AB2_L. 110: (AB2_L+AB2_R)/2. 111: (AB2_L+AB2_R).

BACH0 Register (Bank = 1502)

Index (Absolute)	Mnemonic	Bit	Description
			[3] REG_SDM_DINR_INV. 0: SDM_DINR. 1: – SDM_DINR.
66h (1502CDh)	REG1502CD	7:0	Default: 0x00 Access: R/W
	MUX4_SEL[15:8]	7:0	See description of '1502CCh.'
67h (1502CEh)	REG1502CE	7:0	Default: 0x00 Access: R/W
	HD_SDM_OFFSET[7:0]	7:0	0xCE. HD_SDM_OFFSET [15:0].
67h (1502CFh)	REG1502CF	7:0	Default: 0x00 Access: R/W
	HD_SDM_OFFSET[15:8]	7:0	See description of '1502CEh.'
68h (1502D0h)	REG1502D0	7:0	Default: 0x00 Access: R/W
	HD_SDM_CTRL[7:0]	7:0	0xD0. [15:14] HD_SDM_OFFSET [17:16]. [12] REG_HD_SDM_TEST_EN. HD SDM test enable. 0 = disable. 1 = enable. [11] REG_HD_SDM_SFT_DIS. HD_SDM shift disable. 0 = enable. 1 = disable. [10] REG_HD_DAC_TEST_EN. HD DAC test enable. 0 = disable. 1 = enable. [9] REG_HD_FIX_MSB_EN. HD DWA fixed MSB enable. 0 = disable. 1 = enable. [8:4] REG_HD_FIX_MSB_SEL. HD DWA fixed MSN selection. [3] REG_HD_DITHER_EN. HD SDM dither enable. 0 = disable. 1 = enable. [2:0] REG_HD_DITHER_SEL. HD SDM dither selection.
68h	REG1502D1	7:0	Default: 0x02 Access: R/W

BACH0 Register (Bank = 1502)				
Index (Absolute)	Mnemonic	Bit	Description	
(1502D1h)	HD_SDM_CTRL[15:8]	7:0	See description of '1502D0h.'	
69h (1502D2h)	REG1502D2	7:0	Default: 0x01	Access: R/W
	HD_SDM_CTRL1[7:0]	7:0	0xD2. [15:13] REG_HD_SDM_DINL_SEL. 000: AB1_L. 001: AB1_R. 010: (AB1_L+AB1_R)/2. 011: (AB1_L+AB1_R). 100: AB2_L. 101: AB2_R. 110: (AB2_L+AB2_R)/2. 111: (AB2_L+AB2_R). [12] REG_HD_SDM_DINL_INV. 0: HD_SDM_DINL. 1: – HD_SDM_DINL. [5:4] REG_HD_DWA_LEVEL. 00: 9-level. 01: 17-level. Others: Reserved. [3:2] REG_HD_SDM_LEVEL. 00: 9-level. 01: 17-level. Others: Reserved. [0] REG_HD_SDM_ENABLE. [1] REG_HD_FIRST_ORDER_EN.	
69h (1502D3h)	REG1502D3	7:0	Default: 0x00	Access: R/W
	HD_SDM_CTRL1[15:8]	7:0	See description of '1502D2h.'	
78h (1502F0h)	REG1502F0	7:0	Default: 0x00	Access: R/W
	AU2HDMI_CTRL[7:0]	7:0	0xF0. [15] REG_AU2HDMI_EN. 1: Enable. 0: Disable. [13:12] REG_AU2HDMI_MUX_SEL, select source of AUDIO2HDMI_L/R [31:0]. 00: DMA1 RD. 01: DMA2 RD (reserved). 1x: SRC (sample rate converter output).	
78h (1502F1h)	REG1502F1	7:0	Default: 0x00	Access: R/W
	AU2HDMI_CTRL[15:8]	7:0	See description of '1502F0h.'	

BACH0 Register (Bank = 1502)

Index (Absolute)	Mnemonic	Bit	Description
79h (1502F2h)	REG1502F2	7:0	Default: 0x00 Access: R/W
	AU2HDMI_REF_FS[7:0]	7:0	0xF2. Parameter of the referenced sample signal "AUDIO2HDMI_1K." If (AU2HDMI_REF_FS == 0), Freq. of AUDIO2HDMI_1K is 1KHz. Otherwise, Freq. of AUDIO2HDMI_1K is (48MHz/AU2HDMI_REF_FS).
79h (1502F3h)	REG1502F3	7:0	Default: 0x00 Access: R/W
	AU2HDMI_REF_FS[15:8]	7:0	See description of '1502F2h.'



REGISTER TABLE REVISION HISTORY

Date	Bank	Register
12/01/2020		Created first version